

IT6663/IT6664 Programming Guide

Ver 0.70

Jau-chih.tseng

ITE Tech. Inc.

Last Update Date: 2017/06/20

History

2017/4/18 – Create this document.

2017/6/20 – Revise.

Index

<i>Chap 1 Introduce</i>	1
HARDWARE PROGRAMMING GUIDE.....	2
<i>Chap 2 Initial</i>	2
Read IC Internal Clock	2
Initial Sequence	3
Chip's basic initial sequence	3
<i>Chap 3 HDMI Receiver Configuration</i>	4
EDID Configuration	4
Enable Internal EDID	4
Disable Internal EDID.....	4
Input Clock Detection.....	4
<i>Chap 4 HDMI Splitter Converter Configuration</i>	5
HDMI Splitter signal path select.....	5
Color space register settings	7
Converter matrix register settings.....	7
<i>Chap 5 HDMI Transmitter Configuration</i>	9
Set TMDS Output AFE.....	9
Control GCP	10
SOFTWARE PORTING NOTE	15
<i>Chap 9 Software Export API</i>	15
REGISTER DEFINITION.....	16
<i>Chap 10 HDMI Splitter Convert Registers</i>	16
Bank 0.....	16
Bank 1.....	21
<i>Chap 11 HDMI Rx Registers</i>	23
HDMIRX Register Bank	23
Block 0.....	23
Device Information.....	23
Interrupt.....	23
Device internal status	24
Device Reset/Clock/PowerDown related register.....	25
MHL/HDMI FrontEnd control.....	26
FrontEnd IO Common control.....	29
Video Back End Control Registers	30
Packet Control Registers	30
Device internal Status ReadBack.....	31
Port 0 Internal EDID Register	33
Port 0 HDCP registers	33
Block 1.....	35
HDCP Repeater Return Registers.....	35
Block 2.....	36
SPI Infoframe	36
AVI Infoframe	36
General Purpose Infoframe.....	36
Audio Infoframe	37
MPEG Infoframe.....	37
ACP Packet	37
ISRC Packets.....	37
HDCP AKSV/BKSV/Mi read back	38
Audio N/CTS read back	38
Audio Meta read back	38
Vendor Specific Infoframe	39
Block 3 AutoEQ And AFE related registers.....	40
MHL/HDMI FrontEnd control.....	40
Port 0 CBUS IO control	40
Port 0 EQ1 Read Back.....	40
Port 0 SCDC Registers	41
SCDC Registers.....	42
Block 6 DRM Packet Data.....	42
Dynamic Range and Mastering	42
<i>Chap 12 MHL Registers</i>	43

<i>Chap 13</i>	<i>HDMI TX Register Definitions</i>	50
	HDMI Tx Common Registers	50
	General Registers	50
	TX General Packet Registers	51
	TX Pattern Generator Registers	51
	TX PatGen Video Packet Content Registers	51
<i>Chap 14</i>	<i>HDMI Tx Port0~Port3 Registers</i>	52
	General Registers	52
	System Reset & clock control	52
	System Control Registers	52
	Interrupt Status Registers	53
	Interrupt Enable Registers	54
	DDC and SCDC Control Registers	54
	HDCP Registers	56
	AFE Control Registers	59
	Input Data Processing Registers	60
	Packet FIFO Registers	61
	HDMI Control Registers	62
	Video Record Registers	63
	Misc. Registers	64
<i>Chap 15</i>	<i>CEC Registers in Bank1</i>	66
	CEC Control Registers	66
	CEC Initiator Registers	67
	CEC Follower Registers	67
	CEC Misc. Registers	68

Chap 1 Introduce

IT6664 is a one-to-four way HDMI 2.0 splitter. There are the following components inside:

- HDMI Splitter Converter – I²C address is 0x58 (PCADR pin pulled low) or 0x5A (PCADR pin pulled high).
Following represented as **HDMISPRegXX**.
- HDMI 2.0 Receiver – I²C address is defined in HDMI Converter register 0xF0.
Following represented as **HDMIRxRegXX**.
- HDMI 2.0 TX, Common Controller – I²C address is defined in HDMI Converter register 0xF1.
Following represented as **HDMITxRegXX**.
- HDMI 2.0 TX port 0~port 4 – I²C addresses are defined in HDMI TX common controller register 0x2C~0x2F.
Following represented as **HDMIPort[0|1|2|3]RegXX**.
- MHL controller – I²C address are defined in HDMI Rx register 0x49.
Following represented as **MHLRegXX**.
- Embedded EDID RAM – I²C address are defined in HDMI RX register 0x4B.

Hardware Programming Guide

Chap 2 Initial

Read IC Internal Clock

In order to get accuracy ring clock of the it666x , we will read the calibrated value from chip , the ring clock data read sequence show as below:

Register w/r (i2c address 0x58)	Register offset	data
write	0xFF	0xc3
write	0xFF	0xa5
write	0xF	0x00
write	0x5F	0x04
write	0x5F	0x05
write	0x58	0x12
write	0x58	0x02
write	0x57	0x01
write	0x50	0x00
write	0x51	0x00
write	0x54	0x04
Read	0x61	Data1 (LSB)
Read	0x62	Data2 (MSB)

We can get clock data by calculating “Data2 <<8 + Data1”

for example : 0x54D8 , it means clock is 21720 Hz

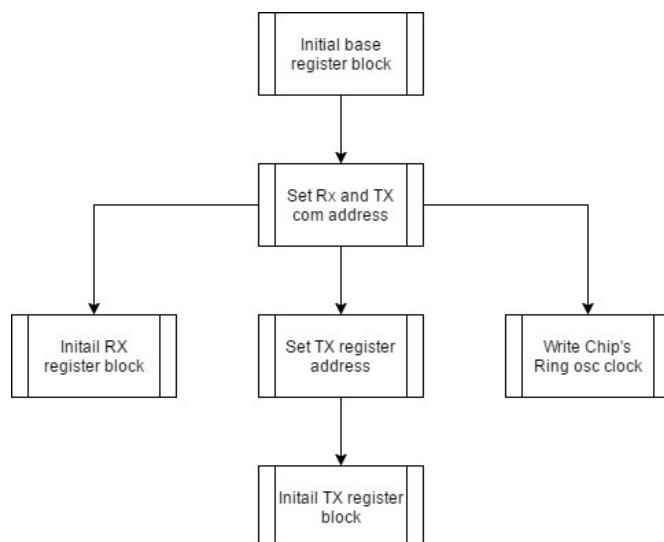
If the ring clock frequency is less than 24MHz, set i2c address 0x58 Reg10[4:0] can increase ring clock frequency. Suggest set reg10[4:0] from ‘01100’ (0x0C) to ‘01110’ (0x0E).

10		7:0			0x6C
	Reg_RING2_PWDB	6	0: power down Ring2(for HP2)	R/W	1
	Reg_RING_PWDB	5	0: power down Ring1(non-recoverable)	R/W	1
	Reg_RING	4:0	Ring frequency parameter Default: 01100 00000: min rate, typically 55% of default 11111: max rate, typically 150% of default	R/W	01100

Initial Sequence

Chip's basic initial sequence

The basic initial sequence you can find in function "it6664_hdmi2sp_initial" at "IT6664_hdmi2sp.c"



Function name	Description
it6664_hdmi2sp_initial	Main initial function
h2rx_ini	Rx's register initial
h2txcom_ini	Tx's comment register initial
h2tx_ini	Tx's register initial
it6664_hdmi2sp_rst	Set Tx's slave address and initial register
h2rx_rst	Rx's function block reset
h2tx_rst	Tx's function block reset

Chap 3 HDMI Receiver Configuration

EDID Configuration

Enable Internal EDID

34	Reg_AutoRCLK	0	1: Auto gate RCLK when PWR5V not detected	R/W	1
	Reg_AutoMRCLK	1	1: Auto gate MHL RCLK when CDSense not detected		
	RegEnHP1AutoPWD	2	1: Auto power down HDCP 1.4 when HDCP2 enabled	R/W	0
	Reserved	3			
	Reg_GateRCLK	4	1: Gate RCLK for backend module	R/W	0
	Reg_GatePCLK	5	1: Gate CLK	R/W	0
	Reg_GateOCLK	6	1: Gate OCLK for MHL module	R/W	0
	Reserved	7			
C5	Reg_P0_DisableShadow	0	1: disable Port 0 Internal EDID	R/W	1
	Reg_P0_EnDetDDC	1	1: Enable monitor DDC detect start	R/W	1
	Reg_P0_EDIDAdrSel	2	1: EDID address A2 ; 0: EDID address A0	R/W	0
	Reserved	3			
	Reg_P0_SoftEDIDRst	4	1: EDID reset	R/W	0
	Reg_P0_EnMultiSeg	5	1: Enable multiple segment	R/W	0
	Reg_P0_EDIDTimeOutOpt	7:6	Timer to adjust EDID Bus Hang	R/W	00

1. Disable Reg_AutoRCLK (RX_Reg34 bit0 set to 0)setting.
2. Write internal EDID to I2C address 0x D8
3. Write EDID' ABCD value to RX_RegC7 and RX_RegC8

Write EDID's checksum to RX_RegC9 and RX_RegCA

C6	Reg_P0_VSDBAdr	7:0	VSDB Offset	R/W	n.a.
C7	Reg_P0_AB	7:0	Port 0 EDID's AB value	R/W	n.a.
C8	Reg_P0_CD	7:0	Port 0 EDID's CD value	R/W	n.a.
C9	Reg_P0_B0Sum	7:0	Port 0 EDID block 1's checksum value	R/W	n.a.
CA	Reg_P0_B1Sum	7:0	Port 0 EDID block2's checksum value	R/W	n.a.

4. Write for EDID i2c slave address

Reg	Register Name	Bit	Bit Description	Default
0x4B	RegEDIDSlaveAdr	7:1	I2C Slave Address for EDID	0x00
	RegEnEDIDSlaveAdr	0	1: Enable access EDID block	

5. Enable internal EDID function. (Set RX_RegC5 bit0 to 0)
6. Enable Reg_AutoRCLK.

Disable Internal EDID

Set RX_RegC5(bit0) to 1 can disable internal EDID function.

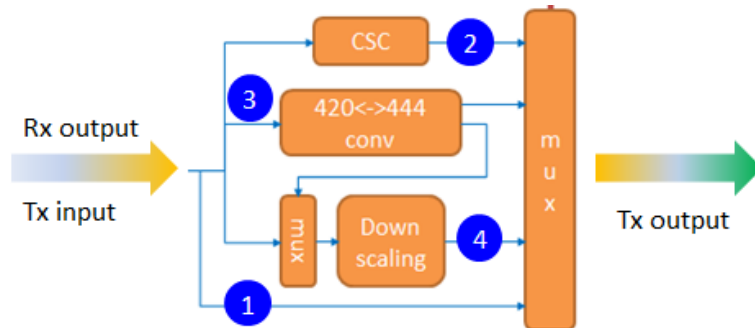
Input Clock Detection

14	P0_IPLL_HS1G	0	Port 0 IPLL clock is higher than 100MHz	RO	
	P0_IPLL_HS1P48G	1	Port 0 IPLL clock is higher than 148MHz	RO	
	P0_IPLL_HS3P4G	2	Port 0 IPLL clock is higher than 340MHz	RO	
	P0_CH0_SymLockSts	3	Port 0 Symbol Lock of DIGCDR_DATA	RO	
	P0_CH1_SymLockSts	4	Port 0 Symbol Lock of DIGCDR_DATA	RO	
	P0_CH2_SymLockSts	5	Port 0 Symbol Lock of DIGCDR_DATA	RO	
	P0_SCDC_CLKRatio	6	Port 0 SCDC Clock Ratio bit	RO	
	P0_SCDC_ScrEn	7	Port 0 SCDC Scramble Enable Bit	RO	

In status register Reg14[2:0], we can judge the input clock is higher or lower than 3G, Reg14[7:6] show the SCDC setting that it6664 received.

Chap 4 HDMI Splitter Converter Configuration

HDMI Splitter signal path select



IT6664 can deal with color space convert of the input signal by selecting paths as shown in the diagram above , and there are four paths as follows:

- Path1 : RX signal bypass
- Path2 : RX signal color space convert
- Path3 : RX signal YUV420 <=> YUV444 convert
- Path4 : RX signal scaling down

We can set Reg0x0D[7:0] at i2c address 0x58 to select which path we want to output for different ports

0D		7:0			0x00
	RegTx3SrcSel	7:6	Tx3 Source selection. See bit 1:0	R/W	00
	RegTx2SrcSel	5:4	Tx2 Source selection. See bit 1:0	R/W	00
	RegTx1SrcSel	3:2	Tx1 Source selection. See bit 1:0	R/W	00
	RegTx0SrcSel	1:0	Tx0 Source selection: 00: Original 01:CSC 10: 420/444 conv 11: Downsample	R/W	00

In Path1 , chip didn't do anything on input signal , so the output signal of it6664 is just the same as the input signal.

In Path2 , chip will do color space convert like RGB => YUV422 , YUV444 => RGB ...etc .

In Path3 , chip will do color space convert for YUV420 <=> YUV444/RGB

In Path4 , chip will do down scale from 3840x2160 to 1920x1280, if input resolution is 4096x2160 , when signal pass through down scaling path ,then we will get 2048x1080P out.

The Path2 settings will show in the table with blue section as below , and the Path3 settings show with yellow section :

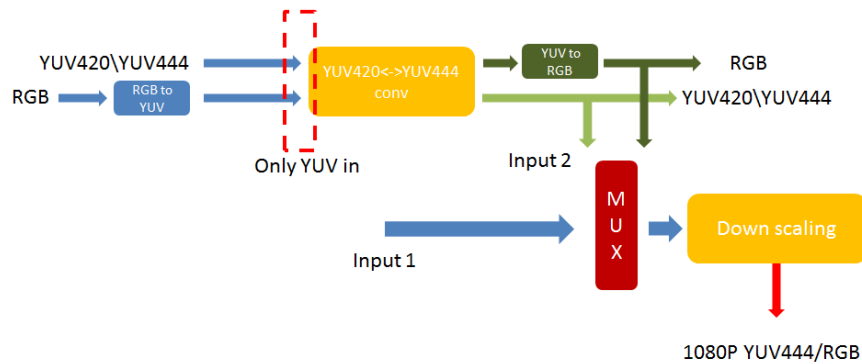
		7:0		R/W	0x40
	RegAutoCSCSel	6	1: CSC direction auto selected with in/out color mode 0: Force		1
6B	Reg_CVOutColMod	5:4	CV path output color mode when Reg0x6C(3)=1	R/W	00
	Reg_OutColMod	3:2	CSC path output color mode	R/W	00
	Reg_CSCSel	1:0	Used when bit6 = 0 00: bypass CSC 10: RGB to YUV	R/W	00

			11: YUV to RGB		
		7:0		R/W	0x10
	RegEn444DBypass	7		R/W	0
	RegEn444CBypass	6		R/W	0
	RegEn420to444	5	Forced when bit3 = 1, else auto set when input is 420	R/W	0
	RegEn444to420	4	Forced when bit3 = 1, else auto set when output is set to 420	R/W	1
6C	RegForce444420	3	1: Force RegEn420to444 or RegEn444to420 0: auto set with in/out color mode(444 out when input is 420, else 420 out)		
	Reg_ForceCVDepth	2	1: force CV block input Depth 0: CV block input depth as RX input	R/W	0
	Reg_CVdepth_set	1:0	See bit 2	R/W	00

Reg6B(6) is RegAutoCSCSel , if Path2 was selected , chip will auto select color space convert by input color space , ex : input YUV , CSC matrix will auto select YUV to RGB.

If we want to force color space convert , the Reg6B(6) should be 0 , and Reg6B(5:4) set RGB(0)/YUV422(1)/YUV444(2) , Reg6B(1:0) select color space matrix.

Reg6C is register setting for Path3 , the default value of Reg6C(3) is 0, chip will auto convert YUV420 <=> RGB , if we want YUV444 <=> YUV240 , Reg6C(3) should be set to 0 and Reg6C(5) / Reg6C(4) should be set to 1 , then set Reg6B(5:4) to YUV444(2).



This diagram describes the relationships of Path3 and Path4 , we can see that there are two inputs can be down scaling to 1080P, one is coming from original signal input and the other one is coming from Path3's output , our hardware limitation is Path4 can't do color space convert again , if one of TX ports connected to a DVI monitor and the original input signal is 4K60 YUV444, we will output 1080P YUV444 and you will see the monitor screen is pink because of DVI monitor did not support YUV444.

Color space register settings

Color space converting table		RGB to YCbCr				YCbCr to RGB			
		RGB to YCbCr 601		RGB to YCbCr 709		YCbCr to RGB 601		YCbCr to RGB 709	
	reg	RGB Limit to YCbCr Limit	RGB Full to YCbCr Limit	RGB Limit to YCbCr Limit	RGB Full to YCbCr Limit	YCbCr Limit to RGB Limit	YCbCr Limit to RGB Full	YCbCr Limit to RGB Limit	YCbCr Limit to RGB Full
Reg_CSCSel[1:0]	6B[1:0]	10	10	10	10	11	11	11	11
Reg_CSCOffset_Signed	6D[6]	0	0	0	0	0	0	0	0
Reg_CSCOffset_YSign	6D[5]	0	0	0	0	0	0	0	0
Reg_CSCOffset_CrSign	6D[4]	0	0	0	0	0	0	0	0
Reg_CSCOffset_CbSign	6E[6]	0	0	0	0	0	0	0	0
Reg_CbOffset	6E[5]	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x00
Reg_YoffSet[7:0]	70	0x00	0x10	0x00	0x10	0x00	0x04	0x00	0x04
Reg_CoffSet[7:0]	71	0x80	0x80	0x80	0x80	0x00	0x00	0x00	0x00
Reg_RGBOffSet[7:0]	72	0x10	0x10	0x10	0x10	0x00	0xA7	0x00	0xA7
Reg_Matrix11V[13:0]	73	0xB2	0x09	0xB8	0xE4	0x00	0x4F	0x00	0x4F
	74	0x04	0x04	0x05	0x04	0x08	0x09	0x08	0x09
Reg_Matrix12V[13:0]	75	0x65	0x0E	0xB4	0x77	0x6B	0x81	0x55	0xBA
	76	0x02	0x02	0x01	0x01	0x3A	0x39	0x3C	0x3B
Reg_Matrix13V[13:0]	77	0xE9	0xC9	0x94	0x7F	0x50	0xDD	0x88	0x4B
	78	0x00	0x00	0x00	0x00	0x3D	0x3C	0x3E	0x3E
Reg_Matrix21V[13:0]	79	0x93	0x0F	0x4A	0xD0	0x00	0x4F	0x00	0x4F
	7A	0x3C	0x3D	0x3C	0x3C	0x08	0x09	0x08	0x09
Reg_Matrix22V[13:0]	7B	0x18	0x84	0x17	0x83	0xF5	0xC4	0x51	0x57
	7C	0x04	0x03	0x04	0x03	0x0A	0x0C	0x0C	0x0E
Reg_Matrix23V[13:0]	7D	0x55	0x6D	0x9F	0xAD	0x02	0x01	0x00	0x02
	7E	0x3F	0x3F	0x3F	0x3F	0x00	0x00	0x00	0x00
Reg_Matrix31V[13:0]	7F	0x49	0xAB	0xD9	0x4B	0x00	0x4F	0x00	0x4F
	80	0x3D	0x3D	0x3C	0x3D	0x08	0x09	0x08	0x09
Reg_Matrix32V[13:0]	81	0x9F	0xD1	0x10	0x32	0xFD	0xFD	0x00	0xFE
	82	0x3E	0x3E	0x3F	0x3F	0x3F	0x3F	0x00	0x3F
Reg_Matrix33V[13:0]	83	0x18	0x84	0x17	0x84	0xDA	0x1F	0x84	0xE8
	84	0x04	0x03	0x04	0x03	0x0D	0x10	0x0E	0x10

Converter matrix register settings

Color space converting table		RGB to YCbCr		YCbCr to RGB	
		RGB to YCbCr 601 RGB Full to YCbCr Full (limit to limit)	RGB to YCbCr 709 RGB Full to YCbCr Full (limit to limit)	YCbCr to RGB 601 YCbCr Full to RGB Full (limit to limit)	YCbCr to RGB 709 YCbCr Full to RGB Full (limit to limit)
	reg				
Reg_CSCSel[1:0]	6B[1:0]	10	10	11	11
Reg_QuanL4b	6D[6]	1	1	1	1
Reg_CRCBLimit	6D[5]	1	1	0	0
Reg_ColorClip	6D[4]	0	0	0	0
Reg_CSCOffset_Signed	6E[6]	0	0	0	0
Reg_CSCOffset_YSign	6E[5]	0	0	0	0
Reg_CSCOffset_CrSign	6E[4]	0	0	0	0
Reg_CSCOffset_CbSign	6E[3]	0	0	0	0
Reg_CbOffset	88	0x00	0x00	0x00	0x00
Reg_YoffSet[7:0]	89	0x00	0x00	0x00	0x00
Reg_CoffSet[7:0]	8A	0x80	0x80	0x00	0x00
Reg_RGBOffSet[7:0]	8B	0x10	0x10	0x00	0x00
Reg_Matrix11V[13:0]	93	0xB2	0xB8	0x00	0x00
	94	0x04	0x05	0x08	0x08
Reg_Matrix12V[13:0]	95	0x65	0xB4	0x6B	0x55
	96	0x02	0x01	0x3A	0x3C
Reg_Matrix13V[13:0]	97	0xE9	0x94	0x50	0x88
	98	0x00	0x00	0x3D	0x3E

Reg_Matrix21V[13:0]	99	0x93	0x4A	0x00	0x00
	9A	0x3C	0x3C	0x08	0x08
Reg_Matrix22V[13:0]	9B	0x18	0x17	0xF5	0x51
	9C	0x04	0x04	0x0A	0x0C
Reg_Matrix23V[13:0]	9D	0x55	0x9F	0x02	0x00
	9E	0x3F	0x3F	0x00	0x00
Reg_Matrix31V[13:0]	9F	0x49	0xD9	0x00	0x00
	A0	0x3D	0x3C	0x08	0x08
Reg_Matrix32V[13:0]	A1	0x9F	0x10	0xFD	0x00
	A2	0x3E	0x3F	0x3F	0x00
Reg_Matrix33V[13:0]	A3	0x18	0x17	0xDA	0x84
	A4	0x04	0x04	0x0D	0x0E

To implement Path1~Path4 setting, we need to write color space matrix register and converter matrix register to i2c address 0x58 , the register offset and register value list in table as above.

The color space matrix registers should be write to HDMI convert registers(0x58) from 0x70 to 0x84 , and the color convert matrix of Path3 should be write from 0x93 to 0xA4.

Chap 5 HDMI Transmitter Configuration

Set TMDS Output AFE

For following HDMI register for each port.

83	Reg_XP_SPARE[2:0]	2:0	XP_SPARE[2:0]: Reserved	RW	000
	Reg_XP_HS2	3	0: HDMI1.4 mode, 1: HDMI2.0 mode	RW	0
		7:4			
84	Reg_XP_EC1	0	when input PCLK05 frequency < 100MHz	RW	0
	Reg_XP_ER0	1	set XP_EC1=1	RW	0
	Reg_XP_GAINBIT	2	set XP_ER0=1 set XP_GAINBIT=0	RW	1
	Reg_XP_ENI	3	1: the charge pump current of TXPLL is increased.	RW	0
	Reg_XP_DEK	4	0: will increase TXPLL KVCO	RW	0
	Reg_XP_PWDI	5	0: TXPLL current bias is not in power-down mode. 1: TXPLL current bias in power-down mode, no current source available for all other blocks such as IPLL/TXPLL driver/ring osc... Normally, XP_PWDI should always be set to 0.	RW	0
	Reg_XP_PWDPLL	6	Power down signal for TXPLL 0: Normal operation 1: TXPLL is power downed	RW	0
	Reg_XP_RESETB	7	Low-active reset signal for TXPLL 0: TXPLL is reset 1: Normal operation	RW	1
85	Reg_XP_TEST[7:0]	7:0	Test purpose usages Reg_XP_TEST 0: ENAC Reg_XP_TEST 1: DISRC Reg_XP_TEST 2: ENCTPI Reg_XP_TEST 3: DER Reg_XP_TEST 4: DEI Reg_XP_TEST 5: ENVC Reg_XP_TEST 6: XPIG Reg_XP_TEST 7: IPIG	RW	0x00
86	Reg_XP_BYPASS	0	Input register, default 0, only used in test mode useful only when PRDIV2=0 and FB DIV2=0 When BYPASS=1, Directly bypass input reference to TXVCLK	RW	0
	Reg_XP_ER1	1	TXPLL_ER1=0 will not increase filter resistance TXPLL_ER1=1 will increase filter resistance	RW	0
	Reg_XP_SNPFD	2	0: enable lv PFD, 1: enable hv PFD	RW	0
	Reg_XP_ALPWDB	3	XP_ALPWDB=0, txpll and tipll pvcc10 power is off XP_ALPWDB=1, txpll and tipll pvcc10 power is on Note: do not set XP_ALPWDB=0 and XP_PWDPLL=0	RW	1
		7:4			
87	Reg_DRV_ISW[4:0]	4:0	Output data channel current level setting Default: 3mA (always on) ISW0: +1.25mA ISW1: +2.5mA ISW2: +5mA ISW3: +10mA ISW4: +4.5mA	RW	00101
		7:5			
88	Reg_DRV_RST	0	Reset signal for HDMI_TX_DRV 1: all flip-flops in the transmitter are reset.	RW	1
	Reg_DRV_PWD	1	Power down signal for HDMI_TX_DRV 1: all flip-flops in the transmitter are reset while all other analog parts are powered off.	RW	0
	Reg_DRV_HS	2	0: VCLK05 slower than 81MHz 1: VCLK05 higher than 81MHz When 1, setting 4-to-1 MUX to high-speed operation mode.	RW	0
	Reg_DRV_PDRXDET	3	0: Normal operation 1: Power down monitor sense (RX detection) circuits	RW	0
	Reg_DRV_PISW[1:0]	5:4	Data channel pre-driver current level setting	RW	01
	Reg_DRV_PISWC[1:0]	7:5	Clock channel pre-driver current level setting	RW	00

89	Reg_DRV_RTTERM[4:0]	4:0	Termination value. This value will impact HDMI differential signals slew rate, swing and eye diagram. The Reg_DRV_TERMON shall be 1 to have this termination active. Note: [4:3] is reserved for ECO option	RW	00000
	Reg_DRV_TERMON	5	Reserved, this bit must be 0 for normal operation. 0: transmitter termination resistors are disabled 1: transmitter termination resistors are enabled	RW	0
	Reg_DRV_PDB_REXTIN	6	1: internal Rext(10K), 0: external Rext(10K)	RW	1
	Reg_DRV_NOPE	7	0: enable pre-emphasis ,1: disable pre-emphasis function	RW	1
8A	Reg_DRV_TPPE[3:0]	0	Pre-emphasis filter register: TPPE0=0 CCTRL is low TPPE0=1 CCTRL is high	RW	0
		1	TPPE1=0 disable pre-emphasis, enable normal differential driver TPPE1=1 enable pre-emphasis	RW	0
		2	TPPE2=0 disable pre-emphasis current source TPPE2=1 enable pre-emphasis current source	RW	0
		3	TPPE3=1 enable pre-driver pre-emphasis current source TPPE3=0 disable pre-driver pre-emphasis current source	RW	0
	Reg_DRV_IRS	4	R channel current source fine tuning	RW	0
	Reg_DRV_IGS	5	G channel current source fine tuning	RW	0
	Reg_DRV_IBS	6	B channel current source fine tuning	RW	0
	Reg_DRV_ICS	7	C channel current source fine tuning	RW	0
8B	Reg_DRV_ISW_C[4:0]	4:0	Output clock channel current level setting Default: 3mA (always on) ISWC0: +1.25mA ISWC1: +2.5mA ISWC2: +5mA ISWC3: +10mA ISWC4: +4.5mA	RW	00001
	Reg_DRV_ISWP	5	Data channel pre-emphasis option current	RW	0
	Reg_AFEP_RBSSEL	6	0: PRBS7, 1: PRBS15	RW	0
	Reg_AFE_ENPRBS	7	0: Normal operation 1: Enable AFE PRBS function, used for test only.	WP	0

Setting for Different Video Clock (TMDS character rate / 10) (for HDMI 2.0 mode, TMDS clock x 4, other as TMDS clock).

VCLK	VCLK < 100MHz	VCLK > 100MHz
Reg84[2:0]	'011'	'100'

ISWC4

VCLK	VCLK < 162MHz	VCLK > 162MHz
Reg88[4]	'0'	'1'

VCLK	VCLK < 150MHz	150MHz ~ 340MHz	340MHz ~ 375 MHz	> 375MHz
reg89[5]	0	1	1	1
reg89[4:0]	00000	00001	00101	00101
reg87[4:0]	00011	01001	01110	01110
reg8B[3:0]	0011	01001	01011	01110
reg8A[3:0]	0000	0000	0000	0000
reg89[7]	1	0	0	0

Output

reg88[1:0] = '00'.

Control GCP

C1	RegAVMute	0	Set AVMute 0: not AVMute, 1: AVMute AVMute <= Rx AVMute or RegC1[0] = '1' ;	RW	1
	RegEnTx2VSync	1	Tx2V FIFO output sync for 1-to-2 Splitter	RW	0

			0: disable, 1: enable		
	RegNoDefPhase	2	0: Not assign initial Phase Indication in Deep Color mode 1: Assign initial Phase Indication in Deep Color mode	RW	0
	RegPhaseReSync	3	0: Normal operation 1: Re-sync Phase Indication	RW	0
	RegColorDepth[2:0]	6:4	HDMI output color depth 000: 8-bit per color component 100: 8-bit per color component 101: 10-bit per color component 110: 12-bit per color component	RW	000
	RegAutoColorDepth	7	1: Auto ColorDepth from input	RW	1
C3	RegSSCPPos	0	HDMI2.0 SSCP position 0: after VDE, 1: after VSync	RW	0
	RegSSCPOpt	1	HDMI2.0 SSCP option when SSCP position is after VDE 0: before packet, 1: after packet	RW	0
	RegSSCPTest	2	HDMI2.0 SSCP test 0: SSCP per frame, 1: SSCP per line	RW	0
	RegEnSSCPTrgPkt	3	HDMI2.0 SSCP end to trigger packet per frame 0: disable, 1: enable	RW	0
	RegPktGenCtrlEn	4	Enable General Control packet 0: disable, 1: enable	RW	0
	RegPktGenCtrlRpt	5	Repeat General Control packet 0: send once, 1: one for each field	RW	0
		7:6			

regC3 = '11', GCP can output for activate the HDMI controlling.

regC1[0] = '1' : overwrite the AVMMute as set then the input AVMMute.

regC1[7] = '1', overwrite the color format of output on regC1[6:4].

Software Porting Note

Chap 9 Software Export API

In reference code , we need two basic I2C functions to write register values

1.i2c_read_byte

(unsigned char address, unsigned char offset, unsigned char byteno, unsigned char *p_data, unsigned char device)

2. i2c_write_byte

(unsigned char address, unsigned char offset, unsigned char byteno, unsigned char *p_data, unsigned char device)

Register Definition

Chap 10 HDMI Splitter Convert Registers

I²C address is 0x58 (PCADR='0') or 0x5A (PCADR='1')

Register block bank is in register 0xF[0], where reglxx means the register in block/bank 1 accessed with I²C sub-address xx.

HDMI Rx Common register I²C address and HDMI Tx Common register I²C address are defined in the register 0xF0/0xF1 of this I²C address registers.

F0	RegRxComSlvAdr	7:0	RX PCCmd port Address Bit0: enable	R/W	0x70
F1	RegTxComSlvAdr	7:0	TX common PCCmd port Address Bit0: enable	R/W	0x96

R/W: read/write register

RO: read only register

W1C: write '1' clear status

W1P: write '1' to generate 1T trigger pulse

Bank 0

Reg	Register Name	Bit	Definition	Type	Default Value
Device Information					
00	VendorID[7:0]	7:0	Vendor ID Low Byte	RO	0x54
01	VendorID[15:8]	7:0	Vendor ID High Byte	RO	0x49
02	DeviceID[7:0]	7:0	Device ID Low Byte	RO	0x64
03	DeviceID[15:8]	7:0	Device ID High Byte	RO	0x66
04	ID	7:4		RO	0xA
Interrupt					
05	Rsvd	7			
	RCVINTB	6	Converter interrupt B(HDMI)	RO	
	RCVINTA2	5	Converter interrupt A(HDCP2)	RO	
	RRxIntSts	4	Rx Interrupt	RO	
	RP3TXIntOut	3	Tx port3 Interrupt	RO	
	RP2TXIntOut	2	Tx port2 Interrupt	RO	
	RP1TXIntOut	1	Tx port1 Interrupt	RO	
	RP0TXIntOut	0	Tx port0 Interrupt	RO	
06					
	RSA_failInt	6	1: RX RSA operation failed	W1C	
	RSHA1_doneInt	5	1: HP2 engine SHA1 command completed	W1C	
	RHP2_UpdtInt	4	1: RX HDCP2 Authentication updated	W1C	
	RHP2_EncChgInt	3	1: RX HDCP2 Encen change detected	W1C	
	RHP2_OffDetInt	2	1: RX HDCP2 EncEn/Dis off detected	W1C	
	RHP2Auth_DoneInt	1	1: RX HDCP2 Authentication done	W1C	
	RHP2Auth_StartInt	0	1: RX HDCP2 Authentication start received	W1C	
07		7:0			
	RFwTimer1Int	5	1: Timer0 reach interrupt(used with 0x1D)	W1C	
	RFwTimer0Int	4	1: Timer0 reach interrupt(used with 0x1C)	W1C	
	RDSFIFOErrInt	3	1: Down scale path FIFO fail	W1C	
	RCVFIFOErrInt	2	1: 420/444 path FIFO fail	W1C	
	RCSCFIFOErrInt	1	1: CSC path FIFO fail	W1C	
	RORIFIFOErrInt	0	1: Original path FIFO fail	W1C	
08		3:0			0x00
	RegEnTxPort3	3	1: Tx port3 enabled 0: disabled	R/W	0
	RegEnTxPort2	2	1: Tx port2 enabled 0: disabled	R/W	0
	RegEnTxPort1	1	1: Tx port1 enabled 0: disabled	R/W	0

	RegEnTxPort0	0	1: Tx port0 enabled 0: disabled	R/W	0
0A		7:0		R/W	0x00
	RegORst	7	1: Reset Random generation block	R/W	0
	RegGateNHRCLK	6	1: power down RX RCLK	R/W	0
	RegGateCVRCLK	5	1: power down Converter RCLK	R/W	0
	RegORIRXPRst	4	1: Reset original RX side PCLK	R/W	0
	RegORITXPRst	3	1: Reset original TX side PCLK	R/W	0
	RegRXHP2Rst	2	1: Reset RX HDCP2	R/W	0
	RegCVRst	1	1: Reset Converter RCLK Register and logic	R/W	0
	RegSoftRstAll	0	1: Reset all chip	R/W	0
0B		7:0		R/W	0x00
	RegDSTXPRst	7	1: Reset Down sample FIFORD clock domain	R/W	0
	RegDSWPRst	6	1: Reset Down sample output clock domain	R/W	0
	RegDSRXPRst	5	1: Reset Down sample input clock domain	R/W	0
	RegCVTXPRst	4	1: Reset Conv FIFORD clock domain	R/W	0
	RegCVWPRst	3	1: Reset Conv output PCLK	R/W	0
	RegCVRXPRst	2	1: Reset Conv input PCLK	R/W	0
	RegCSCTXPRst	1	1: Reset CSC TX side PCLK	R/W	-0
	RegCSCRXPRst	0	1: Reset CSC RX side PCLK	R/W	0
0C		7:2			0x00
	RegCVdoRst	3	1: Reset all fifo path	R/W	0
	RegKeyreset	2	1: Reset HDCP key memory	R/W	0
0D		7:0			0x00
	RegTx3SrcSel	7:6	Tx3 Source selection. See bit 1:0	R/W	00
	RegTx2SrcSel	5:4	Tx2 Source selection. See bit 1:0	R/W	00
	RegTx1SrcSel	3:2	Tx1 Source selection. See bit 1:0	R/W	00
	RegTx0SrcSel	1:0	Tx0 Source selection: 00: Original 01: CSC 10: 420/444 conv 11: Downsample	R/W	00
0E		7:0			0x20
	RegCmdHoldT	7:4	Hold time cnt of I2C Cmd	R/W	0010
	RegRCLKFreqsel	1:0	RCLK freq from RingOSC 00: div2 01: div4 10: div8	R/W	00
0F		1:0			0x00
	RegBlocksel	0	Block select of Conv register	R/W	0
10		7:0			0x6C
	Reg_RING2_PWDB	6	0: power down Ring2(for HP2)	R/W	1
	Reg_RING_PWDB	5	0: power down Ring1(non-recoverable)	R/W	1
	Reg_RING	4:0	Ring frequency parameter Default: 01100 00000: min rate, typically 55% of default 11111: max rate, typically 150% of default	R/W	01100
11		7			0x00
	regEn100msCnt	7	1: Enable 100ms timing calibration 0: End calibration	R/W	0
12	rclk100mscnt[7:0]	7:0	100 ms calibration result[7:0]	RO	
13	rclk100mscnt[15:8]	7:0	100 ms calibration result[15:8]	RO	
14	rclk100mscnt[23:16]	7:0	100 ms calibration result[23:16]	RO	
15	HWRevision[3:0]	3:0	Hardware Revision Number	RO	0x0
18	CVIntMask0	6:0	Interrupt MASK of Reg0x06	R/W	2F
19	CVIntMask1	5:0	Interrupt MASK of Reg0x07	R/W	0F
1C		7:0			0x9E
	RegTimer0IntUnit	7	Timer0 interrupt unit 1: 100 ms 0: 10 ms	R/W	1
	RegTimer0IntNum	6:0	Timer0 time number(*100/10ms)	R/W	0011110
1D		7:0			0x9E
	RegTimer1IntUnit	7	Timer1 interrupt unit 1: 100 ms 0: 10 ms	R/W	1
	RegTimer1IntNum	6:0	Timer1 time number(*100/10ms)	R/W	0011110
1E	Reg1usTimeInt	5:0	1us period number(used with calibration)	R/W	0x14
1F	RegTimeFit	7:0	1us period floating number(used with calibration)		0x00
2B	Reg_DSMax_Value_YRGB[7:	7:0	Max. output value of Down sampling Y	R/W	0xFF

	0]				
2C	Reg_DSMin_Value_YRGB [7:0]	7:0	Min. output value of Down sampling Y	R/W	0x00
2D		7:0		R/W	0x0F
	Reg_DSMin_Value_YRGB[11:8]	7:4	See Reg0x2C	R/W	0x0
	Reg_DSMax_Value_YRGB [11:8]	3:0	See Reg0x2B	R/W	0xF
2E	Reg_DSMax_Value_CRGB[7:0]	7:0	max. value of C	R/W	0xFF
2F	Reg_DSMin_Value_CRGB [7:0]	7:0	min. value of C	R/W	0x00
30		7:0		R/W	0x0F
	Reg_DSMin_Value_CRGB[11:8]	7:4		R/W	0x0
	Reg_DSMax_Value_CRGB [11:8]	3:0		R/W	0xF
31				R/W	0x00
		7	No use	R/W	0
	Reg_SelDSTG	1	1: use FW set value as DS input parameter(reg21-25) 0: HW auto set	R/W	0
	Reg_SelDSSrc	0	1: use FW set value as DS output parameter(reg26-41) 0: HW auto set	R/W	0
32	Reg_DSTG_HFP [7:0]	7:0	FW set HFP of DS block	R/W	N/A
33		7:0		R/W	N/A
	Reg_DSTG_HSPOL	7	FW set HSPOL of DS block	R/W	N/A
	Reg_DSTG_VSPOL	6	FW set VSPOL of DS block	R/W	N/A
	Reg_DSTG_HFP [11:8]	5:0		R/W	N/A
34	Reg_DSTG_HSW [7:0]	7:0	FW set HSW of DS block	R/W	N/A
35	Reg_DSTG_HSW [13:8]	5:0		R/W	N/A
36	Reg_DSTG_HBP [7:0]	7:0	FW set HBP of DS block	R/W	N/A
37	Reg_DSTG_HBP [13:8]	5:0		R/W	N/A
38	Reg_DSTG_HDEW [7:0]	7:0	FW set HDEW of DS block	R/W	N/A
39	Reg_DSTG_HDEW [13:8]	5:0		R/W	N/A
3A	Reg_DSTG_VFP [7:0]	7:0	FW set VFP of DS block	R/W	N/A
3B	Reg_DSTG_VFP [13:8]	5:0		R/W	N/A
3C	Reg_DSTG_VSW [7:0]	7:0	FW set VSW of DS block	R/W	N/A
3D	Reg_DSTG_VSW [13:8]	5:0		R/W	N/A
3E	Reg_DSTG_VBP [7:0]	7:0	FW set VBP of DS block	R/W	N/A
3F	Reg_DSTG_VBP [13:8]	5:0		R/W	N/A
40	Reg_DSTG_VDEW [7:0]	7:0	FW set VDEW of DS block	R/W	N/A
41	Reg_DSTG_VDEW [13:8]	5:0		R/W	N/A
4E		7:0		R/W	0xF0
	RegEnDSAutoRst	7	1: Enable autoreset of DS FIFO	R/W	1
	RegEnCVAutoRst	6	1: Enable autoreset of CV FIFO	R/W	1
	RegEnCSCAutoRst	5	1: Enable autoreset of CSC FIFO	R/W	1
	RegEnORIAutoRst	4	1: Enable autoreset of Original FIFO	R/W	1
	RegDSFIFORst	3	1: Manually Reset DS path FIFO	R/W	0
	RegCVFIFORst	2	1: Manually Reset CV path FIFO	R/W	0
	RegCSCFIFORst	1	1: Manually Reset CSC path FIFO	R/W	0
	RegORIFIFORst	0	1: Manually Reset Original path FIFO	R/W	0
67	RegSetoutdepth	3:0	Force TX output color depth (only non-8 bit to 8 bit is applicable)	R/W	
68		7:0	See 0x67	R/W	0x00
	RegDSoutDepth	7:6	DS path	R/W	00
	RegCVoutDepth	5:4	CV path	R/W	00
	RegCSCoutDepth	3:2	CSC path	R/W	00
	RegOrioutDepth	1:0	Original path	R/W	00
69		7:0		R/W	0x00
	RegForceTXVCLK	5	1: force TXVCLK selection 0: TXVCLK selected by in/out color depth	R/W	0
	RegForceTXPCLK	4	1: force TXPCLK selection 0: TXPCLK selected by TXSrcSel	R/W	0
	RegP3VCLKSel	3	1: use PCLK3 as TXVCLK3 0: use VCLK3	R/W	0
	RegP2VCLKSel	2	1: use PCLK2 as TXVCLK2 0: use VCLK2	R/W	0

	RegP1VCLKSel	1	1: use PCLK1 as TXVCLK1 0: use VCLK1	R/W	0
	RegP0VCLKSel	0	1: use PCLK0 as TXVCLK0 0: use VCLK0	R/W	0
6B		7:0		R/W	0x40
	RegAutoCSCSel	6	1: CSC direction auto selected with in/out color mode 0: Force		1
	Reg_CVOutColMod	5:4	CV path output color mode when Reg0x6C(3)=1	R/W	00
	Reg_OutColMod	3:2	CSC path output color mode	R/W	00
	Reg_CSCSel	1:0	Used when bit6 = 0 00: bypass CSC 10: RGB to YUV 11: YUV to RGB	R/W	00
6D		2:0		R/W	0x00
	Reg_QuanL4B	2	1: set video color quantization Max. to (2^N)-1 0: set video color quantization Max. to 255*2^(N-8) (Where N is the color depth 8, 10, or 12)	R/W	0
	Reg_CRCBLimit	1	0: CbCr full range 0x00~ 0xFF 1: CbCr limit range 0x10~ 0xF0	R/W	0
	Reg_ColorClip	0	0: RGB full range, YUV limited range 1: RGB limit range, YUV full range	R/W	0
6E		7:0		R/W	0x00
	Reg_CSCOffset_Signed	6	CSC Path parameters	R/W	0
	Reg_CSCOffset_YSign	5		R/W	0
	Reg_CSCOffset_CRSign	4		R/W	0
	Reg_CSCOffset_CBSign	3		R/W	0
6F	Reg_CBOffset		CSC Path parameters	R/W	0x00
70	Reg_YOffset		CSC Path parameters	R/W	0x10
71	Reg_COffset		CSC Path parameters	R/W	0x80
72	RegRGBOffset		CSC Path parameters	R/W	0x00
73	Reg_Matrix11V [7:0]	7:0	CSC Path Matrix	R/W	0xB2
74	Reg_Matrix11V [13:8]	5:0		R/W	0x04
75	Reg_Matrix12V [7:0]	7:0		R/W	0x64
76	Reg_Matrix12V [13:8]	5:0		R/W	0x02
77	Reg_Matrix13V [7:0]	7:0		R/W	0xE9
78	Reg_Matrix13V [13:8]	5:0		R/W	0x00
79	Reg_Matrix21V [7:0]	7:0		R/W	0x93
7A	Reg_Matrix21V [13:8]	5:0		R/W	0x1C
7B	Reg_Matrix22V [7:0]	7:0		R/W	0x16
7C	Reg_Matrix22V [13:8]	5:0		R/W	0x04
7D	Reg_Matrix23V [7:0]	7:0		R/W	0x56
7E	Reg_Matrix23V [13:8]	5:0		R/W	0x1F
7F	Reg_Matrix31V [7:0]	7:0		R/W	0x49
80	Reg_Matrix31V [13:8]	5:0		R/W	0x1D
81	Reg_Matrix32V [7:0]	7:0		R/W	0x9F
82	Reg_Matrix32V [13:8]	5:0		R/W	0x1E
83	Reg_Matrix33V [7:0]	7:0		R/W	0x16
84	Reg_Matrix33V [13:8]	5:0		R/W	0x04
88	Reg_CVYOffset		CV path parameters	R/W	0x10
89	Reg_CVCOffset		CV path parameters	R/W	0x80
8A	Reg_CVRGBOffset		CV path parameters	R/W	0x00
8B	Reg_CVCbOffset		CV path parameters	R/W	0x00
90		7:0	See description on bit 0	R/W	0x00
	RegP1_XP_ENCKD4	7		R/W	0
	RegP1_XP_FBDIV2	6		R/W	0
	RegP1_XP_PRDIV2	5		R/W	0
	RegSetP1XPDIV	4		R/W	0
	RegP0_XP_ENCKD4	3		R/W	0
	RegP0_XP_FBDIV2	2		R/W	0
	RegP0_XP_PRDIV2	1		R/W	0
	RegSetP0XPDIV	0	1: Force XPLL Mul/Div parameters with FW 0: parameters auto set by in/out XP_PRDIV2 XP_ENCKD4 XP_FBDIV2 1X: 0 0 0 Div2: 1 0 0 Div4: 1 1 0 Mul2: 0 0 1	R/W	0

91		7:0	See description on reg0x90	R/W	0x00
	RegP3_XP_ENCKD4	7		R/W	0
	RegP3_XP_FBDIV2	6		R/W	0
	RegP3_XP_PRDIV2	5		R/W	0
	RegSetP3XPDIV	4		R/W	0
	RegP2_XP_ENCKD4	3		R/W	0
	RegP2_XP_FBDIV2	2		R/W	0
	RegP2_XP_PRDIV2	1		R/W	0
	RegSetP2XPDIV	0		R/W	0
93	Reg_CVMatrix11V [7:0]	7:0	CV path CSC Matrix	R/W	0xB2
94	Reg_CVMatrix11V [13:8]	5:0		R/W	0x04
95	Reg_CVMatrix12V [7:0]	7:0		R/W	0x64
96	Reg_CVMatrix12V [13:8]	5:0		R/W	0x02
97	Reg_CVMatrix13V [7:0]	7:0		R/W	0xE9
98	Reg_CVMatrix13V [13:8]	5:0		R/W	0x00
99	Reg_CVMatrix21V [7:0]	7:0		R/W	0x93
9A	Reg_CVMatrix21V [13:8]	5:0		R/W	0x1C
9B	Reg_CVMatrix22V [7:0]	7:0		R/W	0x16
9C	Reg_CVMatrix22V [13:8]	5:0		R/W	0x04
9D	Reg_CVMatrix23V [7:0]	7:0		R/W	0x56
9E	Reg_CVMatrix23V [13:8]	5:0		R/W	0x1F
9F	Reg_CVMatrix31V [7:0]	7:0		R/W	0x49
A0	Reg_CVMatrix31V [13:8]	5:0		R/W	0x1D
A1	Reg_CVMatrix32V [7:0]	7:0		R/W	0x9F
A2	Reg_CVMatrix32V [13:8]	5:0		R/W	0x1E
A3	Reg_CVMatrix33V [7:0]	7:0		R/W	0x16
A4	Reg_CVMatrix33V [13:8]	5:0		R/W	0x04
EE		7:0		R/W	0x21
	Regcmd_deglitch	5	1: Enable deglitch on PCCMD	R/W	1
	RegIntrOutType	4	Interrupt pin output type 1: open drain 0: push pull	R/W	0
	RegIntPol	3	Interrupt pin output polarity 1: low active 0: high active	R/W	0
	RegINT_SR	2	Interrupt IO slew rate	R/W	0
	RegINT_DRV	1:0	Interrupt IO driving ST	R/W	01
EF		7:0		R/W	0x11
	RegPCCMD_SR	3	PCCMD IO Slew rate	R/W	0
	RegPCCMD_SMT	2		R/W	0
	RegPCCMD_DRV	1:0	PCCMD IO Driving strength	R/W	01
F0	RegRxComSlvAdr	7:0	RX PCCmd port Address Bit0: enable	R/W	0x70
F1	RegTxComSlvAdr	7:0	TX common PCCmd port Address Bit0: enable	R/W	0x96

Bank 1

Reg	Register Name	Bit	Definition	Type	Default Value
110		7:0		R/W	0xC1
	RegEnRxHDCP2	6	1: Enable RX HDCP2 (clearing this bit makes HDCP port reg 0x50 shows no HDCP2 support)	R/W	1
	RegHP2EnHP2Rpt	3	1: Enable RX HDCP2 repeater function	R/W	0
	RegHP2selauthdone	0	1: auth done after EKS 0: auth done after V and M completed	R/W	1
111		7:0		R/W	0x00
	RHP2Reauthreq	1	1: FW triggering HDCP2 Reauthentication	W1P	0
	RKSVList_Ready	0	1: when down stream KSV list is ready, write to this bit to trigger V' calculation and message generation	W1P	0
118	RegHP2RxInfo[7:0]		HDCP2 Rxinfo (must be filled when Repeater is enabled)	R/W	0x00
119	RegHP2RxInfo[15:8]			R/W	0x00
11A	RegHP2RxStatus[7:0]		HDCP2 RxStatus (must be filled when Repeater is enabled)	R/W	0x00
11B	RegHP2RxStatus[15:8]			R/W	0x00
11C	RegHP2RxStatusMask	2:0	Select the way to report status	R/W	0x00
11D		7:0		R/W	0x00
	RegHP2FiltRdErr	3	1: filter message read error 0: abort once a read error detected	R/W	0
	RegHP2FWRxcap	2	1: Rx cap filled to SRAM by FW 0: auto filling	R/W	0
	RegHP2FWKSV	1	1: KSV List filled to SRAM by FW 0: auto filling	R/W	0
	RegHP2FWRx	0	1: RRx filled to SRAM by FW 0: auto filling	R/W	0
11E		7:0		R/W	0x00
	RegEnHP1SHA1	5	1: Enable HDCP2 engine to calculate SHA1	R/W	0
	RegSHA1OFFSET	4:0	SHA1 begin address (of KSV FIFO)	R/W	00000
11F		7:0		R/W	0x00
	RHP1SHA1Trg	7	1: Trigger SHA1 calculation	W1P	0
	RegSHA1Sel	6:4	SHA1 source select 000: TX 0 ; 001: TX 1 010: TX 2 ; 011: TX 3 100: RX	R/W	000
120		7:0		R/W	0x00
	HDCP2Int_Mask	6:3	Mask of RX HDCP2 event interrupt	R/W	0000
121		7:0	HDCP2 interrupt(Masked)	W1C	
	Rpt_str_manage	7	Repeater stream management message received	W1C	
	Rpt_Send_ack	6	Repeater Send Ack message received	W1C	
	SKE_Send	5	SKE message received	W1C	
	LC_init	4	LC initial message received	W1C	
	RSA_Fail	3	RSA operation fail	W1C	
	Storekm	2	Stored Km received	W1C	
	No_storekm	1	Non-stored Km received	W1C	
	AKE_init	0	AKE initial message received	W1C	
122		7:0		W1C	
	RECCReauth	7	HW Reauth issued	W1C	
	Msg_Rderr	6	Message Read Error detected	W1C	
	M Read done	5	M' message read done	W1C	
	KSV/V' Read done	4	V' message read done	W1C	
	L' Read Done	3	L' message read done	W1C	
	pair Rd Done	2	Pairing message read done	W1C	0
	H Rd Done	1	H' message read done	W1C	
	CERT Rd Done	0	CERT message read done	W1C	
123		7:0			
	RHP2RSA_Fail	7	1: last RSA command failed	RO	
	Msg_Status	6:5		RO	
	RHP2_Encen_flag	4	1: HDCP2 Encryption is on	RO	
	REnHDCP2	2	1: HDCP2 is enabled(by AKE init)	RO	0
	HDCP_off	1	1: HDCP2 goes off interrupt	W1C	

	HDCP_ON	0	1: HDCP2 goes on interrupt	W1C	
12C ~ 128	RKSVFIFORdData[39:0]	7:0	KSV Address auto increment when reg12C is read	RO	
12D		7:0		R/W	0x00
	RegEnKSVFIFORd	7	1: Enable FW reading KSV entries to reg128~reg12C	R/W	
	RKSVFIFORdIdx	4:0	Read Address of KSV FIFO	R/W	
155 ~ 152	RSHA1CalH0[31:0]	31:0	SHA1 output Value from HASH block(when SHA1 calculation is triggered)	RO	
159 ~ 156	RSHA1CalH1[31:0]	31:0		RO	
15D ~ 15A	RSHA1CalH2[31:0]	31:0		RO	
161 ~ 15E	RSHA1CalH3[31:0]	31:0		RO	
165 ~ 162	RSHA1CalH4[31:0]	31:0		RO	
170		7:0			0x04
	RBKSVRxVld	4	1: RX BKSv valid	RO	
173		7:0			0x00
	RRomBIST_fail	7	Key ROM BIST fail	RO	
	RRomBIST_done	6	Key ROM BIST done	RO	
	RRdrom_fail	3	Read: Read ROM fail	RO	
	RRomBIST_Trg	2	1: Trigger Key ROM BIST	W1P	
	RRdROM_Trg	1	1: Trigger Key ROM Read	W1P	
	RRdBKSV_Trg	0	1: Trigger read BKSv	W1P	
174		7:0			0x00

Chap 11 HDMI Rx Registers

HDMI Rx register I²C address is defined in HDMI Converter register 0xF0. Default value is 0x70.

MHL Register I²C Address is in reg49
EDID Register I²C Address is in reg4B.

Register Type Note:

W1C : Write 1 Clear

W1P : Write 1 generate pulse

RO : Read Only

R/W : Read and Write

HDMIRX Register Bank

Block 0

Reg	Register Name	Bit	Definition	Type	Default Value
Device Information					
00	VendorID[7:0]	7:0	Vendor ID Low Byte	RO	0x54
01	VendorID[15:8]	7:0	Vendor ID High Byte	RO	0x49
02	DeviceID[7:0]	7:0	Device ID Low Byte	RO	0x64
03	DeviceID[15:8]	7:0	Device ID High Byte	RO	0x66
04	RevID	7:0	Device Revision ID	RO	0xA0
Interrupt					
05	P0_PWR5V_Chg	0	Port 0 power 5V state change Interrupt	W1C	
	P0_RxCLKOn_Det	1	Port 0 Rx Clock on detect Interrupt	W1C	
	P0_RxCLK_StbChg	2	Port 0 Rx Clock stable change Interrupt	W1C	
	P0_RxCLKChg_Det	3	Port 0 Rx Clock change detect Interrupt	W1C	
	P0_HDMIMode_Chg	4	Port 0 HDMI/DVI mode change Interrupt	W1C	
	P0_ECCErrTrg	5	Port 0 ECC error Interrupt	W1C	
	P0_DeSkewErr	6	Port 0 deskew error interrupt	W1C	
06	P0_H2VSkewFail	7	Port 0 H2V FIFO Fail interrupt	W1C	
	P0_Chx_SymLockChg	0	Port 0 Symbol lock change interrupt	W1C	
	P0_CH0_SymFIFORst	1	Port 0 Channel 0 Symbol lock FIFO auto reset interrupt	W1C	
	P0_CH1_SymFIFORst	2	Port 0 Channel 1 Symbol lock FIFO auto reset interrupt	W1C	
	P0_CH2_SymFIFORst	3	Port 0 Channel 2 Symbol lock FIFO auto reset interrupt	W1C	
	P0_Ch0_SymLockRst	4	Port 0 Channel 0 Symbol lock auto reset interrupt	W1C	
	P0_Ch1_SymLockRst	5	Port 0 Channel 1 Symbol lock auto reset interrupt	W1C	
	P0_Ch2_SymLockRst	6	Port 0 Channel 2 Symbol lock auto reset interrupt	W1C	
07	P0_FSMFail	7	Port 0 HDMI FSM fail interrupt	W1C	
	P0_CH0_LagErr	0	Port 0 Channel 0 Deskew Lag error interrupt. Cleared when write reg 0x07,D[2:0]/=000	W1C	
	P0_CH1_LagErr	1	Port 0 Channel 1 Deskew Lag error interrupt. Cleared when write reg 0x07,D[2:0]/=000	W1C	
	P0_CH2_LagErr	2	Port 0 Channel 2 Deskew Lag error interrupt. Cleared when write reg 0x07,D[2:0]/=000	W1C	
	P0_MHLMODEChg	3	Port 0 MHL/HDMI mode change Interrupt	W1C	
	P0_SAREQDoneInt	4	Port 0 auto SAR EQ Done Interrupt	W1C	
	P0_SAREQFailInt	5	Port 0 auto SAR EQ Fail Interrupt	W1C	
	P0_EQDoneInt	6	Port 0 auto EQ Done Interrupt	W1C	
08	P0_EQFailInt	7	Port 0 auto EQ Fail Interrupt	W1C	
	P0_SCDC_CLKRatioChgInt	0	Port 0 SCDC Clock Ratio bit Change Interrupt	W1C	
	P0_SCDC_ScrEnChgInt	1	Port 0 SCDC Scramble Enable Bit Change Interrupt	W1C	

	P0_HDMI2ScrambleChg_Det	2	Port 0 HDMI2.0 Scramble Status Change	W1C	
	P0_EDIDBusHangSts	3	Port 0 EDID Bus Hang Interrupt	W1C	
	P0_CAOF_Done	4	Port 0 Calibration Done Interrupt	W1C	
	P0_CAOF_Fail	5	Port 0 Calibration Fail Interrupt	W1C	
	P0_AutoEQUpd	6	Port 0 Auto EQ Update	W1C	
	Reserved	7			
09	P0_HDCP_Auth_Start	0	Port 0 HDCP authentication start Interrupt	W1C	
	P0_HDCP_Auth_Done	1	Port 0 HDCP authentication done Interrupt	W1C	
	P0_HDCP_EncChgSts	2	Port 0 HDCP Encryption change interrupt	W1C	
	P0_HDCP_OffDetSts	3	Port 0 HDCP Off detected interupt	W1C	
	P0_RDetDDCAct	4	1: Port 0 DDC active; 0: Port 0 DDC Idle	RO	
	P0_REDIDBusHang	5	1: Port 0 EDID bus hang 0: Port 0 EDID bus OK	RO	
	Reserved	7:6			
0F	Reg_Block_Sel	2:0	Register block select	R/W	000
	Reserved	6:3		RO	
	Reserved	7			
10	VidAbnormalSts	0	Video Abnormal interrupt,	W1C	
	SCDTChg	1	Video Stable State Change Interrupt	W1C	
	VideoModeChg	2	Input Video Mode Change Interrupt	W1C	
	TimerInt	3	Timer count to setting value Interrupt	W1C	
	PktSetMute	4	Set Mute Packet is received Interrupt	W1C	
	PktLeftMute	5	Left Mute Packet is received Interrupt	W1C	
	AutoAudioMute	6	Audio Auto Mute Interrupt	W1C	
	AudioFIFOError	7	Audio FIFO error Interrupt	W1C	
11	ISRC1_Det	0	ISRC1 packet detect Interrupt	W1C	
	ISRC2_Det	1	ISRC2 packet detect Interrupt	W1C	
	GenPkt_Det	2	General packet detect Interrupt	W1C	
	CD_Det	3	Color Depth change Interrupt	W1C	
	NoAVI_Rcv	4	No AVI InfoFrame is received	W1C	
	NoAud_Rcv	5	No Audio InfoFrame is received	W1C	
	NoGenPkt_Rcv	6	No GenPkt is received	W1C	
	NoGenPkt2_Rcv	7	No AVI packet is received	W1C	
12	NewAVIPkt_Det	0	New AVI InfoFrame detect Interrupt	W1C	
	NewMPEGPkt_Det	1	New MPEG Packet detect Interrupt	W1C	
	NewSPDPkt_Det	2	New SPD Packet detect Interrupt	W1C	
	NewACPPkt_Det	3	New ACP Packet detect Interrupt	W1C	
	NewAudioPkt_Det	4	New Audio Packet detect Interrupt	W1C	
	NewDMRPkt_Det	5	New Dynamic Range InformFrame detect Interrupt	W1C	
	R3D_ValidChgInt	6	VSIF 3D valid change Interrupt	W1C	
	RVidParaChgInt	7	Video parameter change interrupt, check reg1A/reg1B for detail information.	W1C	

Device internal status

13	P0_PWR5V_Det	0	Port 0 power 5V detect	RO	
	P0_HDMI_Mode	1	Port 0 HDMI(1)/DVI(0) mode	RO	
	P0_RxCLK_Detect	2	Port 0 Rx clock detect	RO	
	P0_RxCLK_Valid	3	Port 0 Rx clock is valid	RO	
	P0_PxCLK_Stable	4	Port 0 Rx clock is stable	RO	
	P0_IPLL_LOCK	5	Port 0 IPLL is locked	RO	
	P0_MHLMode	6	Port 0 MHL(1)/HDMI(0) mode	RO	
	P0_RxSymbolLock	7	Symbol Lock of H2V output	RO	
14	P0_IPLL_HS1G	0	Port 0 IPLL clock is higher than 100MHz	RO	
	P0_IPLL_HS1P48G	1	Port 0 IPLL clock is higher than 148MHz	RO	
	P0_IPLL_HS3P4G	2	Port 0 IPLL clock is higher than 340MHz	RO	
	P0_CH0_SymLockSts	3	Port 0 Symbol Lock of DIGCDR_DATA	RO	
	P0_CH1_SymLockSts	4	Port 0 Symbol Lock of DIGCDR_DATA	RO	
	P0_CH2_SymLockSts	5	Port 0 Symbol Lock of DIGCDR_DATA	RO	
	P0_SCDC_CLKRatio	6	Port 0 SCDC Clock Ratio bit	RO	
	P0_SCDC_ScrEn	7	Port 0 SCDC Scramble Enable Bit	RO	
15	P0_SCDC_RREnable	0	Port 0 SCDC RR Enable Bit	RO	
	P0_SCDC_Scramble Status	1	Port 0 HDMI2.0 Scramble_Status	RO	
	Reserved	5:2			

	P0_EDIDDDCIde	6	1: Port 0 EDID Idle 0: Port 0 EDID active	RO	
	P0_HDCPDDCIde	7	1: Port 0 HDCP DDC Idle 0: Port 0 HDCP DDC Active	RO	
19	Reserved	3:0			
	OLOCK	4	OPLL is locked	RO	
	OPLL_HS	5	OPLL in High Speed	RO	
	MHL3DFreSeq	6	Receive MHL frame sequence 3D InfoFrame	RO	
	SCDT	7	Video is stable	RO	
1A	VidParaChang_Sts[7:0]	7:0	Bit0: H Front Porch Change Bit1: H Sync Width Change Bit2: H Back Porch Change Bit3: H DE Width change Bit4: H 2ndVSync Rise Location Change Bit5: V Front Porch Change Bit6: V Sync Width Change Bit7: V Back Porch Change	RO	
1B	VidParaChang_Sts[10:8]	2:0	Bit8: VDE Width Change Bit9: 2 nd VSync Front Porch Change Bit10: Interlaced Mode Change	RO	
	PixRptErrInt	3	Receive error Pixel Repeating Information interrupt	W1C	
	PixRpt	5:4	00: Receive PixRepeat 0 01: Receive PixRepeat 2 11: Receive PixRepeat 4	RO	
	Reserved	7:6		RO	

Device Reset/Clock/PowerDown related register

20	Reg_GateGMRCLK	0	1: Gate RCLK for MHL GMRCLK domain	R/W	0
	Reg_GateNMRCLK	1	1: Gate RCLK for MHL NMRCLK domain	R/W	0
	Reg_GateERCLK	2	1: Gate RCLK for internal EDID R/W	R/W	0
	Reserved	3		R/W	0
	Reg_GateNH2RCLK	4	1: Enable RCLK for Registers	R/W	0
	Reg_GateP0HP1RCLK	5	1: Gate RCLK for port0 HDCP1.x	R/W	0
	Reserved	7:6	1: Gate RCLK for port1 HDCP1.x	R/W	0
21	Reg_GateAGRCLK	0	1: Gate RCLK for port 0 internal EDID I2C	R/W	0
	Reserved	7:1			
22	RegVdoRst	0	1: Video logic reset	R/W	0
	Reserved	1			
	RegEnIntRst	2	1: Interrupt logic reset	R/W	0
	RegRegRst	3	1: Register block reset	R/W	0
	Reserved t	7:4			
23	Reg_P0_SWRst	0	1: Port 0 all logic reset	R/W	0
	Reg_P0_HDCPRst	1	1: Port 0 HDCP logic reset	R/W	0
	Reg_P0_CDRDataRst	2	1: Port 0 CDR Data logic reset	R/W	0
	Reg_P0_DCLKRst	3	1: Port 0 CLKD10 domain logic reset	R/W	0
	Reg_P0_EQRst	4	1: Port 0 EQ logic reset	R/W	0
	Reg_P0_EnAutoRst	5	1: Enable Port 0 Auto reset	R/W	0
	Reg_P0_EnHDCPRst	6	1: Enable Port 0 HDCP	R/W	0
	Reg_P0_EnCKRst	7	1: Enable Port 0 Auto reset when clock is not stable	R/W	0
24	Reserved	0			
	Reg_OPLLrSt	1	1: Port 0 OPLL reset	R/W	0
	Reg_P0_IPLLrSt	2	1: Port 0 IPLL reset	R/W	0
	RegORst	3	1: MHL OCLK domain logic reset	R/W	0
	RegDDCRst	4	1: MHL DDC logic reset	R/W	0
	RegMSCRst	5	1: MHL MSC logic reset	R/W	0
	RegLinkRst	6	1: MHL link logic reset	R/W	0
	RegDisvrSt	7	1: MHL discovery logic reset	R/W	0
25	Reg_P0_BRST_RCVR	0	1: Port 0 AFE channel 0 decoder reset	R/W	0
	Reg_P0_GRST_RCVR	1	1: Port 0 AFE channel 1 decoder reset	R/W	0
	Reg_P0_RRST_RCVR	2	1: Port 0 AFE channel 2 decoder reset	R/W	0
	Reg_P0_PWDTOP	3	1: Port 0 all digital logic and APLL power down	R/W	0
	Reserved	4			
	Reg_P0_IPLLPWDI	5	1: Port 0 IPLL current bias power down	R/W	0
	Reg_P0_IPLLPWD	6	1: Port 0 IPLL power down	R/W	0
	Reg_P0_OPLLPWD	7	1: Port 0 OPLL power down	R/W	0

26	Reg_P0_ETCPWD	0	1: Port 0 clock differential pair power down	R/W	0
	Reg_P0_CH0PWD	1	1: Port 0 channel 0 differential pair power down	R/W	0
	Reg_P0_CH1PWD	2	1: Port 0 channel 1 differential pair power down	R/W	0
	Reg_P0_CH2PWD	3	1: Port 0 channel 2 differential pair power down	R/W	0
	Reg_P0_CH0PWDRTERM	4	1: Port 0 channel 0 differential pair termination power down	R/W	0
	Reg_P0_CH1PWDRTERM	5	1: Port 0 channel 1 differential pair termination power down	R/W	0
	Reg_P0_CH2PWDRTERM	6	1: Port 0 channel 2 differential pair termination power down	R/W	0
27	Reg_P0_CLKPWDRTERM	7	1: Port 0 Clock differential pair termination power down	R/W	0
	Reg_P0_PWDAFEAll	0	1: Port 0 All AFEs power down	R/W	0
	Reg_P0_EQPWD	1	1: Port 0 AFE EQ power down	R/W	0
	Reg_P0_CH0CLKD4PWD	2	1: Port 0 channel 0 CLKD4 power down	R/W	0
	Reg_P0_CH1CLKD4PWD	3	1: Port 0 channel 1 CLKD4 power down	R/W	0
	Reg_P0_CH2CLKD4PWD	4	1: Port 0 channel 2 CLKD4 power down	R/W	0
	Reg_P0_CH0CLKD10PWD	5	1: Port 0 channel 0 CLKD10 power down	R/W	0
28	Reg_P0_CH1CLKD10PWD	6	1: Port 0 channel 1 CLKD10 power down	R/W	0
	Reg_P0_CH2CLKD10PWD	7	1: Port 0 channel 2 CLKD10 power down	R/W	0
	Reg_P0_EnRxAutoPwDAFE	0	1: Enable Port 0 RXAFE auto power down	R/W	0
	Reg_P0_EnRxAutoPwDLL	1	1: Enable Port 0 PLL auto power down	R/W	0
	Reg_P0_EnAutoPwCLKD4	2	1: Enable Port 0 CLKD4 auto power down	R/W	0
	Reg_P0_EnAutoPwCLKD10	3	1: Enable Port 0 CLKD10 auto power down	R/W	0
	Reg_P0_EnAutoPwCDREQ	4	1: Enable Port 0 CDR EQ auto power down	R/W	0
29	Reg_P0_EnAutoPwTERM	5	1: Enable Port 0 termination auto power down	R/W	0
	Reg_P0_EnAutoPwTCLK	6	1: Enable Port 0 TCLK auto power down	R/W	0
	Reg_P0_EnAutoPwBTERM	7	1: Enable Port 0 B Channel termination auto power down	R/W	1
	Reg_P0_DISMHLPWD	0	1: Disable MHL mode auto power down	R/W	0
	Reg_P0_IgnClkRdyPwD	1	1: Ignore clock ready signal	R/W	0
	Reg_P0_CKValidSel	2	Clock valid condition selector. 1: Clock detect ; 0: Clock valid	R/W	0
	Reg_P0_RSetCLK	3	1: SW force IPLL speed mode and Input clock ready	R/W	0
2A	Reg_P0_RDetClkRdy	4	1: Input Clock Ready when Reg29h(3)=1	R/W	0
	Reg_P0_RDetClkHigh	5	1: IPLL in high speed mode when Reg29h(3)=1	R/W	0
	Reg_P0_InvCLKD10I	6	1: Inverse Port 0 input CLKD10I	R/W	1
	Reg_P0_InvHCLK	7	1: Inverse Port 0 input HCLK	R/W	1
	Reg_P0_InvCOFCLK	0	1: Inverse Port 0 COFCLK	R/W	0
	Reg_P0_GateEQRCLK	1	1: Disable Port 0EQRCLK	R/W	0
	Reg_P0_GateHP1TMDCLK	2	1: Disable Port 0 HDCP1 TMDCLK	R/W	0
2B	Reg_P0_GateTMDCLK	3	1: Disable Port 0 TMDCLK	R/W	0
	Reg_P0_GateWDGRCLK	4	1: Disable Port 0 WatchDog CLK	R/W	0
	Reg_P0_GateHCLK	5	1: Disable Port 0 HDCLK	R/W	0
	Reg_P0_CAOFRST	6	1: reset CAOF blocks 0: normal operation	R/W	0
	Reserved	7			
	Reserved	7:0			
	Reserved	7:0			
33	Reserved	7:0			
	Reg_AutoRCLK	0	1: Auto gate RCLK when PWR5V not detected	R/W	1
	Reg_AutoMRCLK	1	1: Auto gate MHL RCLK when CDSense not detected	R/W	0
	RegEnHP1AutoPwD	2	1: Auto power down HDCP 1.4 when HDCP2 enabled	R/W	0
	Reserved	3			
	Reg_GateRCLK	4	1: Gate RCLK for backend module	R/W	0
	Reg_GatePCLK	5	1: Gate CLK	R/W	0
34	Reg_GateOCLK	6	1: Gate OCLK for MHL module	R/W	0
	Reserved	7			
MHL/HDMI FrontEnd control					
38	Reg_P0_SCVFiltSel	0	0: disable correct 1-bit SCV code error 1: enable correct 1-bit SCV code error	R/W	0
	Reg_P0_ITogSel	1	0: Disable LFSR scramble 1: Enable LFSR scramble	R/W	0
	Reserved	2	Must be 0 at normal case		
	Reg_P0_BLKDeSkew	3		R/W	0
	Reg_P0_FiltCTRL4B	4	0: not filter CTRL[3:0] 1: filter CTRL[3:0]	R/W	1
	Reg_P0_SetEnLFSR	5	0: Enable LFSR depend on SCDC Scramble_Status	R/W	0

			1: Set LFSR scramble using Reg_EnLFSR		
	Reg_P0_EnLFSR	6	0: Disable LFSR scramble 1: Enable LFSR scramble	R/W	0
	Reg_P0_H2LockOpt	7	0: normal operation 1: speed up at simulation/test mode	R/W	0
39	Reserved	0			
	Reserved	1			
	Reserved	2			
	Reg_P0_BypassFIFO	3	1: bypass 10bit FIFO 0: normal	R/W	0
	Reg_P0_CDRDbgSel	5:4	CDR debug selector	R/W	00
	Reserved		Must be 0 at normal case		
	Reg_P0_SymlockOpt	7		R/W	0
3A	Reserved	7:0			
3B	Reserved	5:0	Bit0: must be 1 Bit1: must be 1 Bit2: must be 0 Bit3: must be 1 Bit4: Reserved Bit5: Reserved		000011
	Reserved	7:6			
3C	Reg_P0_EnIPLockChk	0	1: Enable Check IPLL_LOCK when CLKD5 FIFO auto-reset is enabled.	R/W	1
	Reg_P0_EnCLKVldChk	1	1: Enable Check CLK_Valid when CLKD5 FIFO auto-reset is enabled.	R/W	1
	Reg_P0_IPStableTime	3:2	1: IPLL stable time select 01: 50us ; 10: 100us ; others: 75us	R/W	00
	Reg_P0_EnPLLBufRst	4	1: Enable CLKD5 Data FIFO auto-reset	R/W	1
	Reg_P0_HWBypassFIFO	5	1: Enable HW Bypass CLKD5 Data FIFO	R/W	1
	Reg_EnSWHCLKSel	6	1: enable force HCLK source select	R/W	0
	Reg_HCLKSel	7	HCLK source select when reg3C(6)=1	R/W	0
3D	Reg_P0_FixTek3D	0	0: Rx FSM data island trailing guard-band always ends after 2T. 1: Rx FSM data island trailing guard-band ends only if blank starts.	R/W	0
	Reg_P0_deskewdone	1	0: the signal deskew_done is always asserted after DE observed unless reset. 1: the signal deskew_done is deasserted if symbol lock is de-asserted.	R/W	0
	Reg_P0_filtalign	2	0: Channel skew edge changes once if any DE position change observed. 1: Channel skew edge changes only if 2 identical DE position change observed.	R/W	0
	Reg_P0_EnTERC4Crt	3	1: enable 1-bit auto-correction of certain TERC4 code errors.	R/W	0
	Reg_P0_EnCtrlCrt	4	1: enable 1-bit auto-correction of certain control code errors.	R/W	0
	Reg_P0_FiltDguard	6:5	00: no filter, will enter hdmi mode once "data preamble + data guard band" is seen. 01: enter hdmi mode when second data preamble + guard band sequence is seen. 10: enter hdmi mode when third data preamble + guard band sequence is seen. 11: enter hdmi mode when fourth data preamble + guard band sequence is seen.	R/W	00
	Reg_P0_EnVHDMI	7	1: enable detection of video format with "video preamble and video guard band" but without "data preamble and data guard band"; Such video will be regarded as "HDMI" mode if this bit is enabled.	R/W	0
3E	Reg_P0_FiltDE	0	1: deglitch 1T blank period 0: no deglitch	R/W	0
	Reg_P0_ENVdoComp	1	1: enable video code-err comparison	R/W	0
	Reg_P0_EnLockDisp	2	1: enable disparity-lock when doing video code comparison	R/W	0
	Reg_P0_DEDisp0	3		R/W	0
	Reserved	6:4	1: enable video code-err comparison	R/W	0
	P0_RVidCompLockFail	7		W1C	
3F	Reserved	7:0			
40	Reg_P0_HDMI Mode	0	1: HDMI/DVI mode value when reg40h(1)=1	R/W	0

41	Reg_P0_HDMI ModeOVRT	1	1: Enable HDMI/DVI mode over-write at MHLCORE/HDMICORE module	R/W	0
	Reg_P0_Swap_Pol	2	1: swap input 10bit word's polarity of H2V output	R/W	0
	Reg_P0_Swap_RB	3	1: swap input channel 0 and channel 2 of H2V output	R/W	0
	Reserved	7:4		R/W	0
	RegBCHThresh[2:0]	2:0	Threshold to raise BCHError interrupt	R/W	100
	Reg_P0_HoldECCErr	3	1: receive correct pkt will not decrease eccerrcnt	R/W	0
	Reg_P0_KeepErr	4	1: keep error status and counter when clock change to not valid	R/W	0
	Reg_P0_DisBCH	5	1: disable BCH correction	R/W	0
42	Reserved	7:6			
	Reg_P0_ForceMode	0	1: Force H2V data mode	R/W	0
	Reg_P0_ModeSel	1	H2V data mode select when reg31h(0)=1 1: MHL mode ; 0: HDMI mode	R/W	0
	Reg_P0_EnH2VCheck	2	1: enable H2V blank check	R/W	1
	Reg_P0_H2VFIFORst	3	1: H2VFIFO reset	R/W	0
	Reg_P0_EnH2VAutoRst	4	1: enable H2VFIFO auto reset	R/W	1
	Reg_P0_TimeOut_En	5	1: Enable FSM state go to lost after 4096T not seeing DE; 0:	R/W	1
	Reg_P0_IPLockDlySel[1:0]	7:6	IP lock delay when Reg_P0_UseIPLOCK='0' 00: IP lock after receive 32x256 watch dog clock 01: IP lock after receive 64x256 watch dog clock 10: IP lock after receive 128x256 watch dog clock 11: IP lock after receive 256x256 watch dog clock	R/W	10
43	Reg_P0_EnUpdAvg	0	1: Watchdog will use average value for PCLK sample	R/W	1
	Reg_P0_UseIPLOCK	1	1: Watchdog reply IP lock when AFE IPL Lock s 0: Watchdog reply IP lock after receiving certain number of watchdog clock	R/W	1
	Reg_P0_RCLKDeltaSel	3:2	Watchdog PCLK Delta value select 00: 1%; 01: 2%; 10: 3%; 11:5%	R/W	00
	Reg_P0_EnClkDiffMin	4	1: the minimum delta value will be 2	R/W	0
	P0_WDog RxCLK PreDiv2	5		RO	00
	P0_WDog RxCLK PreDiv4	6		RO	
	P0_WDog RxCLK PreDiv8	7		RO	
	Reserved				
44	Reg_P0_WCLKHighNum	5:0	IPLL_HS1G threshold	R/W	100001
	Reg_HDMI2DetTsel	7:6		R/W	00
45	Reg_P0_WCLKValidNum	7:0	PxCK_4DOG valid threshold	R/W	0xA6
46	RegWCLKHighNumB	5:0	IPLL_HS1P48Gthreshold	R/W	010110
	RegWCLKHighExt[0]	6	IPLL_HS1P48G threshold	R/W	0
	Reserved	7			
47	RegWCLKHighNumC	5:0	IPLL_HS3P4G threshold 1: TMDS clock >=340MHz 0: TMDS clock < 340 MHz	R/W	001010
	RegWCLKHighExt[2:1]	7:6	IPLL_HS3P4G threshold	R/W	10
48	P0_TMDCLKSpeed	7:0	TMDCLK = RCLK*256/ TMDCLKSpeed	RO	
49	Reg_P0_MHLPortAdrEn	0	1: enable PC I2C access MHL block	R/W	0
	Reg_P0_MHLPortAdr	7:1	I2C Slave Addresss for MHL block	R/W	1100000
4B	RegEnEDIDSlaveAdr	0	1: Enable access EDID block	R/W	0
	RegEDIDSlaveAdr	7:1	I2C Slave Address for EDID	R/W	1010100
4C	RegCDDGSEL	1:0		R/W	00
	RegVBUSDGSe[	4:2		R/W	000
	Reserved	5		R/W	0
	Reserved	6			
	Reserved	7			
4D	Reg_PCI2CHoldT	3:0	PC interface I2C hold time setting	R/W	0010
	Reserved	4			
	Reg_I2CDegLitch	5	DDC I2C deglitch	R/W	1
	Reg_I2CFiltType	6	DDC I2C filter type	R/W	0
	Reg_I2CFiltTap	7	DDC I2C filter tap	R/W	0
4E	RegDebugSel	7:0			
4F	Reserved	7:0			

FrontEnd IO Common control					
50	Reg_PCSDA_Slew	7	Slew rate setting for I2C pins 0: Fast, 1: Slow	R/W	1
	Reserved	6			
	Reg_PCSDA_ST	5:4	Current strength of command I2C pins 00: 2.5mA, 01: 5mA, 10: 7.5mA, 11: 10mA	R/W	10
	Reg_DDCSDA_Slew	3	Slew rate of HDCP I2C pins 0: Fast ; 1: Slow	R/W	1
	Reg_DDC_SMT	2	DDC port SmithTrig	R/W	0
	Reg_DDCSDA_ST	1:0	Current strength of HDCP I2C pins 00: 2.5mA, 01: 5mA, 10: 7.5mA, 11: 10mA	R/W	10
51	Reserved	7			
	RegPS_CntlValue	6	ENVBUS output value when reg5Bh[5:4]=10	R/W	0
	RegPS_CntlMode	5:4	00: ENVBUS IO will output inversed renvbus 01: ENVBUS IO will output renvbus 10: ENVBUS IO will output RegPS_CntlValue 11: ENVBUS IO will output inversed renvbus	R/W	00
	Reg_ENVBUS_Slew	3	ENVBUS IO Slew Rate setting	R/W	1
	Reserved	2			
	Reg_ENVBUS_ST	1:0	ENVBUS IO Strength setting	R/W	01
52	Reserved	3:0		R/W	0
	Reg_RxPWR5V_PD	4	RXPWR5V IO pull-dn setting	R/W	0
	Reg_RxPWR5V_SMT	5	RXPWR5V IO schmitt-trigger setting	R/W	0
	Reg_CDSENSE_PD	6	CDSENSE IO pull-dn setting	R/W	0
	Reg_CDSENSE_SMT	7	CDSENSE IO schmitt-trigger setting	R/W	0
53	Reg_P0_IntEn[7:0]	7:0	Interrupt enable for Reg0x05 1: Enable 0: Disable	R/W	0x00
54	Reg_P0_IntEn[15:8]	7:0	Interrupt enable for Reg0x06 1: Enable 0: Disable	R/W	0x00
55	Reg_P0_IntEn[23:16]	7:0	Interrupt enable for reg0x07 1: Enable 0: Disable	R/W	0x00
56	Reg_P0_IntEn[31:24]	7:0	Interrupt enable for reg0x08 1: Enable 0: Disable	R/W	0x00
57	Reg_P0_IntEn[39:32]	7:0	Interrupt enable for reg0x09 1: Enable 0: Disable	R/W	0x00
5D	Reg_BK_IntEn[7:0]	7:0	Interrupt enable for reg0x10 1: Enable 0: Disable	R/W	0x00
5E	Reg_BK_IntEn[15:8]	7:0	Interrupt enable for reg0x011 1: Enable 0: Disable	R/W	0x00
5F	Reg_BK_IntEn[23:16]	7:0	Interrupt enable for reg0x012 1: Enable 0: Disable	R/W	0x00
60	RegIntPol	0	Interrupt pin output polarity 1: low active 0: high active	R/W	1
	RegIntrOutType	1	Interrupt pin output type 1: open drain 0: push pull	R/W	1
	RegEnHDMIInt	2	1: Enable HDMI interrupt	R/W	1
	RegEnSetInt	3	1: Enable MHL Set Interrupt	R/W	1
	RegEnCBUSInt	4	1: Enable MHL CBus Interrupt	R/W	1
	RegEnIntOut	5	1: Enable Interrupt IO output	R/W	0
	Reserved	6			
	RegEn_Debug	7	1: enable debug output to IO	R/W	0
61	Reserved	7:0			
62	Reserved	7:0			

Video Back End Control Registers

67	Reg_En_DefPhase	0	1: Enable GCP default_phase setting	R/W	1
	Reg_DisDefRst	1	1: Disable default_phase reset	R/W	0
	Reg_EnAutoRstCD	2	1: enable CD auto reset	R/W	1
	Reg_RstCD	3	1: Reset CD_conv module	R/W	0
	Reg_En_MuteCD	4	1: enable CD_conv mute	R/W	0
	Reserved	7:5			
68	RegFNumSel	0	Select video detect stable frame number 0: 2 1: 3	R/W	0
	RegAVMuteSel	1	1: gating avmute in video detect module	R/W	0
	RegEnSyncMon	2	1: Enable H/V Sync monitor at video detect module	R/W	1
	RegEnDEMon	3	1: Enable DE monitor at video detect module	R/W	1
	RegMonHTotalVld	4	Used at video detect module	R/W	1
	RegEnHGlitchDet	5	1: enable HSync Glitch detection	R/W	0
	RegVFrontPorchSel	7:6	Used at video detect module	R/W	01
69	RegHSyncVldRng	4:0	HSync rising edge valid range	R/W	00000
	RegSTBMode	6:5	Video stable condition selector 01: clock stable ; 10: H stable ; others: V stable	R/W	00
	Reserved	7		R/W	00
6A	Reg_SETPixRPT[1:0]	1:0	Pixel repeat manual setting	R/W	0
	Reg_ForcePixRPT	2	1: enable manual pixel repeat setting	R/W	0
	Reg_CDDetOpt	3	CD_Det interrupt select 0: CD different and CD/= "0000" 1: CD different	R/W	0
	Reg_CDFlowOpt	4	CD flow option (only for debug)	R/W	1
	Reserved	7:5			
6D	Reserved	5:0			
	RegEnHW3DfreSeq	6	Auto enable re-generate DE function when receive MHL frame sequential 3D VIC 1: enable ; 0: disable	R/W	1
	RegEn3DfreSeq	7	Re-generate DE when receive MHL frame sequential 3D 1: enable ; 0: disable	R/W	0

Packet Control Registers

76	Reg_PktRec_Type_H	7:0	Decide which kind of packet to be fully recorded on General PKT register	R/W	0x0A
77	Reg_PktRec_Type	7:0	Decide which kind of packet to be fully recorded on General PKT register	R/W	0x83
78	Reg_ACP_Rec_Type	7:0	The packet type will be recoded to ACP register	R/W	0x04
79	Reg_ISRC1_Rec_Type	7:0	The packet type will be recoded to ISRC1 register	R/W	0x05
7A	Reg_ISRC2_Rec_Type	7:0	The packet type will be recoded to ISRC2 register	R/W	0x06
7B	RegPktFlag_Ctrl 6: DRM 5: ACP 4: General 3: SPD 2: Audio 1: MPEG 0: AVI	6:0	0: interrupt only when first receiving or a new value is updated. 1: interrupt each time a packet is received.	R/W	000000
	Reserved	7			
7C	Reg_NewPkt_Sel 3: ISRC2 2: ISRC1 1: ACP 0: General Packet	3:0	For each bit 1: latch unsupport packet data 0: latch normal data	R/W	0000
	Reg_ISRC2_HL	4	ISRC2 registers latch data selector	R/W	0
	Reg_GenPkt_HL	5	GenPkt register latch data selector	R/W	0
	Reg_PktSetClr	7:6	Packet set clear	R/W	00
	Reserved				
7D	Reg_CHNum	3:0	Channel number setting when Reg_ForceCh=1	R/W	0000
	Reg_ForceCH	4	1: force channel number	R/W	0
	Reg_LayoutNum	5	Layout value setting when Reg_ForceLayout=1	R/W	0
	Reg_ForceLayout	6	1: force layout value	R/W	0
	CLR_FramePkt_Err	7	Clear the error register asserted by packets that is received	R/W	0

			not in a manner directed by spec		
7E	Reg_PktType_Mask[7:0]	7:0	Received packet type mask[7:0]	R/W	0xFF
7F	Reg_PktType_Mask[11:8]	3:0	Received packet type mask[11:8]	R/W	1111
	Reserved	7:4			
Audio Control Registers					
80	Reserved	7:0			
81	RegDis_sdm	0	1: disable sdm	R/W	0
	RegDien	1	1: enable sdm dither	R/W	0
	RegForce_AVMute	2	Software forced-AVMute mode	R/W	0
	RegAVMute_Value	3	AVMute value when software AVMute is enable	R/W	0
	RegAud_Info_Force	4	Force Audio setting from Aud info frame	R/W	0
	RegDis_False_DE	5	Disable false DE output	R/W	1
	RegForce_FS	6	Force Audio FS mode	R/W	0
	Reg_I2SCEn	7	1: enable compressed but non-HBR audio output from I2S path	R/W	0
82	Reserved	7:0			
83~88	Reserved	7:0			
89	Reserved	7:0			
8A	RegFS_Set[5:0]	5:0	Software set sampling frequency xx0000: 44.1KHz ; xx0010: 48KHz xx0011: 32KHz xx1000: 88.2KHz xx1010: 96KHz xx1100: 176.4KHz ; xx1110: 192KHz 000101: 384KHz xx1001: 768KHz 001011: 64KHz 101011: 128KHz 011011: 256KHz 111011: 512KHz 110101: 1024KHz	R/W	0010
	RegAVMuteOpt[1:0]	7:6	AVMute control option Debug only Bit6: 0 :software force avmute(reg81,D2) will not reset Pkt dec 1: software force avmute(reg81,D2) will reset Pkt dec Bit7: 0: avmute will not clear Pkt_set 1: avmute clear Pkt_set	R/W	00
8B	Reserved	7:0			
8C	Reserved	7:0			
8E	Reserved	7:0			
8E	Reserved	7:0			
8F	RegCmd_DegLitch	0	PC I2C deglitch	R/W	1
	RegCmd_FiltType	1	PC I2C filter type	R/W	0
	RegCmd_FiltTap	2	PC I2C filter tap	R/W	0
	Reserved	7:3			
90	RegTimerCntSet	6:0	Timer period setting	R/W	0000000
	RegTimerUnit	7	Enable timer	W1P	

Device internal Status ReadBack

96	AVI infoframe pkt error	0	Error Means it's ever transmitted but vanished in at least a certain period 1: AVI Infoframe pkt error.	RO	
	Audio infoframe pkt error	1	Error Means it's ever transmitted but vanished in at least a certain period 1: Audio Infoframe pkt error.	RO	
	GCP FramePkt_Err error	2	Error Means it's ever transmitted but vanished in at least a certain period 1: GCP pkt error.	RO	
	GMP FramePkt_Err error	3	Error Means it's ever transmitted but vanished in at least a certain period 1: GMP pkt error.	RO	

	RSetIntSts	4	MHL Set interrupt status	RO	
	RCBusIntSts	5	MHL CBus interrupt status	RO	
	RHDMIInt	6	HDMI interrupt status	RO	
	RCECIntSts	7	CEC interrupt status	RO	
97	Reserved	7			
	RHDMIIntGroup[6:0]	6:0	Indicate Interrupt occur in which group	RO	
98	VidModeChg	0	Indicate if a video mode change occurs	RO	
	VidInterlaceMode	1	Indicate video is in interlaced mode	RO	
	VidField	2	Video field number in interlaced mode	RO	
	PxVideoStable	3	Indicate if video signal is stable	RO	
	GCP_CD_Active	7:4	Color depth decoder value 4: 24bits ; 5: 30bits ; 6: 36bits ; 7: 48bits	RO	
99	PIXCLKSpeed[7:0]	7:0	PIXCLK = RCLK*255/ PIXCLKSpeed	RO	
9A	PIXCLKSpeed[10:8]	2:0	PIXCLK = RCLK*255/ PIXCLKSpeed	RO	
	Reserved	6:4			
	RegEnPCntUpd	7	'0': Stop Pixel Clock Count Update to Read '1': Enable Pixel Clock Count Updating.	R/W	1
9B	Vid_HTotal[7:0]	7:0	The total pixel count of one line[7:0]	RO	
9C	Vid_HTotal[13:8]	5:0	The total pixel count of one line[13:8]	RO	
	Reserved	7:6		RO	
9D	Vid_HDEW[7:0]	7:0	The active pixel count of one line[7:0]	RO	
9E	Vid_HDEW[13:8]	5:0	The active pixel count of one line[13:8]	RO	
	Reserved	7:6			
9F	Vid_HSW[7:0]	7:0	The width of HSync [7:0]	RO	
A0	Vid_HFPH[7:0]	7:0	The width of HSync front porch [7:0]	RO	
A1	Vid_HSW[8]	0	The width of HSync [8]	RO	
	Reserved	3:1			
	Vid_HFPH[11:8]	7:4	The width of HSync front porch [11:8]	RO	
A2	Vid_VTotal[7:0]	7:0	The total line count of a field [7:0]	RO	
A3	Vid_VTotal[13:8]	5:0	The total line count of a field [11:8]	RO	
	Reserved	7:6			
A4	Vid_VDEW[7:0]	7:0	The active line count of a field [7:0]	RO	
A5	Vid_VDEW[13:8]	5:0	The active line count of a field [7:0]	RO	
	Reserved	7:6			
A6	Vid_VSW[7:0]	7:0	The width of VSync [7:0]	RO	
A7	Vid_VFPH[7:0]	7:0	The width of VSync front porch[7:0]	RO	
A8	Vid_VSW[8]	0	The width of VSync [8]	RO	
	Reserved	3:1			
	Vid_VFPH[11:8]	7:4	The width of VSync front porch [11:8]	RO	
A9	Vid_HBP[7:0]	7:0	The width of HSync back porch [7:0]		
AA	Vid_HBP[10:8]	2:0	The width of HSync back porch [10:8]		
	P0_AVMute	3	Port 0 is in AVmute state	RO	
	Reserved	4			
	P0_HSync_In_Pol	5	Port 0 HSync input polarity	RO	
	P0_VSync_In_Pol	6	Port 0 VSync input polarity	RO	
	Reserved	7			
AB	Vid_HVR2nd[7:0]	7:0	The offset [7:0] of 2nd Vsync rise when interlaced mode		
AC	Vid_HVR2nd[13:8]	5:0	The offset [13:8] of 2nd Vsync rise when interlaced mode		
	Reserved	7:6			
AD	Vid_VFP2nd[7:0]	7:0	The width of 2nd VSync front porch[7:0]		
AE	Vid_VBPH[7:0]	7:0	The width of VSync back porch [7:0]		
AF	Vid_VFP2nd[11:8]	3:0	The width of 2nd VSync front porch[11:8]		
	Vid_VBPH[10:8]	6:4	The width of VSync back porch [10:8]		
	Reserved	7			
B0	Pix_Rep	3:0	Video input status 0000: no repetition ; 0001: pixel sent 2 times 0011: pixel sent 4 times	RO	
	Aud3DOn	4	0: not 3D audio 1: 3D audio	RO	
	MultAudOn	5	0: not Multi-Stream Audio 1: Multi-Stream Audio	RO	
	HBRAudio	6	Audio type is High Bit Rate	RO	
	DSDAudio	7	Audio type is DSD	RO	
B1	Audio_CH_Valid[5:0]	5:0	Audio channels valid Flag	RO	
	Audio_Layout	6	Audio layout is 1 or 0	RO	
	Audio_ON	7	Audio is ON	RO	

B2	Audio_CH_Status[7:0]	7:0	Audio Channel status decoder value[7:0]	RO	
B3	Audio_CH_Status[15:8]	7:0	Audio Channel status decoder value[15:8]	RO	
B4	Audio_CH_Status[23:16]	7:0	Audio Channel status decoder value[23:16]	RO	
B5	Audio_CH_Status[31:24]	7:0	Audio Channel status decoder value[31:24]	RO	
B6	Audio_CH_Status[39:32]	7:0	Audio Channel status decoder value[39:32]	RO	
B7	Pkt_Type_R[7:0]	7:0	Received packet type low byte Pkt_type is cleared each time it's read	RO	
B8	Pkt_Type_R[11:8]	3:0	Received packet type high byte	RO	
	Reserved	7:4			
B9	P0_Receive_Err	7:0	[7]: 2-bit BCH error is detected and packet is discarded. [6]: 1-bit BCH error is detected and packet is discarded. [5:4]: A decode error is detected on channel 2 [3:2]: A decode error is detected on channel 1 [1:0]: A decode error is detected on channel 0	W1C	
BA	P0_AFE_RegDbg	4:0	AFE debug signal read back 4: phase lock 3: audio pll lock 2:0: SDET_Normal	RO	
	P0_Video_Err	7:5	Video code error	W1C	
BB	P0_Ch0Code_Err	7:0	Error counter of code error	W1C	
BC	P0_Ch1Code_Err	7:0	Error counter of code error	W1C	
BD	P0_Ch2Code_Err	7:0	Error counter of code error	W1C	
BE	P0_Pkt_Err	7:0	Packet data error counter	W1C	
BF	P0_ErrCrt	7:0	[7:6]: reserved [5:3]: TercErrCrt for CH2 to CH0 [2:0]: CtrErrCrt for CH2 to CH0	W1C	
C0	P0_CH0_Lag	2:0	CH0 Deskew lag read back	RO	
	P0_CH1_Lag	5:3	CH1 Deskew lag read back	RO	
	Reserved	7:6			
C1	P0_CH2_Lag	2:0	CH2 Deskew lag read back	RO	
	Reserved	7:3			
C2	MCLKSEL_P	2:0	MCLKSel_P read back	RO	
	Reserved	3			
	MCLKSEL_D	6:4	MCLKSel_D read back	RO	
	Reserved	7			

Port 0 Internal EDID Register

C5	Reg_P0_DisableShadow	0	1: disable Port 0 Internal EDID	R/W	1
	Reg_P0_EnDetDDC	1	1: Enable monitor DDC detect start	R/W	1
	Reg_P0_EDIDAdrSel	2	1: EDID address A2 ; 0: EDID address A0	R/W	0
	Reserved	3			
	Reg_P0_SoftEDIDRst	4	1: EDID reset	R/W	0
	Reg_P0_EnMultiSeg	5	1: Enable multiple segment	R/W	0
	Reg_P0_EDIDTimeOutOpt	7:6	Timer to adjust EDID Bus Hang	R/W	00
C6	Reg_P0_VSDBAdr	7:0	VSDB Offset	R/W	n.a.
C7	Reg_P0_AB	7:0	Port 0 EDID's AB value	R/W	n.a.
C8	Reg_P0_CD	7:0	Port 0 EDID's CD value	R/W	n.a.
C9	Reg_P0_B0Sum	7:0	Port 0 EDID block 1's checksum value	R/W	n.a.
CA	Reg_P0_B1Sum	7:0	Port 0 EDID block2's checksum value	R/W	n.a.

Port 0 HDCP registers

CB	Reg_P0_HDCP_StableCnt	1:0	Will influence the interrupt auth_done. If not"00", auth_done only raised when some vsyncs pass without reauth seen.	R/W	00
	Reg_P0_HDCP_En	2	1: Enable HDCP	R/W	1
	Reg_P0_HDCP_A0Pin	3	1: Select 0x76 as DDC address. (default is 0x74)	R/W	0
	Reg_P0_FastHDCPMode	4	HDCP Bcap's FAST bit (Bcaps[4]) setting	R/W	0
	Reg_P0_HoldCntNum	6:5	HDCP DDC SCL hold time counter value	R/W	01
	Reg_P0_MasterSel	7	1: PC ; 0: HDCP	R/W	0
CC	Reserved	7:0			
CD	Reserved	7:0			
CE	Reg_P0_UpdRiSel	0	1: Ri value always update	R/W	0

			0: Ri value update only when deskew done		
	Reg_P0_SelAVMute	1	1: EESS_avmute will always mute	R/W	0
	Reg_P0_En_PwrOn_RdBKSV	2	1: Enable reading BKSV on power-on	R/W	1
	Reg_P0_EnHDCPOff	3	1: enable HDCP OFF function, if continuous 16 fields not receive enable description, then auth_state change to un-auth.	R/W	0
	Reg_P0_HDCPCtrl 7: Enable repeater 6: KSV ready 5: Enable feature 1.1 4: Enable Fast authentication	7:4	Enable repeater function Set KSV ready when IT6536 is repeater Enable HDCP 1.1 feature Enable Fast authentication	R/W	0011
CF	HDCP_ROMBIST_Trg	0	ROMBIST function trigger signal.	W1P	
	P0_HDCP_RdKSV_Trg	1	Software re_read BKSV trigger signal.	W1P	
	P0_RdROM_Trg	2		W1P	
	P0_AbortDDC_Trg	3	Abort DDC trigger signal.	W1P	
	P0_HDCP_Ri_Rec	4	Indicate HDCP TX had readback Ri, when write will be clear to 0	W1C	
	P0_HDCP_Encen_Flag	5	Port 0 HDCP Encryption Enable	RO	
	ROMDecFail	6	SIPROM decode failed	RO	
	P0_HDCP_OESSIdle	7	1: OESS Idle	RO	
D0	P0_HDCP_On_Flag	0	1: Encryption on going	RO	
	P0_HDCP_KeyRd_Fail_Flag	1	1: Read key from OTP was failed	RO	
	P0_HDCP_EESS	2	1: HDCP operates in EESS mode	RO	
	P0_HDCP_AdvCipher	3	HDCP Ainfo's ENABLE_1.1_FEATURES bit	RO	
	P0_HDCP_Auth_State	7:4	Show the HDCP authentication status	RO	
D1	P0_HDCP_Status 7: Pj ever read 6: V'H4 ever read 5: Ri ever read 4: R0 ever read 3: AKSV ever written 2: An ever written 1: BKSV ever read 0: HDCP port ever read	7:0	Bit[3:0] are cleared only by reset Bit[7:4] are cleared by re-authentication and reset	RO	
D2	Reg_P0_BCap7B	0	HDCP BCap bit 7 value	R/W	0
	Reserved	7:1			
D3	Reserved	7:0			
D4	Reserved	7:0			
D5	Reserved	7:0			
D6	Reserved	5:0			
	RPort0_EnHDCP	6	1 when HDCP2 is active	RO	
	Reserved	7			
D7~E1	Reserved	7:0			
E2	Reg_EnHDCP2	0			
	ReSelUnAuthSht	1		R/W	0
	Reg_HP2ToutSel	3:2		R/W	00
	Reg_SetHP2ToutReauth	4		R/W	0
	Reg_DisUnAuthenEncEn	5			0
	Reg_EnECCDnHDCP	6			0
	RKSVList_Ready	7	KSV list ready, 1T pulse when write this bit high	R/W	0
E3	Reg_HDCP2Ver	7:0	HDCP2.2 Version	R/W	0x04
F1	RegHPMaxEcc[7:0]	7:0		R/W	0x64
F2	Reserved	7:0			
F3	Reserved	7:0			
F4	Reserved	7:0			
F5	Reserved	7:0			
F6	Reg_SHA1_Trg	0	1 T pulse when write this bit high	R/W	0
	Reserved	6:1			0
	Reg_ForceHP2	7		R/W	0
F7	Reserved	7:0			
F8	Register Write Protect Password			R/W	n.a.
F9	TimeStamp_EN	7		R/W	0
	TimeStamp_Value	6:0		RO	

Block 1

HDCP Repeater Return Registers					
110	P0_BSTATUS[7:0]	7:0	Port 0 HDCP BStatus[7:0]	R.W	0x00
111	P0_BSTATUS[11:8]	3:0	Port 0 HDCP BStatus[11:8]	R.W	0000
	P0_HDMI Mode	4		RO	
	P0_BSTATUS[15:13]	7:5	Port 0 HDCP BStatus[11:8]	R.W	000
112	Reserved	7:0			
113	Reserved	7:0			
114	SHA1_H0[7:0]	7:0	HDCP SHA1_H0[7:0]	R.W	n.a.
115	SHA1_H0[15:8]	7:0	HDCP SHA1_H0[15:8]	R.W	n.a.
116	SHA1_H0[23:16]	7:0	HDCP SHA1_H0[23:16]	R.W	n.a.
117	SHA1_H0[31:24]	7:0	HDCP SHA1_H0[31:24]	R.W	n.a.
118	SHA1_H1[7:0]	7:0	HDCP SHA1_H1[7:0]	R.W	n.a.
119	SHA1_H1[15:8]	7:0	HDCP SHA1_H1[15:8]	R.W	n.a.
11A	SHA1_H1[23:16]	7:0	HDCP SHA1_H1[23:16]	R.W	n.a.
11B	SHA1_H1[31:24]	7:0	HDCP SHA1_H1[31:24]	R.W	n.a.
11C	SHA1_H2[7:0]	7:0	HDCP SHA1_H2[7:0]	R.W	n.a.
11D	SHA1_H2[15:8]	7:0	HDCP SHA1_H2[15:8]	R.W	n.a.
11E	SHA1_H2[23:16]	7:0	HDCP SHA1_H2[23:16]	R.W	n.a.
11F	SHA1_H2[31:24]	7:0	HDCP SHA1_H2[31:24]	R.W	n.a.
120	SHA1_H3[7:0]	7:0	HDCP SHA1_H3[7:0]	R.W	n.a.
121	SHA1_H3[15:8]	7:0	HDCP SHA1_H3[15:8]	R.W	n.a.
122	SHA1_H3[23:16]	7:0	HDCP SHA1_H3[23:16]	R.W	n.a.
123	SHA1_H3[31:24]	7:0	HDCP SHA1_H3[31:24]	R.W	n.a.
124	SHA1_H4[7:0]	7:0	HDCP SHA1_H4[7:0]	R.W	n.a.
125	SHA1_H4[15:8]	7:0	HDCP SHA1_H4[15:8]	R.W	n.a.
126	SHA1_H4[23:16]	7:0	HDCP SHA1_H4[23:16]	R.W	n.a.
127	SHA1_H4[31:24]	7:0	HDCP SHA1_H4[31:24]	R.W	n.a.
128~1AB	Reserved	7:0			
1AC	Reserved	7:0			
1AD	Reserved	7:0			

Block 2

SPI Infoframe

210	SRC_Ver	7:0	SPD infoFrame Version	RO	
211	SRC_PB25	7:0	SPK infoFrame Data Byte 25	RO	

AVI Infoframe

212	AVI_Length	7:0	AVI infoFrame Length	RO	
213	AVI_Version	7:0	AVI infoFrame Version	RO	
214	AVI_PB0	7:0	AVI infoFrame Data Byte 0	RO	
215	AVI_PB1	7:0	AVI infoFrame Data Byte 1	RO	
216	AVI_PB2	7:0	AVI infoFrame Data Byte 2	RO	
217	AVI_PB3	7:0	AVI infoFrame Data Byte 3	RO	
218	AVI_PB4	7:0	AVI infoFrame Data Byte 4	RO	
219	AVI_PB5	7:0	AVI infoFrame Data Byte 5	RO	
21A	AVI_PB6	7:0	AVI infoFrame Data Byte 6	RO	
21B	AVI_PB7	7:0	AVI infoFrame Data Byte 7	RO	
21C	AVI_PB8	7:0	AVI infoFrame Data Byte 8	RO	
21D	AVI_PB9	7:0	AVI infoFrame Data Byte 9	RO	
21E	AVI_PB10	7:0	AVI infoFrame Data Byte 10	RO	
21F	AVI_PB11	7:0	AVI infoFrame Data Byte 11	RO	
220	AVI_PB12	7:0	AVI infoFrame Data Byte 12	RO	
221	AVI_PB13	7:0	AVI infoFrame Data Byte 13	RO	
222	AVI_PB14	7:0	AVI infoFrame Data Byte 14	RO	
223	AVI_PB15	7:0	AVI infoFrame Data Byte 15	RO	

General Purpose Infoframe

224	GenPkt_HB0	7:0	General Packet Header Byte 0	RO	
225	GenPkt_HB1	7:0	General Packet Header Byte 1	RO	
226	GenPkt_HB2	7:0	General Packet Header Byte 2	RO	
227	GenPkt_PB0	7:0	General Packet Data Byte 0	RO	
228	GenPkt_PB1	7:0	General Packet Data Byte 1	RO	
229	GenPkt_PB2	7:0	General Packet Data Byte 2	RO	
22A	GenPkt_PB3	7:0	General Packet Data Byte 3	RO	
22B	GenPkt_PB4	7:0	General Packet Data Byte 4	RO	
22C	GenPkt_PB5	7:0	General Packet Data Byte 5	RO	
22D	GenPkt_PB6	7:0	General Packet Data Byte 6	RO	
22E	GenPkt_PB7	7:0	General Packet Data Byte 7	RO	
22F	GenPkt_PB8	7:0	General Packet Data Byte 8	RO	
230	GenPkt_PB9	7:0	General Packet Data Byte 9	RO	
231	GenPkt_PB10	7:0	General Packet Data Byte 10	RO	
232	GenPkt_PB11	7:0	General Packet Data Byte 11	RO	
233	GenPkt_PB12	7:0	General Packet Data Byte 12	RO	
234	GenPkt_PB13	7:0	General Packet Data Byte 13	RO	
235	GenPkt_PB14	7:0	General Packet Data Byte 14	RO	
236	GenPkt_PB15	7:0	General Packet Data Byte 15	RO	
237	GenPkt_PB16	7:0	General Packet Data Byte 16	RO	
238	GenPkt_PB17	7:0	General Packet Data Byte 17	RO	
239	GenPkt_PB18	7:0	General Packet Data Byte 18	RO	
23A	GenPkt_PB19	7:0	General Packet Data Byte 19	RO	
23B	GenPkt_PB20	7:0	General Packet Data Byte 20	RO	
23C	GenPkt_PB21	7:0	General Packet Data Byte 21	RO	
23D	GenPkt_PB22	7:0	General Packet Data Byte 22	RO	
23E	GenPkt_PB23	7:0	General Packet Data Byte 23	RO	
23F	GenPkt_PB24	7:0	General Packet Data Byte 24	RO	
240	GenPkt_PB25	7:0	General Packet Data Byte 25	RO	
241	GenPkt_PB26	7:0	General Packet Data Byte 26	RO	
242	GenPkt_PB27	7:0	General Packet Data Byte 27	RO	

Audio Infoframe

243	Audio_Version	7:0	Audio infoFrame Version	RO	
244	Audio_PB0	7:0	Audio infoFrame Data Byte 0	RO	
245	Audio_PB1	7:0	Audio infoFrame Data Byte 1	RO	
246	Audio_PB2	7:0	Audio infoFrame Data Byte 2	RO	
247	Audio_PB3	7:0	Audio infoFrame Data Byte 3	RO	
248	Audio_PB4	7:0	Audio infoFrame Data Byte 4	RO	
249	Audio_PB5	7:0	Audio infoFrame Data Byte 5	RO	
24A	Audio_Length	7:0	Audio infoFrame Length	RO	

MPEG Infoframe

24B	MPEG_Version	7:0	MPEG infoFrame Version	RO	
24C	MPEG_Length	7:0	MPEG infoFrame Length	RO	
24D	MPEG_PB0	7:0	MPEG infoFrame Data Byte 0	RO	
24E	MPEG_PB1	7:0	MPEG infoFrame Data Byte 1	RO	
24F	MPEG_PB2	7:0	MPEG infoFrame Data Byte 2	RO	
250	MPEG_PB3	7:0	MPEG infoFrame Data Byte 3	RO	
251	MPEG_PB4	7:0	MPEG infoFrame Data Byte 4	RO	
252	MPEG_PB5	7:0	MPEG infoFrame Data Byte 5	RO	

ACP Packet

253	ACP_HB0	7:0	ACP packet Header Byte 0	RO	
254	ACP_Type	7:0	ACP packet Header Byte 1	RO	
255	ACP_HB2	7:0	ACP packet Header Byte 2	RO	
256	ACP_PB0	7:0	ACP packet Data Byte 0	RO	
257	ACP_PB1	7:0	ACP packet Data Byte 1	RO	
258	ACP_PB2	7:0	ACP packet Data Byte 2	RO	
259	ACP_PB3	7:0	ACP packet Data Byte 3	RO	
25A	ACP_PB4	7:0	ACP packet Data Byte 4	RO	
25B	ACP_PB5	7:0	ACP packet Data Byte 5	RO	
25C	ACP_PB6	7:0	ACP packet Data Byte 6	RO	
25D	ACP_PB7	7:0	ACP packet Data Byte 7	RO	
25E	ACP_PB8	7:0	ACP packet Data Byte 8	RO	
25F	ACP_PB9	7:0	ACP packet Data Byte 9	RO	
260	ACP_PB10	7:0	ACP packet Data Byte 10	RO	
261	ACP_PB11	7:0	ACP packet Data Byte 11	RO	
262	ACP_PB12	7:0	ACP packet Data Byte 12	RO	
263	ACP_PB13	7:0	ACP packet Data Byte 13	RO	
264	ACP_PB14	7:0	ACP packet Data Byte 14	RO	
265	ACP_PB15	7:0	ACP packet Data Byte 15	RO	
266	ACP_PB16	7:0	ACP packet Data Byte 16	RO	
267	ACP_PB17	7:0	ACP packet Data Byte 17	RO	
268	ACP_PB18	7:0	ACP packet Data Byte 18	RO	
269	ACP_PB19	7:0	ACP packet Data Byte 19	RO	
26A	ACP_PB20	7:0	ACP packet Data Byte 20	RO	
26B	ACP_PB21	7:0	ACP packet Data Byte 21	RO	
26C	ACP_PB22	7:0	ACP packet Data Byte 22	RO	
26D	ACP_PB23	7:0	ACP packet Data Byte 23	RO	
26E	ACP_PB24	7:0	ACP packet Data Byte 24	RO	
26F	ACP_PB25	7:0	ACP packet Data Byte 25	RO	
270	ACP_PB26	7:0	ACP packet Data Byte 26	RO	
271	ACP_PB27	7:0	ACP packet Data Byte 27	RO	

ISRC Packets

272	ISRC1_HB0	7:0	ISRC1 packet Header Byte 0	RO	
273	ISRC1_HB1	7:0	ISRC1 packet Header Byte 1	RO	
274	ISRC1_HB2	7:0	ISRC1 packet Header Byte 2	RO	
275	ISRC1_PB0	7:0	ISRC1 packet Data Byte 0	RO	
276	ISRC1_PB1	7:0	ISRC1 packet Data Byte 1	RO	
277	ISRC1_PB2	7:0	ISRC1 packet Data Byte 2	RO	

278	ISRC1_PB3	7:0	ISRC1 packet Data Byte 3	RO	
279	ISRC1_PB4	7:0	ISRC1 packet Data Byte 4	RO	
27A	ISRC1_PB5	7:0	ISRC1 packet Data Byte 5	RO	
27B	ISRC1_PB6	7:0	ISRC1 packet Data Byte 6	RO	
27C	ISRC1_PB7	7:0	ISRC1 packet Data Byte 7	RO	
27D	ISRC1_PB8	7:0	ISRC1 packet Data Byte 8	RO	
27E	ISRC1_PB9	7:0	ISRC1 packet Data Byte 9	RO	
27F	ISRC1_PB10	7:0	ISRC1 packet Data Byte 10	RO	
280	ISRC1_PB11	7:0	ISRC1 packet Data Byte 11	RO	
281	ISRC1_PB12	7:0	ISRC1 packet Data Byte 12	RO	
282	ISRC1_PB13	7:0	ISRC1 packet Data Byte 13	RO	
283	ISRC1_PB14	7:0	ISRC1 packet Data Byte 14	RO	
284	ISRC1_PB15	7:0	ISRC1 packet Data Byte 15	RO	
285	ISRC2_HB0	7:0	ISRC2 packet Header Byte 0	RO	
286	ISRC2_HB1	7:0	ISRC2 packet Header Byte 1	RO	
287	ISRC2_HB2	7:0	ISRC2 packet Header Byte 2	RO	
288	ISRC2_PB0	7:0	ISRC2 packet Data Byte 0	RO	
289	ISRC2_PB1	7:0	ISRC2 packet Data Byte 1	RO	
28A	ISRC2_PB2	7:0	ISRC2 packet Data Byte 2	RO	
28B	ISRC2_PB3	7:0	ISRC2 packet Data Byte 3	RO	
28C	ISRC2_PB4	7:0	ISRC2 packet Data Byte 4	RO	
28D	ISRC2_PB5	7:0	ISRC2 packet Data Byte 5	RO	
28E	ISRC2_PB6	7:0	ISRC2 packet Data Byte 6	RO	
28F	ISRC2_PB7	7:0	ISRC2 packet Data Byte 7	RO	
290	ISRC2_PB8	7:0	ISRC2 packet Data Byte 8	RO	

HDCCP AKSV/BKSV/Mi read back

29A	P0_AKSV[7:0]	7:0	HDCCP AKSV ReadBack [7:0]	RO	
29B	P0_AKSV[15:8]	7:0	HDCCP AKSV ReadBack [15:8]	RO	
29C	P0_AKSV[23:16]	7:0	HDCCP AKSV ReadBack [23:16]	RO	
29D	P0_AKSV[31:24]	7:0	HDCCP AKSV ReadBack [31:24]	RO	
29E	P0_AKSV[39:32]	7:0	HDCCP AKSV ReadBack [39:32]	RO	
29F	P0_BKSV[7:0]	7:0	HDCCP BKSV value readback [7:0]	RO	
2A0	P0_BKSV[15:8]	7:0	HDCCP BKSV value readback [15:8]	RO	
2A1	P0_BKSV[23:16]	7:0	HDCCP BKSV value readback [23:16]	RO	
2A2	P0_BKSV[31:24]	7:0	HDCCP BKSV value readback [31:24]	RO	
2A3	P0_BKSV[39:32]	7:0	HDCCP BKSV value readback [39:32]	RO	
2A4	P0_M0[7:0]	7:0	HDCCP Mi value readback [7:0]	RO	
2A5	P0_M0[15:8]	7:0	HDCCP Mi value readback [15:8]	RO	
2A6	P0_M0[23:16]	7:0	HDCCP Mi value readback [23:16]	RO	
2A7	P0_M0[31:24]	7:0	HDCCP Mi value readback [31:24]	RO	
2A8	P0_M0[39:32]	7:0	HDCCP Mi value readback [39:32]	RO	
2A9	P0_M0[47:40]	7:0	HDCCP Mi value readback [47:40]	RO	
2AA	P0_M0[55:48]	7:0	HDCCP Mi value readback [55:48]	RO	
2AB	P0_M0[63:56]	7:0	HDCCP Mi value readback [63:56]	RO	

Audio N/CTS read back

2BE	N[19:12]	7:0	Audio N parameter decoder value [19:12]	RO	
2BF	N[11:4]	7:0	Audio N parameter decoder value [11:4]	RO	
2C0	N[3:0]	3:0	Audio N parameter decoder value [3:0]	RO	
	CTS[3:0]	7:4	Audio CTS parameter decoder value [3:0]	RO	
2C1	CTS[19:12]	7:0	Audio CTS parameter decoder value [19:12]	RO	
2C2	CTS[11:4]	7:0	Audio CTS parameter decoder value [11:4]	RO	

Audio Meta read back

2C4	AudMeta_HB1	7:0	Audio Meta packet header byte 1	RO	
2C5	AudMeta_HB2	7:0	Audio Meta packet header byte 2	RO	
2C6	AudMeta_PB0	7:0	Audio Meta packet data byte 0	RO	
2C7	AudMeta_PB1	7:0	Audio Meta packet data byte 1	RO	
2C8	AudMeta_PB2	7:0	Audio Meta packet data byte 2	RO	
2C9	AudMeta_PB3	7:0	Audio Meta packet data byte 3	RO	

2CA	AudMeta_PB4	7:0	Audio Meta packet data byte 4	RO	
2CB	AudMeta_PB5	7:0	Audio Meta packet data byte 5	RO	
2CC	AudMeta_PB6	7:0	Audio Meta packet data byte 6	RO	
2CD	AudMeta_PB7	7:0	Audio Meta packet data byte 7	RO	
2CE	AudMeta_PB8	7:0	Audio Meta packet data byte 8	RO	
2CF	AudMeta_PB9	7:0	Audio Meta packet data byte 9	RO	
2D0	AudMeta_PB10	7:0	Audio Meta packet data byte 10	RO	
2D1	AudMeta_PB11	7:0	Audio Meta packet data byte 11	RO	
2D2	AudMeta_PB12	7:0	Audio Meta packet data byte 12	RO	
2D3	AudMeta_PB13	7:0	Audio Meta packet data byte 13	RO	
2D4	AudMeta_PB14	7:0	Audio Meta packet data byte 14	RO	
2D5	AudMeta_PB15	7:0	Audio Meta packet data byte 15	RO	
2D6	AudMeta_PB16	7:0	Audio Meta packet data byte 16	RO	
2D7	AudMeta_PB17	7:0	Audio Meta packet data byte 17	RO	
2D8	AudMeta_PB18	7:0	Audio Meta packet data byte 18	RO	
2D9	AudMeta_PB19	7:0	Audio Meta packet data byte 19	RO	

Vendor Specific Infoframe

2DA	Vendor_Ver		Vendor Specific InfoFrame version	RO	
2DB	Vendor_Length		Vendor Specific InfoFrame length	RO	
2DC	Vendor_PB0		Vendor Specific InfoFrame Data Byte 0	RO	
2DD	Vendor_PB1		Vendor Specific InfoFrame Data Byte 1	RO	
2DE	Vendor_PB2		Vendor Specific InfoFrame Data Byte 2	RO	
2DF	Vendor_PB3		Vendor Specific InfoFrame Data Byte 3	RO	
2E0	Vendor_PB4		Vendor Specific InfoFrame Data Byte 4	RO	
2E1	Vendor_PB5		Vendor Specific InfoFrame Data Byte 5	RO	
2E2	Vendor_PB6		Vendor Specific InfoFrame Data Byte 6	RO	
2E3	Vendor_PB7		Vendor Specific InfoFrame Data Byte 7	RO	
2E4	Vendor_PB8		Vendor Specific InfoFrame Data Byte 8	RO	
2E5	Vendor_PB9		Vendor Specific InfoFrame Data Byte 9	RO	
2E6	Vendor_PB10		Vendor Specific InfoFrame Data Byte 10	RO	
2E7	Vendor_PB11		Vendor Specific InfoFrame Data Byte 11	RO	
2E8	Vendor_PB12		Vendor Specific InfoFrame Data Byte 12	RO	
2E9	Vendor_PB13		Vendor Specific InfoFrame Data Byte 13	RO	
2EA	Vendor_PB14		Vendor Specific InfoFrame Data Byte 14	RO	
2EB	Vendor_PB15		Vendor Specific InfoFrame Data Byte 15	RO	
2EC	Vendor_PB16		Vendor Specific InfoFrame Data Byte 16	RO	
2ED	Vendor_PB17		Vendor Specific InfoFrame Data Byte 17	RO	
2EE	Vendor_PB18		Vendor Specific InfoFrame Data Byte 18	RO	
2EF	Vendor_PB19		Vendor Specific InfoFrame Data Byte 19	RO	
2F0	Vendor_PB20		Vendor Specific InfoFrame Data Byte 20	RO	
2F1	Vendor_PB21		Vendor Specific InfoFrame Data Byte 21	RO	
2F2	Vendor_PB22		Vendor Specific InfoFrame Data Byte 22	RO	
2F3	Vendor_PB23		Vendor Specific InfoFrame Data Byte 23	RO	
2F4	Vendor_PB24		Vendor Specific InfoFrame Data Byte 24	RO	
2F5	Vendor_PB25		Vendor Specific InfoFrame Data Byte 25	RO	
2F6	Vendor_PB26		Vendor Specific InfoFrame Data Byte 26	RO	
2F7	Vendor_PB27		Vendor Specific InfoFrame Data Byte 27	RO	

Block 3 AutoEQ And AFE related registers

MHL/HDMI FrontEnd control					
320	Reg_P0_B_CS	2:0	B_CS initial value	R/W	011
	Reg_P0_G_CS	5:3	G_CS initial value	R/W	011
	Reg_P0_EnEQFailRty	6	1: enable EQ retry when EQ failed	R/W	0
	Reg_P0_EnClkStb2TrgEQ	7	1: enable clock stable to trigger AUTOEQ	R/W	0
321	Reg_P0_R_CS	2:0	R_CS initial value	R/W	011
	Reserved	3	Must be 0	R/W	0
	Reg_P0_SameRSCS	4	1: apply CH0 RS/CS to CH1 and CH2 0: : CH0/CH1/CH2 use individual RS/CS value	R/W	0
	Reserved	7:5			
326	Reserved	4:0			
	Reg_P0_UseBValue	5	1: all 3 channels use RS value of CH_B 0: CH_B/CH_G/CH_R use individual RS values	R/W	0
	Reserved	7:6			
327	Reg_P0_B_RS	6:0	RS value of CH_B if Reg_P0_OvWrRSCS = '1'	R/W	0111111
	Reg_P0_B_OvWrRS	7	1: Manually set RS value 0: RS value decided by autoEQ	R/W	0
328	Reg_P0_G_RS	6:0	RS value of CH_G if Reg_P0_OvWrRSCS = '1'	R/W	0111111
	Reg_P0_G_OvWrRS	7	1: Manually set RS value 0: RS value decided by autoEQ	R/W	0
329	Reg_P0_R_RS	6:0	RS value of CH_R if Reg_P0_OvWrRSCS = '1'	R/W	0111111
	Reg_P0_R_OvWrRS	7	1: Manually set RS value 0: RS value decided by autoEQ	R/W	0
32A	Reg_P0_LSRMIN	3:0	RS minimum value during AUTOEQ for TMDSClk ≤ 100MHz	R/W	0011
	Reg_P0_HSRMIN	7:4	RS minimum value during AUTOEQ for TMDSClk > 100MHz	R/W	0000
32B	Reg_P0_EQCKStbSel	0	1: clock valid and clock stable 0: only clock stable	R/W	1
	Reserved	7:1			
32C~339	Reserved	7:0			
33A	Reg_P0_CDR2X	0		R/W	0
	Reg_P0_ENHEQ	1		R/W	1
	Reg_P0_ENMEQ	2		R/W	0
	Reserved	7:3			
33B	Reserved	7:0			
Port 0 CBUS IO control					
3AD	Reg_CBUS_ST0	0	CBUS I/O driving strength[0]	R/W	0
	Reg_CBUS_ST1	1	CBUS I/O driving strength[1]	R/W	1
	Reserved	2			
	Reg_CBUS_SLEW	3	CBUS I/O slew rate control 0: fast 1: slow	R/W	1
	Reg_CBUS_SMT	4	1: enable CBUS schmitt trigger	R/W	0
	Reg_CBUS_SPD	5	CBUS pull down signal	R/W	0
	Reserved	7:6	Must be 00	R/W	00
3AE	Reg_V18V0	0		R/W	0
	Reg_V18V1	1		R/W	0
	Reg_CBUSR1K0	2		R/W	0
	Reg_CBUSR1K1	3		R/W	0
	Reg_CBUSR1K2	4		R/W	0
	Reg_CBUSR100K0	5		R/W	0
	Reg_CBUSR100K1	6		R/W	0
	Reg_CBUSR100K2	7		R/W	0
Port 0 EQ1 Read Back					
3D5	P0_Rec_B_RS	6:0	AutoEQ calculated B_RS value	RO	
	Reserved	7		RO	

3D6	P0_Rec_G_RS	6:0	AutoEQ calculated G_RS value	RO	
	Reserved	7		RO	
3D7	P0_Rec_R_RS	6:0	AutoEQ calculated R_RS value	RO	
	Reserved	7			
3D8	P0_Rec_B_CS	2:0	AutoEQ calculated B_CS value	RO	
	P0_Rec_G_CS	5:3	AutoEQ calculated G_CS value	RO	
	Reserved	7:6			
3D9	P0_Rec_R_CS	2:0	AutoEQ calculated R_CS value	RO	
	Reserved	7:3		RO	

Port 0 SCDC Registers

3E3	Reg_P0_SCDCRROPT[2:0]	2:0	[2:1] : RR SDA hold time 00: 5 us 01: 10 us 10: 15 us 11: use Reg_SCDCDevicID2B&"00" [0] : RR Request issue time 0: N-0.1 ms 1: N-0.5 ms	R/W	010
	Reserved	7:3			0
3E4	Reg10msErrCnt[7:0]	7:0		R/W	
3E5	Reg10msErrCnt[9:8]	1:0		R/W	
	Reg_P0_SETScrbEn	2	When Reg_P0_OvWrSCDC = '1' OverWrite Scramble_Enable, does not care SCDC setting	R/W	0
	Reg_P0_SETCLKRatio	3	When Reg_P0_OvWrSCDC = '1' OverWrite TMDs_Bit_Clock_Ratio	R/W	0
	Reg_P0_OvWrSCDC	4		R/W	0
	Reserved	5		R/W	0
	Reg_P0_SinkStandby	6		R/W	0
	Reserved	7		R/W	0
3E6	Reg_P0_SinkSCDCVer	7:0		R/W	0x01
3E7	Reg_P0_SrcSCDCVer	7:0		RO	
3E8	P0_RR_Enable	0	1: Source Support Read Request 0: Source does not support Read Request	RO	
	P0_Scramble_Enable	1	1 :Sink Detect Scramble 0: Sink does not detect scramble	RO	
	P0_TMDs_Bit_Clock_Ratio	2	0: 1/10 1: 1/40	RO	
	P0_Scramble_Status	3	1:	RO	
	P0_RSinkUpdateFlag[0]: Status_Update_Sts	4		RO	
	P0_RSinkUpdateFlag[1]: CED_Update_Sts	5		RO	
	P0_RSinkUpdateFlag[2]: RR_Test_Sts	6		RO	
	Reserved	7			
3E9	P0_SCDC_Clock_Detected	0		RO	
	P0_SCDC_Ch0_Locked	1		RO	
	P0_SCDC_Ch1_Locked	2		RO	
	P0_SCDC_Ch2_Locked	3		RO	
	Reg_P0_IntRdSel[2:0]	6:4	P0_CED Record[15:0] Register Read Back select 000: SCDC reg0x50/reg0x51 010: SCDC reg0x52/reg0x53 100: SCDC reg0x54/reg0x55 Others: SCDC reg0x56/ SCDC reg0xC0	R/W	000
	Reg_P0_IntClrCEDReg/ Reg_P0_IntRdCEDReg	7	1: Write this bit high to generate 1T pulse Reg_P0_IntClrCEDReg to clear CED error counter 0: Write this bit low to generate 1T pulse Reg_P0_IntRdCEDReg to latch CED error counter	W1P	
3EA	P0_CED Record[7:0]	7:0		RO	
3EB	P0_CED Record[15:0]	7:0		RO	

SCDC Registers

3EC	Reg_SCDCOUI_1B	7:0	SCDC reg0xD0	R/W	n.a.
3ED	Reg_SCDCOUI_2B	7:0	SCDC reg0xD1	R/W	n.a.
3EE	Reg_SCDCOUI_3B	7:0	SCDC reg0xD2	R/W	n.a.
3EF	Reg_SCDCDevicID1B	7:0	SCDC reg0xD3	R/W	n.a.
3F0	Reg_SCDCDevicID2B	7:0	SCDC reg0xD4	R/W	n.a.
3F1	Reg_SCDCDevicID3B	7:0	SCDC reg0xD5	R/W	n.a.
3F2	Reg_SCDCDevicID4B	7:0	SCDC reg0xD6	R/W	n.a.
3F3	Reg_SCDCDevicID5B	7:0	SCDC reg0xD7	R/W	n.a.
3F4	Reg_SCDCDevicID6B	7:0	SCDC reg0xD8	R/W	n.a.
3F5	Reg_SCDCDevicID7B	7:0	SCDC reg0xD9	R/W	n.a.
3F6	Reg_SCDCDevicID8B	7:0	SCDC reg0xDA	R/W	n.a.
3F7	Reg_SCDCDevicID9B	7:0	SCDC reg0xDB	R/W	n.a.
3F8	Reg_SCDCDevicID10B	7:0	SCDC reg0xDC	R/W	n.a.
3F9	Reg_SCDCDevicID11B	7:0	SCDC reg0xDD	R/W	n.a.
3FA	Reg_SCDCMSpec0B	7:0	SCDC reg0xDE	R/W	n.a.
3FB	Reg_SCDCMSpec1B	7:0	SCDC reg0xDF	R/W	n.a.
3FC	Reg_SCDCMSpec2B	7:0	SCDC reg0xE0	R/W	n.a.
3FD	Reg_SCDCMSpec3B	7:0	SCDC reg0xE1	R/W	n.a.

Block 6 DRM Packet Data

Dynamic Range and Mastering

610	DRM_HB0	7:0	DRM InfoFrame header byte 0	RO	
611	DRM_HB1	7:0	DRM InfoFrame header byte 1	RO	
612	DRM_HB2	7:0	DRM InfoFrame header byte 2	RO	
613	DRM_PB0	7:0	DRM InfoFrame data byte 0	RO	
614	DRM_PB1	7:0	DRM InfoFrame data byte 1	RO	
615	DRM_PB2	7:0	DRM InfoFrame data byte 2	RO	
616	DRM_PB3	7:0	DRM InfoFrame data byte 3	RO	
617	DRM_PB4	7:0	DRM InfoFrame data byte 4	RO	
618	DRM_PB5	7:0	DRM InfoFrame data byte 5	RO	
619	DRM_PB6	7:0	DRM InfoFrame data byte 6	RO	
61A	DRM_PB7	7:0	DRM InfoFrame data byte 7	RO	
61B	DRM_PB8	7:0	DRM InfoFrame data byte 8	RO	
61C	DRM_PB9	7:0	DRM InfoFrame data byte 9	RO	
61D	DRM_PB10	7:0	DRM InfoFrame data byte 10	RO	
61E	DRM_PB11	7:0	DRM InfoFrame data byte 11	RO	
61F	DRM_PB12	7:0	DRM InfoFrame data byte 12	RO	
620	DRM_PB13	7:0	DRM InfoFrame data byte 13	RO	
621	DRM_PB14	7:0	DRM InfoFrame data byte 14	RO	
622	DRM_PB15	7:0	DRM InfoFrame data byte 15	RO	
623	DRM_PB16	7:0	DRM InfoFrame data byte 16	RO	
624	DRM_PB17	7:0	DRM InfoFrame data byte 17	RO	
625	DRM_PB18	7:0	DRM InfoFrame data byte 18	RO	
626	DRM_PB19	7:0	DRM InfoFrame data byte 19	RO	
627	DRM_PB20	7:0	DRM InfoFrame data byte 20	RO	
628	DRM_PB21	7:0	DRM InfoFrame data byte 21	RO	
629	DRM_PB22	7:0	DRM InfoFrame data byte 22	RO	
62A	DRM_PB23	7:0	DRM InfoFrame data byte 23	RO	
62B	DRM_PB24	7:0	DRM InfoFrame data byte 24	RO	
62C	DRM_PB25	7:0	DRM InfoFrame data byte 25	RO	
62D	DRM_PB26	7:0	DRM InfoFrame data byte 26	RO	

Chap 12 MHL Registers

Note: Need to set HDMIRx reg34[0] as '0' to read the register while 5V not present.

Address is in HDMIRx reg49.

Reg	Name	Bit	Definition	Type	Default
00	RegEnCBusDbgMode	7	CBUS TX/RX queue debug mode 0: disable, 1: enable	R/W	0
	Reserved	6:4			
	RegEnCBusDbgCtrl[3:0]	3	1: Enable Reserved packet debug	R/W	0
		2	1: Enable MSC packet debug	R/W	0
		1	1: Enable Vendor-specific packet debug	R/W	0
		0	1: Enable DDC packet debug	R/W	0
01	RegEnCBusDeGlitch	7	CBUS input de-glitch 0: disable, 1: enable	R/W	0
	RegOSCDivSel[2:0]	6:4	OSCLK divide counter selection	R/W	100
	Reg100msTOAdj[1:0]	3:2	00: 100ms, 01: 99ms, 10: 101ms, 11: reserved	R/W	00
	RegEnCBusNack	1	Enable CBUS Nack bit protocol	R/W	0
	RegEn100msCnt	0	Enable 100ms calibration counter	R/W	0
04	RCBusDDCHangInt	7	CBUS DDC I2C BUS Hang interrupt Write '1' to clear this interrupt	W1C	
	RCBusWrBurstInt	6	MSC channel receive WRITE_BURST interrupt Write '1' to clear this interrupt	W1C	
	RCBusWrStatInt	5	MSC channel receive WRITE_STAT interrupt Write '1' to clear this interrupt	W1C	
	RCBusMSCRxMSGInt	4	MSC channel receive MSC_MSG interrupt Write '1' to clear this interrupt	W1C	
	RCBusRxPktFailInt	3	CBUS Link Layer receive a packet fail interrupt 1: when reg15h[3:0] /= 0	RO	
	RCBusRxPktDoneInt	2	CBUS Link Layer receive a packet done interrupt Write '1' to clear this interrupt	W1C	
	RCBusTxPktFailInt	1	CBUS Link Layer send a packet fail interrupt 1: when reg15h[7:4] /= 0	RO	
	RCBusTxPktDoneInt	0	CBUS Link Layer send a packet done interrupt Write '1' to clear this interrupt	W1C	
05	RCBusDDCRpdFailInt	7	CBUS DDC channel responder fail interrupt 1: when reg17h®16h /= 0	RO	
	RCBusDDCRpdDoneInt	6	CBUS DDC channel responder done interrupt Write '1' to clear this interrupt	W1C	
	Reserved	5			
	RCLKModeChgInt	4	RegMHLSts01B[2:0] change interrupt Write '1' to clear this interrupt	W1C	
	RCBusMSCRpdFailInt	3	CBUS MSC channel responder fail interrupt 1: when reg1Bh®1Ah /= 0	RO	
	RCBusMSCRpdDoneInt	2	CBUS MSC channel responder done interrupt Write '1' to clear this interrupt	W1C	
	RCBusMSCReqFailInt	1	CBUS MSC channel requester fail interrupt 1: when reg19h®18h /= 0	RO	
	RCBusMSCReqDoneInt	0	CBUS MSC channel requester done interrupt Write '1' to clear this interrupt	W1C	
06	RVBusChgInt	7	VBUS detection status change interrupt Write '1' to clear this interrupt	W1C	
	RDCapRdyChgInt	6	RegMHLSts00B[0] change interrupt Write '1' to clear this interrupt	W1C	
	RmuteChgInt	5	RegMHLSts01B[4] change interrupt Write '1' to clear this interrupt	W1C	
	RPathEnChgInt	4	RegMHLSts01B[3] change interrupt Write '1' to clear this interrupt	W1C	
	RCBusDisvFailInt	3	CBUS discovery process fail interrupt Write '1' to clear this interrupt	W1C	

	RCBusDisvDoneInt	2	CBUS discovery process done interrupt Write '1' to clear this interrupt	W1C	
	RWakeUpFailInt	1	CBUS discovery wakeup fail interrupt Write '1' to clear this interrupt	W1C	
	RWakeUpInt	0	CBUS discovery wakeup interrupt Write '1' to clear this interrupt	W1C	
08	RDDCBusHangMask	7	Interrupt Mask		
	RCBusWrBurstMask	6	0: Enable the corresponding interrupt	R/W	1
	RCBusWrStatMask	5	1: Disable the corresponding interrupt	R/W	1
	RCBusMSCRxMSGMask	4		R/W	1
	RCBusRxPktFailMask	3		R/W	1
	RCBusRxPktDoneMask	2		R/W	1
	RCBusTxPktFailMask	1		R/W	1
	RCBusTxPktDoneMask	0		R/W	1
09	RCBusDDCRpdFailMask	7		R/W	1
	RCBusDDCRpdDoneMask	6			
	Reserved	5		R/W	1
	CLKModeChgMask	4			
	RCBusMSCRpdFailMask	3		R/W	1
	RCBusMSCRpdDoneMask	2		R/W	1
	RCBusMSCReqFailMask	1		R/W	1
	RCBusMSCReqDoneMask	0		R/W	1
0A	RVBusChgMask	7		R/W	1
	RDCapRdyChgMask	6		R/W	1
	RMuteChgMask	5		R/W	1
	RPathEnChgMask	4		R/W	1
	RCBusDisvFailMask	3		R/W	1
	RCBusDisvDoneMask	2		R/W	1
	RWakeUpFailInt	1		R/W	1
	RWakeUpInt	0		R/W	1
0C	RegPPHDCPOpt	7	PackedPixel mode HDCP option 0: use Ch1, 1: use Ch1/Ch2	R/W	0
	Reserved	6:4			
	RegPPHDCPOpt2	3	PackedPixel mode HDCP option 2 for video period 0: use Ch0/Ch1 ; 1: use Ch1/Ch2	R/W	0
	RegClrPathEn	2	FW send PATH_EN{Sink}=0 Write '1' to trigger PATH_EN{Sink}=0 packet	W1P	
	RegSetPathEn	1	FW send PATH_EN{Sink}=1 Write '1' to trigger PATH_EN{Sink}=1 packet	W1P	
	RegEnHWPPathEn	0	HW PATH_EN control 0: Disable, 1: Enable	R/W	0
0E	Reserved	7:4			
	RegVBUS5VTSel[1:0]	3:2	VBUS 5V deglitch time selection 00: 50ms, 01: 75ms, 10: 100ms, 11: 125ms	R/W	01
	Reserved	1:0			
0F	Reserved	7:6			
	RegEnPPGBSwap	5	GuardBand Swap when PackMode	R/W	1
	RegSWRstDDFSM	4	SW Reset Discovery and Disconnect FSM 0: Normal operation, 1: Standby Mode Valid only when RegHWRstDDFSM='0'	R/W	1
	Reserved	3:0			
10	Reserved	7:5			
	RPATH_EN	4	PATH_EN status This is hardware internal status and it is updated after the PATH_EN transmission is done. PATH_EN{Sink}=1 to set status to 1 PATH_EN{Sink}=0 to set status to 0	RO	
	RVBus5Vdet	3	0: VBUS 5V not detected 1: VBUS 5V detected	RO	
	Reserved	2			
	REnVBUS	1	1: Pwr5V is output from VBUS	RO	
	REnCBUS	0	0: CBUS is not in connected state 1: CBUS is in connected state	RO	
12	R100msTimeCnt[7:0]	7:0	100ms counter value	RO	
13	R100msTimeCnt[15:8]	7:0		RO	
14	R100msTimeCnt[23:16]	7:0		RO	
15	RCBusTxPktFailSts[3:0]	7:4	CBUS Link Layer TX packet fail status	W1C	

			Reset when RCBUS_TxPktFailInt is cleared [7]: Reserved [6]: Initiator arbitrate error [5]: Tx packet timeout [4]: Tx packet fail		
	RCBUS_RxPktFailSts[3:0]	3:0	CBUS Link Layer RX packet fail status Reset when RCBUS_RxPktFailInt is cleared [3]: Reserved [2]: Rx parity check fail [1]: Rx packet timeout [0]: Rx packet fail	W1C	
16	RCBUSDDCRpdFailSts[7:0]	7:0	CBUS DDC channel responder fail status Reset when RCBUSDDCRpdFailInt is cleared [7]: TxState/=IDLE, receive new packet [6]: DDC bus hang [5]: transmit packet failed [4]: receive unexpected STOP [3]: 100ms timeout cause by link layer error [2]: 100ms timeout [1]: RxState/=IDLE, receive unexpected packet [0]: RxState=IDLE, receive non-SOF packet	W1C	
17	RCBUSDDCRpdFailSts[15:8]	7:0	[7:1] : Reserved [0]: TxState/=IDLE, receive ddcTxArbLose	W1C	
18	RCBUSMSCReqFailSts[7:0]	7:0	CBUS MSC channel requester fail status Reset when RCBUSMSCReqFailInt is cleared [7]: receive packet before transmit 1st packet [6]: un-enable HW retry and arbitration lose [5]: MSC_MSG Requester receive NACK packet [4]: receive ABORT packet [3]: Re-try threshold exceed [2]: protocol error [1]: 100ms timeout (too few packet) [0]: Incomplete packet	W1C	
19	RCBUSMSCReqFailSts[15:8]	7:0	[7:2]: Reserved [1]: FW mode RxPktFIFO is not empty [0]: FW mode fail	W1C	
1A	RCBUSMSCRpdFailSts[7:0]	7:0	CBUS MSC channel reponder fail status Reset when RCBUSMSCRpdFailInt is cleared [7]: Protocol error [6]: MSC_MSG responder Busy [5]: 100ms timeout (too few packet) [4]: Incomplete packet [3]: receive dPacket in reponder Idle [2]: invalid command [1]: incremental Bad Offset [0]: initial bad Offset	W1C	
1B	Reserved	7:2			
	RCBUSMSCRpdFailSts[9:8]	1:0	CBUS MSC channel responder fail status Reset when RCBUSMSCRpdFailInt is cleared [1]: receive ABORT packet [0]: Re-try threshold exceed	W1C	
1C	Reserved	7:3			
	RMSCMSGBusy	2	RX MSG register is valid	RO	
	RCBUSMSCBusy	1	CBUS MSC channel busy	RO	
	RCBUSDDCBusy	0	CBUS DDC channel busy	RO	
1D	RegDDCErrCode[7:0]	7:0	CBUS DDC channel ERRORCODE	R/W	0x00
1E	RegMSCErrCode[7:0]	7:0	CBUS MSC channel ERRORCODE	R/W	0x00
1F	RSStatus	7:0	CBUS Discovery's FSM status	RO	
28	RegCBUSTimerAdj	7	1: enable CBUS timer adjustment	R/W	0
	RegForceMode	6	1: enable force input HDMI/MHL mode	R/W	0
	RegDongleMode	5	1: Dongle Mode, 0: Receiver Mode	R/W	0
	RegForce_MHLMode	4	Force input Mode when reg28h[6]=1 1: MHL mode, 0: HDMI mode	R/W	0
	RegCBUSFloatAdj	3	Discovery related pulse width select	R/W	0
	RegSState5_EN	2	1: Permit state transition to Sink5_FloatCBUS	R/W	1
	RegForce_Standby	1	1: Force Discovery module to Standby state	R/W	0
	RegVBUS_DET_EN	0	1: enable VBUS detect	R/W	1
29	Reg10usSrcSel	7	0: from Calibration	R/W	0

			1: from Crystal 27MHz		
	RegVBUSDGSEL	6:4	VBUS Deglitch select	R/W	000
	RegCBUSDGSEL	3:2	CBUS Deglitch select	R/W	00
	RegCDDGSEL	1:0	CDSense Deglitch select	R/W	00
2A	RegDisvOpt	7	1: auto reset discovery FSM when discovery state transition from S2/S3 to S1.	R/W	0
	RegWkFailThr	6:5	WakeUp fail counter threshold value	R/W	10
	RegWkPulseSAdj	4	1: Enable WakeUp pulse small adjustment	R/W	0
	RegWkPulseLAdj	3	1: Enable WakeUp pulse large adjustment	R/W	0
	Reg5VStableTSEL	2:1	Power 5V stable time after RDetCDSense goes high. 00: 75ms, 01: 100ms, 10: 125ms, 11:150ms	R/W	01
	RegHWRstDDFSM	0	1: enable HW mode for RstDDFSM 0: enable SW mode for RstDDFSM	R/W	0
2B	Reserved	7:6			
	RegWakeUpFailTogSEL	5:4	Timeout selection for wakeup fail operation	R/W	01
	RegWakeUpFailFlt	3	1: Turn down 1K and 100K when wakeup fail trigger	R/W	1
	RegWakeUpFailTog	2	1: Output HPD when wakeup fail trigger	R/W	0
	RegAutoWakeUpFailTrg	1	1: Enable Auto WakeUp fail trigger	R/W	1
	FWWakeUpFailTrg	0	WakeUp Fail Trigger by FW	W1P	
30	Reg_TParityEn	7	Test parity enable	R/W	0
	Reg_TParity	6	Test parity value	R/W	0
	Reg_RetryMax	5:0	Retry packet number of times	R/W	0x20
31	Reg_CBusOESel	7	Time of Cbus OE selection for driving high 0: 200ns, 1: 300ns	R/W	0
	Reg_EnArbNack	6	Enable arbitration using Nack bit	R/W	0
	Reg_ARB_Bit	5:0	CBUS arbitration Low time	R/W	0x0A
32	Reg_BurstTime	7:0	CBUS requester continue time after ack	R/W	0x0C
33	Reg_PktNumMax	7:0	CBUS initiator output packet maximum number	R/W	0x18
34	Reg_SyncMax	7:0	Sync maximum time	R/W	0x17
35	Reg_SyncMin	7:0	Sync minimum time	R/W	0x07
36	Reg_AckHigh	7:4	ACK high to low time	R/W	0x8
	Reg_AckLow	3:2	ACK low pulse time	R/W	01
	Reg_RxAckSel	1	Rx ACK selection, 0: external, 1: internal	R/W	1
	Reg_EnArbDrvH	0	Enable CBUS driving High at arbitration	R/W	1
37	Reg_PacketEnd	7:0	Follower output packet done	R/W	0x13
38	RegEnDDCStopBusy	7	1: Enable DDC Stop Busy function	R/W	1
	RegRcvAbortRpt	6	1: reply to error status when receive ABORT.	R/W	1
	RegEnRtnAbort	5	1: reply Abort when receive unexpected EOF.	R/W	0
	RegUnexpStopRpy	4	1: reply error to Source when receive unexpected Stop.	R/W	1
	RegEnRptStr	3	1: enable I2C repeat start protocol.	R/W	1
	RegPCRReq	2	0: generate I2C clocks. 1: Abort I2C Bus	R/W	0
	RegPCTrg	1	0 => 1: PC I2C request trigger	R/W	0
	RegMasterSel	0	I2C master select. 1: PC, 0: CBUS	R/W	0
39	RegCmd_Deglitch	7	1: Enable I2C Deglitch	R/W	0
	RegSoftDDCSDA	6	1: DDCSDA controlled by SW	R/W	0
	RegSoftDDCSCL	5	1: DDCSCL controlled by SW	R/W	0
	RegSoftDDC	4	1: Enable DDC access by SW	R/W	0
	RegGenCLKPulse	3:0	Generate DDCSCL number by SW	R/W	0000
3A	RegCmd_FiltTap	7	I2C Deglitch Filter Tap select	R/W	0
	RegCmd_FiltType	6	I2C Deglitch Filter Type select	R/W	0
	RegBusHoldT	5:0	I2C bus hold time	R/W	000011
3B	RegI2CStCnt	7:0	I2C frequency setting	R/W	0x63
3C	Reserved	7:4			
	RegEnI2CHWAbort	3	1: Enable HW I2C Abort when DDCBusHang	R/W	1
	RegScratchPadClr	2	1: clear scratch pad registers to 0	R/W	0
	RegOvWrPPMode	1	1: PPMODE value when enable Over write	R/W	0
	RegEnOvWrPPMode	0	1: Enable over write PPMODE value	R/W	0
3D	RDDCStatus	7:0	[7]: DDC's Arbitration Lose Status [6]: DDC's WaitBus Status [5]: DDC's NAK Status [4]: DDC Done (include SCLGen and Abort) [3]: DDC busy [2]: DDC in WaitBus State [1]: DDC Receive NAK [0]: DDC Normal operation Done	RO	
3E	RDDCDBG	7:0	[7]: DDC Bus Hang	RO	

			[6]: DDC request [5:4] : DDC request command [3:0] : I2C FSM state		
50	RegMSCTxCmd[7:0]	7	WRITE_STAT/SET_INT	W1P	
		6	READ_DEVCAP	W1P	
		5	GET_MSC_ERRORCODE	W1P	
		4	GET_DDC_ERRORCODE	W1P	
		3	CLR_HPD	W1P	
		2	SET_HPD	W1P	
		1	GET_VENDOR_ID	W1P	
		0	GET_STATE	W1P	
51	RegMSCTxCmd[15:8]	7	FW_MODE	W1P	
		6:2	Reserved		
		1	MSC_MSG	W1P	
		0	WRITE_BURST	W1P	
52	RegMSCHwMask[7:0]	7	WRITE_STAT/SET_INT hardware mask 0: handled by HW, 1: handle by FW	R/W	0
		6	READ_DEVCAP hardware mask 0: handled by HW, 1: handle by FW	R/W	0
		5	GET_MSC_ERRORCODE hardware mask 0: handled by HW, 1: handle by FW	R/W	0
		4	GET_DDC_ERRORCODE hardware mask 0: handled by HW, 1: handle by FW	R/W	0
		3	CLR_HPD hardware mask 0: handled by HW, 1: handle by FW	R/W	0
		2	SET_HPD hardware mask 0: handled by HW, 1: handle by FW	R/W	0
		1	GET_VENDOR_ID hardware mask 0: handled by HW, 1: handle by FW	R/W	0
		0	GET_STATE hardware mask 0: handled by HW, 1: handle by FW	R/W	0
53	RegMSCHwMask[15:8]	7	FW_MODE hardware mask 0: enable FW mode, 1: disable FW mode	R/W	1
		6:2	Reserved		
		1	MSC_MSG hardware mask 0: handed by HW, 1: handle by FW	R/W	0
		0	WRITE_BURST hardware mask 0: handled by HW, 1: handle by FW	R/W	0
54	RegMSCTxValue0B[7:0]	7:0	Requester TX value byte 0	R/W	0x00
55	RegMSCTxValue1B[7:0]	7:0	Requester TX value byte 1	R/W	0x00
56	RMSCRxValue0B	7:0	Requester RX value byte 0	RO	
57	RxPktFIFORECnt	7:5	Receive packet FIFO read counter	RO	
	TxPktFIFORECnt	4:2	Transmit packet FIFO read counter	RO	
	TxPktFIFOWECnt	1:0	Transmit packet FIFO write counter	RO	
58	RMSCTxPktFIFOStg[4:0]	7:3	MSC TX packet FIFO stage	RO	
	RMSCTxPktFIFO[10:8]	2:0	[15:8]=high byte for WRITE_BURST (only [10:8] can be Read Back)	R/W	
59	RMSCTxPktFIFO[7:0]	7:0	[10:9]=hdrbit[1:0], [8]=ctrlbit, [7:0]=value[7:0]	R/W	
			Write low byte to push a packet to TxPktFIFO Read low byte to pop a packet from TxPktFIFO	R/W	
5A	RMSCRxPktFIFOStg[4:0]	7:3	MSC RX packet FIFO stage	RO	
	RMSCRxPktFIFO[10:8]	2:0	[10:9]=hdrbit[1:0], [8]=ctrlbit, [7:0]=value[7:0]	RO	
5B	RMSCRxPktFIFO[7:0]	7:0	Read low byte to pop a packet from RxPktFIFO	RO	
5C	RegEnPktFIFOBurst	7	0: Disable TxPktFIFO and RxPktFIFO burst mode 1: Enable TxPktFIFO and RxPktFIFO burst mode	R/W	0
	RegMSCBurstWrID	6	0: ID from TxPktFIFO 1: ID form MHL5E/5F	R/W	0
	RegMSCBurstWrOpt	5	WRITE_BURST command option 0: Write Adopter ID, 1: Not write Adopter ID	R.W	0
	RegMSCEnBurstWr	4	WRITE_BURST command mode 0: single mode, 1: burst mode	R/W	1
	RegEnMSCHwRty	3	MSC hardware retry 0: Disable, 1: Enable	R/W	0
	RegMSCRxUCP2Nack	2	Receive UCP when UCP_RECV_SUPPORT=0 0: return ACK, 1: return NACK	R/W	1

	RMSCRxPktFIFOClr	1	RX packet FIFO clear Write '1' to generate 1T pulse	W1P	
	RMSCTxPktFIFOClr	0	TX packet FIFO clear Write '1' to generate 1T pulse	W1P	
5D	RegVendorID[7:0]	7:0	Vendor ID	R/W	0x00
5E	RegWrBurstIDH[7:0]	7:0	ADOPTER_ID_H of the responder or BURST_ID_H for 3D Burst ID code	R/W	0x00
5F	RegWrBurstIDL[7:0]	7:0	ADOPTER_ID_L of the responder or BURST_ID_L for 3D Burst ID code	R/W	0x00
60	RMSGRxValue0B[7:0]	7:0	MSC_MSG responder RX value byte 0	RO	
61	RMSGRxValue1B[7:0]	7:0	MSC_MSG responder RX value byte 1	RO	
62	RMSGRxValue2B[7:0]	7:0	MSC_MSG responder RX value byte 2	RO	
63	RMSGRxValue3B[7:0]	7:0	MSC_MSG responder RX value byte 3	RO	
64	Reserved	7:5			
	RMSCRxScrPadNum[4:0]	4:0	(Read Only) Receive scratch pad byte number	RO	
66	RegEnMSCMSG2nd	7	0: Disable MSC_MSG 2 nd stage buffer 1: Enable MSC_MSG 2 nd stage buffer	R/W	0
	Reserved	6			
	RegMSCRxRAP2Nack	5	Receive RAP when RAP_RECV_SUPPORT=0 0: return ACK, 1: return NACK	R/W	0
	RegMSCRxRCP2Nack	4	Receive RCP when RCP_RECV_SUPPORT=0 0: return ACK, 1: return NACK	R/W	0
	Reserved	3:2			
	RegBurstWrTOSel	1:0	Burst Write timeout select 00: 320ms, 01: 340ms, 10: 360ms, 11: 400ms	R/W	01
80	RegMHLCap00B	7:0	DEV_STATE[7:0]	R/W	0x00
81	RegMHLCap01B	7:4	MHL_VER_MAJOR[3:0]	R/W	0x2
		3:0	MHL_VER_MINOR[3:0]	R/W	0x0
82	RegMHLCap02B	7	Reserved		
		6:5	PLIM[1:0]	R/W	01
		4	POW	R/W	1
		3:0	DEV_TYPE[3:0]	R/W	0x1
83	RegMHLCap03B	7:0	ADOPTER_ID_H[7:0]	R/W	0x00
84	RegMHLCap04B	7:0	ADOPTER_ID_L[7:0]	R/W	0x00
85	RegMHLCap05B	7:6	Reserved		
		5	SUPP_VGA	R/W	1
		4	SUPP_ISLANDS	R/W	1
		3	SUPP_PPIXEL	R/W	0
		2	SUPP_YCBCR422	R/W	1
		1	SUPP_YCBCR444	R/W	1
		0	SUPP_RGB444	R/W	1
86	RegMHLCap06B	7:2	Reserved		
		1	AUD_8CH	R/W	1
		0	AUD_2CH	R/W	1
87	RegMHLCap07B	7	SUPP_VT	R/W	0
		6:4	Reserved		
		3	VT_GAME	R/W	0
		2	VT_CINEMA	R/W	0
		1	VT_PHOTO	R/W	0
		0	VT_GRAPHICS	R/W	0
88	RegMHLCap08B	7	LD_GUI	R/W	0
		6	LD_SPEAKE	R/W	1
		5	LD_RECORD	R/W	0
		4	LD_TUNER	R/W	0
		3	LD_MEDIA	R/W	0
		2	LD_AUDIO	R/W	0
		1	LD_VIDEO	R/W	0
		0	LD_DISPLAY	R/W	1
89	RegMHLCap09B	7:0	BANDWIDTH[7:0]	R/W	0x0F
8A	RegMHLCap10B	7:5	Reserved		
		4	UCP_RECV_SUPPORT	R/W	1
		3	UCP_SEND_SUPPORT	R/W	1
		2	SP_SUPPORT	R/W	1

		1	RAP_SUPPORT	R/W	1
		0	RCP_SUPPORT	R/W	1
8B	RegMHLCap11B	7:0	DEVICE_ID_H[7:0]	R/W	0x00
8C	RegMHLCap12B	7:0	DEVICE_ID_L[7:0]	R/W	0x00
8D	RegMHLCap13B	7:0	SCRATCHPAD_SIZE	R/W	0x10
8E	RegMHLCap14B	7:4	STAT_SIZE[3:0]	R/W	0x3
		3:0	INT_SIZE[3:0]	R/W	0x3
8F	RegMHLCap15B	7:0	Reserved		0x00
A0	RegMHLInt00B	7:5	(Interrupt Status) Write '1' to clear this interrupt		
		4	3D_REQ (Interrupt Status) Write '1' to clear this interrupt	W1C	
		3	GNT_WRT (Interrupt Status) Write '1' to clear this interrupt	W1C	
		2	REQ_WRT (Interrupt Status) Write '1' to clear this interrupt	W1C	
		1	DSCR_CHG (Interrupt Status) Write '1' to clear this interrupt	W1C	
		0	(Interrupt Status) Write '1' to clear this interrupt		
A1	RegMHLInt01B	7:2	(Interrupt Status) Write '1' to clear this interrupt		
		1	EDID_CHG (Interrupt Status) Write '1' to clear this interrupt	W1C	
		0	(Interrupt Status) Write '1' to clear this interrupt		
A2	RegMHLInt02B	7:0	(Interrupt Status) Write '1' to clear this interrupt		
A3	RegMHLInt03B	7:0	(Interrupt Status) Write '1' to clear this interrupt		
B0	RegMHLSts00B	7:1	Reserved		
		0	DCAP_RDY (Read Only)	RO	0
B1	RegMHLSts01B	7:5	Reserved		
		4	MUTED (Read Only)	RO	0
		3	PATH_EN (Read Only)	RO	0
		2:0	CLK_MODE[2:0] (Read Only) 011: Normal 24-bit mode 010: PackedPixel mode others: reserved	RO	011
B2	RegMHLSts02B	7:0	Reserved		0x00
B3	RegMHLSts03B	7:0	Reserved		0x00
C0	RegMHLPad00B[7:0]	7:0	Scratchpad register byte 0	RO	0x00
C1	RegMHLPad01B[7:0]	7:0	Scratchpad register byte 1	RO	0x00
C2	RegMHLPad02B[7:0]	7:0	Scratchpad register byte 2	RO	0x00
C3	RegMHLPad03B[7:0]	7:0	Scratchpad register byte 3	RO	0x00
C4	RegMHLPad04B[7:0]	7:0	Scratchpad register byte 4	RO	0x00
C5	RegMHLPad05B[7:0]	7:0	Scratchpad register byte 5	RO	0x00
C6	RegMHLPad06B[7:0]	7:0	Scratchpad register byte 6	RO	0x00
C7	RegMHLPad07B[7:0]	7:0	Scratchpad register byte 7	RO	0x00
C8	RegMHLPad08B[7:0]	7:0	Scratchpad register byte 8	RO	0x00
C9	RegMHLPad09B[7:0]	7:0	Scratchpad register byte 9	RO	0x00
CA	RegMHLPad10B[7:0]	7:0	Scratchpad register byte 10	RO	0x00
CB	RegMHLPad11B[7:0]	7:0	Scratchpad register byte 11	RO	0x00
CC	RegMHLPad12B[7:0]	7:0	Scratchpad register byte 12	RO	0x00
CD	RegMHLPad13B[7:0]	7:0	Scratchpad register byte 13	RO	0x00
CE	RegMHLPad14B[7:0]	7:0	Scratchpad register byte 14	RO	0x00
CF	RegMHLPad15B[7:0]	7:0	Scratchpad register byte 15	RO	0x00

Chap 13 HDMI TX Register Definitions

Register Type:

RW: read/write Register

RO: read only Register

WP: read only, write protected by password

W1C: write '1' clear status

W1P: write '1' to generate 1T trigger pulse

HDMI Tx Common Registers

The I²C address of HDMI Tx common registers is defined in HDMI Converter register 0xF1.

General Registers

Reg	Register Name	Bit	Definition	Type	Default
00	Vender ID[7:0]	7:0	Vendor ID	RO	0x54
01	Vender ID[15:8]	7:0			0x49
02	Device ID[7:0]	7:0	Device ID	RO	0x64
03	Device ID[15:8]	7:0			0x66
04	Revision ID[7:0]	7:0	Revision ID	RO	0xA0
Common Register					
05		0	Software RCLK domain reset	RW	0
		1	Software TxPatGen reset	RW	1
	RegSoftH2RSARst	2	Software HDCP2Tx RSA engine reset	RW	0
		7:2			
06					
	RegPCGateRCLK	1	1: power down RCLK	RW	0
	RegEnPGVCLKInv	2	1: Enable TxPatGen VCLK inversion	RW	0
	RegEnPGPCLKInv	3	1: Enable TxPatGen PCLK inversion	RW	0
07 ~ 0E					
0F	P0RefHDMIIntSts	0	Port0 HDMI interrupt status	RO	-
	P1RefHDMIIntSts	1	Port1 HDMI interrupt status	RO	-
	P2RefHDMIIntSts	2	Port2 HDMI interrupt status	RO	-
	P3RefHDMIIntSts	3	Port3 HDMI interrupt status	RO	-
		4	Port0 CEC interrupt status	RO	-
		5	Port1 CEC interrupt status	RO	-
		6	Port2 CEC interrupt status	RO	-
		7	Port3 CEC interrupt status	RO	-
10	RegHPDDGTimeSel[1:0]	1:0	HPD de-glitch time selection	RW	10
	RegRxSenDGTimeSel[1:0]	3:2	RxSen de-glitch time selection	RW	01
		7:4			
11	RegTimeLoMax[7:0]	7:0	Timer Reference	RW	0x00
12	RegTimeLoMax[15:8]	7:0		RW	0x4A
13	RegTimeLoMax[17:16]	1:0		RW	0x2
	Reserved	3:2		WP	00
		5:4			
	RegTimeBaseSel	7:6	Ri/Pj Read Timer Reference switch (only when RegHDCPSyncDet=1) 00: RegTimeLowMax 01: RegTimeLowMax/32 10: RegTimeLowMax/64 11: RegTimeLoxMax/128	RW	0x0
14	RegTimeRetryAuthen[7:0]	7:0	Timer Reference.	RW	0xC8
15	Reserved	0		RW	0

		1			
		2			
		3		RW	0
	Reserved	5:4		RW	01
		6			
	RegVSIF3D_FW	7	1: FW overwrite ISFB Start and ISFB End	RW	0
16	RegISFBStart[7:0]	7:0	MHL 3D FS Mode ISFB start line number	RW	0x---
17	RegISFBStart[12:8]	4:0			
		7:5			
18	RegISFBEnd[7:0]	7:0	MHL 3D FS Mode ISFB end line number	RW	0x---
19	RegISFBEnd[12:8]	4:0			
		7:5			
20					
	RegEnKSVFIFORst	1	1: KSV FIFO reset	RW	0
	RegEnFWKSVFIFO	2	1: enable FW KSV FIFO access (debug mode)	RW	0
		3			
	RegKsvSrcSel[2:0]	6:4	KSV List source selection 000: TX Port0 HDCP1.4 001: TX Port1 HDCP1.4 010: TX Port2 HDCP1.4 011: TX Port3 HDCP1.4 100: TX HDCP2.2 engine 0 101: TX HDCP2.2 engine 1 110: TX HDCP2.2 engine 2 111: TX HDCP2.2 engine 3	RW	000
		7			
21	RegKsvFifoWrPtr[4:0]	4:0	Current KSV List FIFO write pointer When I2C burst write beyond Reg27, the write pointer increases 1 automatically by HW.	RW	00000
		7:5			
22	RKsvFifoWrStg[4:0]	4:0	Current KSV List FIFO write stage Reset by RegEnKsvFifoRst and over-written by FW KSV write	RO	-----
		7:5			
23	RegKsvFifoWrData[7:0]	7:0	KSV write data	RW	0x--
24	RegKsvFifoWrData[15:8]	7:0	(write to Reg27 will trigger a request to write KSV data to SRAM when	RW	0x--
25	RegKsvFifoWrData[23:16]	7:0	RegEnFwKsvFifo=1)	RW	0x--
26	RegKsvFifoWrData[31:24]	7:0	Note: when I2C burst write beyond Reg27, HW rewind offset to Reg23.	RW	0x--
27	RegKsvFifoWrData[39:32]	7:0		RW	0x--
HDCP2ARB Register					
28	RRCCalBusy	0	HDCP2ARB RC Cal engine calculation busy	RO	
	RRSABusy	1	HDCP2ARB RSA engine calculation busy	RO	
	RSHA256Busy	2	HDCP2ARB SHA256 engine calculation busy	RO	
		7:3			
29 ~ 2B					
2C	RegP0TxSlvAdr[7:0]	7:0	TX Port0 HDMI slave address, bit[0] control the enable function	RW	0x98
2D	RegP1TxSlvAdr[7:0]	7:0	TX Port1 HDMI slave address, bit[0] control the enable function	RW	0x9A
2E	RegP2TxSlvAdr[7:0]	7:0	TX Port2 HDMI slave address, bit[0] control the enable function	RW	0x9C
2F	RegP3TxSlvAdr[7:0]	7:0	TX Port3 HDMI slave address, bit[0] control the enable function	RW	0x9E

TX General Packet Registers

Reg	Name	Bit	Definition	Type	Default
30	RegPktNull0Hdr[7:0]	7:0	Null Packet HB0	RW	0x--
31	RegPktNull1Hdr[7:0]	7:0	Null Packet HB1	RW	0x--
32	RegPktNull2Hdr[7:0]	7:0	Null Packet HB2	RW	0x--
33	RegPktNull00PB[7:0]	7:0	Null Packet PB00	RW	0x--
34	RegPktNull01PB[7:0]	7:0	Null Packet PB01	RW	0x--
~	~		~		
4D	RegPktNull26PB[7:0]		Null Packet PB26		
4E	RegPktNull27PB[7:0]	7:0	Null Packet PB27	RW	0x--
4F					

FF	PassWord	7:0	0xC3/0xA5 on, 0xFF off	RW	0x--
----	----------	-----	------------------------	----	------

Chap 14 HDMI Tx Port0~Port3 Registers

HDMI Tx Port 0~3 use the same registers but different I²C addresses, which defined in HDMI Tx common register 0x2C~0x2F.

Register banks are selected using Reg0F[1:0].
When Reg05[0]=0, Register bank 0 is active.
When Reg05[9]=1, Register bank 1 is active.

Be careful to switch to the accurate Register bank when you program Registers.

CEC:

Before accessing CEC Register, RegEnCRCLK in RegFC[0] must be set 1 to enable CEC function.

General Registers

System Reset & clock control					
01	RegSoftRRst	0	Software reset	RW	0
	RegSoftVRst	1	Software VCLK domain reset	RW	1
	RegSoftPRst	2	Software PCLK domain reset	RW	1
		3			
	RegStoreKmRst	4	HDCP 2.2 stored Km reset	RW	0
	RegHDCP_Rst	5	HDCP function reset	RW	0
		6			
02		7			
	RegGateRCLK	0	1: power down RCLK	RW	0
	RegGateCRCLK	1	1: power down CRCLK for CEC function	RW	0
	RegEnVCLKInv	2	1: Enable latch VCLK inversion	RW	0
	RegEnPCLKInv	3	1: Enable latch PCLK inversion	RW	0
		4			
		5			
03		6			
		7			
	RHPDetect	0	Hot Plug detect status 0: plug off, 1: plug on	RO	-
	RxSenDetect	1	RxSense detect status 0: not detect, 1: detected	RO	-
	TxVidStable	2	Video input status: 0: unstable video input, 1: stable video input	RO	-
	VTMDSStable	3	TMDs stable	RO	-
	TxCCLK_Stable	4	Pixel clock stable	RO	-
05		5			
		6			
	XP_LOCK	7	TMDs clock PLL lock status 0: XPLL is unlock, 1: XPLL lock	RO	-
		0	0: Bank 0, Reg00h~ Regffh 1: Bank 0, Reg100h~ Reg1ffh for CEC use	RW	0
	RefHDMIIntSts	1	HDMI interrupt status	RO	-
		2	CEC interrupt status	RO	-
		7:3			
06	RPCLKCnt[7:0]	7:0	Pixel clock frequency counter	RO	0x--
07		3:0		RO	0x-
		6:4	Extra pixel clock division counter	RW	000
		7	Enable pixel clock counter	RW	0
System Control Registers					
08	RegEnHDMIInt	0	HDMI interrupt group to external INT pin 0: Disable, 1: Enable	RW	1
		1			
	RegForceRxOn	2	0: Normal operation, 1: Force HDP/RxSen ON	WP	0

	Reg_ForceVOut	3	Force Video Stable 0: Normal, 1: Force TxVideo Stable	WP	0
	RegForceTMDSStb	4	Force TMDs stable 0: Disable, 1: Enable	WP	0
		5			
		6			
		7			
0A					
0B					
0C	RegPktEventSel[3:0]	3:0	Packet event selection for interrupt 0000: GenCtrl packet 0001: Null packet 0010: Reserved 0011: AVI packet 0100: Reserved 0101: Reserved 0110: GBD packet 0111: VS packet 1000: Reserved 1001: Reserved 1010~1111: reserved	RW	0000
		7:4			
0D	RegTimerIntNum[6:0]	6:0	User defined timer interrupt count number	RW	0x1E
	RegTimerIntUnit	7	0: unit=10ms, 1: unit=100ms	RW	1
0E	RegPktEnFIFOBIST	0	1: enable TX packet SRAM BIST	RW	0
	RegPktLFSRBISTOpt	1	TX packet SRAM BIST option	RW	0
	RegEnFIFOBIST_rc	2	1: enable HDCP2.2 RC SRAM BIST	RW	0
	RegLFSRBISTOpt_rc	3	HDCP2.2 RC SRAM BIST option	RW	0
	RPktFIFOBISTDone	4	1: TX packet SRAM BIST done	RO	-
	RPktFIFOBISTFail	5	1: TX packet SRAM BIST fail	RO	-
	RFIFOBISTDone_rc	6	1: HDCP2.2 RC SRAM BIST done	RO	-
	RFIFOBISTFail_rc	7	1: HDCP2.2 RC SRAM BIST fail	RO	-
0F	RegEnFIFOBIST_msg	0	1: enable HDCP2.2 message SRAM BIST	RW	0
	RegLFSRBISTOpt_msg	1	HDCP2.2 message SRAM BIST option	RW	0
	RegEnFIFOBIST_s	2	1: enable HDCP2.2 signature SRAM BIST	RW	0
	RegLFSRBISTOpt_s	3	HDCP2.2 signature SRAM BIST option	RW	0
	RFIFOBISTDone_msg	4	1: HDCP2.2 message SRAM BIST done	RO	-
	RFIFOBISTFail_msg	5	1: HDCP2.2 message SRAM BIST fail	RO	-
	RFIFOBISTDone_s	6	1: HDCP2.2 signature SRAM BIST done	RO	-
	RFIFOBISTFail_s	7	1: HDCP2.2 signature SRAM BIST fail	RO	-

Interrupt Status Registers

10	RHPDChgInt	0	HPD change interrupt	W1C	0
	RRxSenChgInt	1	RxSen change interrupt	W1C	0
	RDDCCmdFailInt	2	DDC command fail interrupt Reset by (Reg_MasterSel=1 or RegEnDDCMasterArb=1) and RDDC_Req=0xF	RO	-
	RDDCFIFOErrInt	3	DDC FIFO error interrupt Reset by (Reg_MasterSel=1 or RegEnDDCMasterArb=1) and RDDC_Req=0x9	RO	-
	RDDCFIFOValidInt	4	DDC FIFO valid interrupt Reset by reading DDC 32-stage FIFO to empty	RO	-
		5			
	RVSyncEventInt	6	VSynC event interrupt	W1C	0
	RVidStableChgInt	7	Video stable change interrupt	W1C	0
11	RAuthFailInt	0	HDCP authentication fail interrupt	W1C	0
	RAuthDoneInt	1	HDCP authentication done interrupt	W1C	0
	RKSVListChkInt	2	HDCP KSV list check interrupt	W1C	0
		3			
		4			
		5			
	RPktEventInt	6	Packet event done interrupt (Refer to RegPktEventSel[3:0])	W1C	0
	RFwTimeTrgInt	7	User timer interrupt	W1C	0
12	RExtIntSts[7:0]	0	HDMI 2.0 SCDC update flag interrupt	RO	-
		1	HDMI 2.0 SCDC detect read request interrupt	W1C	0

		2	Video input FIFO error interrupt	W1C	0
		3	DDC bus hang interrupt Reset by (Reg_MasterSel=1 or RegEnDDCMasterArb=1) and RDDC_Req=0xA	RO	-
		4	HDCP Ri check done interrupt	W1C	0
		5	HDCP Pj check done interrupt	W1C	0
		6	Video parameter change interrupt	W1C	0
		7	HDCP 1.2 sync. detection fail interrupt	W1C	0
		0	HDCP2.2 RHDCP2RepWaitTxMgm_Int	W1C	0
13	RExtIntSts[15:8]	1	HDCP 2.2 RHDCP2RxIDSRMChk_Int	W1C	0
		2	HDCP 2.2 RREAUTH_REQ	W1C	0
		3	HDCP 2.2 RHDCP2StrmMgmFail_Int	W1C	0
		4	HDCP 2.2 RHDCP2ListSRMChk_Int	W1C	0
		5	HDCP2.2 RHDCP2RepWaitTxEks_Int	W1C	0
		6	HDCP2.2 RHDCP2KSVChkDone_Int	W1C	0
		7	HDCP2.2 RHDCP2KSVWrDone_Int	W1C	0
14	RExtIntSts[23:16]	0	Packet FIFO RPktFifoErrInt	W1C	0
		1	Packet FIFO RPktDataErrInt	W1C	0
		2	Packet FIFO RPktAud3DErrInt	W1C	0

Interrupt Enable Registers

18	RegEnHPDChgInt	0	Enable of the corresponding interrupts from Reg10 to Reg14 0: disable this interrupt 1: enable this interrupt. RegEnHDCP2KSVChkDone_Int <= reg1bh[6]	RW	1
	RegEnRxSenChgInt	1		RW	1
	RegEnDDCCmdFailInt	2		RW	0
	RegEnDDCFIFOErrInt	3		RW	0
	RegEnDDCFIFOValidInt	4		RW	0
		5		RW	0
	RegEnVSyncEventInt	6		RW	0
	RegEnVidStableChgInt	7		RW	0
19	RegEnAuthFailInt	0		RW	0
	RegEnAuthDoneInt	1		RW	0
	RegEnKSVListChkInt	2		RW	0
		3		RW	0
		4		RW	0
		5		RW	0
	RegEnPktEventInt	6		RW	0
	RegEnFwTimeTrgInt	7		RW	0
1A	RegEnExtInt[7:0]	7:0		RW	0
1B	RegEnExtInt[15:8]	7:0		RW	0
1C	RegEnExtInt[23:16]	7:0		RW	0
1D					
1E					
1F					

DDC and SCDC Control Registers

Reg	Register Name	Bit	Definition	Type	Default
28	Reg_MasterSel	0	Switch FW/HW to access DDC port 0: HW or HW+FW when RegEnDDCMasterArb=1, 1: FW only	RW	0
	RegSoftDDCSDA	1	Software DDCSDA control	RW	1
	RegSoftDDCSCL	2	Software DDCSCL control	RW	1
	RegSoftDDC	3	1: Enable software DDC mode	RW	0
	RegGenCLKPulse[3:0]	7:4	DDC abort CLK pulse	RW	0x9
29	RDDC_Header[7:0]	7:0	PC DDC request slave address: 0x74 when access Rx HDCP 0xA0 when access Rx EDID 0xA8 when access RX SCDC (HDMI 2.0)	RW	0x--
2A	RDDC_ReqOffSet[7:0]	7:0	Register address	RW	0x--
2B	RDDC_ReqByte[7:0]	7:0	Register R/W byte number	RW	0x--
2C	RDDC_ReqByte[9:8]	1:0		RW	00

		7:2			
2D	RDDC_Segment[7:0]	7:0	EDID segment	RW	0x--
2E	RDDC_Req[3:0]	3:0	PC DDC request command 0x0: Sequential Burst Read 0x1: Sequential Burst Write 0x2: Link check read (HDCP hardware mode) 0x2: SCDC update read when Reg11=0xA8 (HDMI 2.0) 0x3: EDID read 0x4: AKSV write 0x5: Ainfo write 0x6: An write 0x7: EDID write 0x9: DDC FIFO clear 0xA: GenerateSCL clock pulse 0xF: Abort DDC command.	RW	0x0
	RDDCArbState[1:0]	5:4	DDC arbitrator state 00: Idle, 01: HDCP, 10: SCDC, 11: FW	RO	--
	DDC_SCL	6	DDC SCL pin status	RO	-
	DDC_SDA	7	DDC SDA pin status	RO	-
2F	RDDC_Status[7:0]	0	RDDC_HW_Stus_Done 0: HW DDC is not complete, 1: HW DDC transfer is complete	RO	0x--
		1	RDDC_FIFOEmpty		
		2	RDDC_FIFOFull		
		3	RDDC_Stus_ArbiLose 1: DDC has something error		
		4	RDDC_Stus_WaitBus 1: DDC has something error		
		5	RDDC_Stus_NoACK 1: DDC has something error		
		6	RDDC_Active		
		7	RDDC_FW_Stus_Done 0: FW DDC is not complete, 1: FW DDC transfer is complete		
30	RDDC_RdWrFIFO	7:0	Read/Write DDC FIFO content There are 32-stage read/write DDC FIFO.	RW	0x--
31					
32	Reg_HDCPHeader	7:0	HDCP slave address from HDCP Engine	RW	0x74
33					
34	Reg_BusHoldT[5:0]	5:0	DDC Bus start/stop setup/hold time requirement	RW	0x03
	Reserved	6		RW	0
	Reg_FWDDCFIFOAcc	7	1: Enable FW DDC FIFO access mode	RW	1
35	RegDDCSpeed[1:0]	1:0	DDC Master Clock 00: DDC clock using 75K 01: DDC clock using 125K 10: DDC clock using 312K 11: simulation only	RW	00
		3:2			
	RegDDCMasterRst	4	DDC master function reset 0: normal operation, 1: reset	RW	0
	RegDDCCmdFailOpt	5	DDC command fail option 0: disable timeout check, 1: enable timeout check	RW	1
	RegDDCBusHangTOSel[1:0]	7:6	DDC bus hang timeout selection 00: 64ms, 01: 128ms, 10: 192ms, 11: 256ms	RW	00
36	RDDC_FIFOSTg[5:0]	5:0	DDC FIFO stage 000000: FIFO empty 100000: FIFO full	RO	0x--
	RDDC_FIFOOverRd	6	DDC FIFO over-read (clear by DDC FIFO clear command)	RO	-
	RDDC_FIFOOverWr	7	DDC FIFO over-write (clear by DDC FIFO clear command)	RO	-
37	RegDisDDCFIFO	0	1: disable DDC FIFO	RW	0
38	reserved				
39	reserved				
3A	RegEnFlagPolling	0	SCDC update flag polling when video is stable 0: disable, 1: enable	RW	0
	RegEnH2DetRR	1	HDMI2.0 Read Request detection 0: disable, 1: enable	RW	0

	RegFlagPollSel[1:0]	3:2	00: VSync Event, 01: 100ms, 10: 150ms, 11: 200ms	RW	00
	RegEnFlagUpdRd	4	Update read of SCDC update flag 0: disable, 1: enable	RW	0
	RegSCDCFlagByte	5	0: 1-byte (0x10), 1: 2-byte (0x10~0x11)	RW	0
	RegDisH2Auto	6	Disable HDMI2.0 auto-mode of SCDC 0x20[1:0] 0: auto mode, 1: manual mode	RW	0
	RegEnDDCMasterArb	7	DDC master arbitration function 0: disable, 1: enable	RW	1
3B					
3C					
3D					
3E	RSCDCUpdFlag[7:0]	7:0	Read '1' when the corresponding SCDC Update Flag is '1'	RO	0x00
3F	RSCDCUpdFlag[15:8]	7:0	Clear to '0' when FW write '1' to the corresponding bit at SCDC Update Flag	RO	0x00

HDCP Registers

Reg	Register Name	Bit	Definition		Default
40	Reg_AnGen	0	Write this bit 1 to start Cipher Hardware generating a random number for HDCP1.4. Write 0 to stop the Cipher Hardware. The generated Random number can be read back from VgenAn[63:0]	RW	0
	RAuthen_CS[6:0]	7:1	Authentication FSM current state When RegHDCP2Sel=0, this Authen_CS is for HDCP1.4 When RegHDCP2Sel=1, this Authen_CS is for HDCP2.2	RO	0x--
41	RegCPDesired	0	Write 1 to enable HDCP	RW	0
	RegAEnable1p1Feature	1	Enable HDMI Tx HDCP1.1 Feature	RW	0
	RegHDCPSyncDet	2	Enable HDCP 1.2 SyncDetect	RW	0
	RegKeepOutOpt	3	HDCP keepout option	RW	1
	RBKSVWrTrg	4	1: Trigger BKSV write to KSV FIFO	W1P	
		7:5			
42	RAuthen_Fire	0	Write 1 to start HDCP1.4/2.2 authentication process	W1P	-
	RList_Done	1	Write this bit when processing KSV list check interrupt event Write 1 when KSV FIFO list check pass	W1P	-
	RList_Fail	2	Write this bit when processing KSV list check interrupt event Write 1 when KSV FIFO list check fail	W1P	-
	RInitLFSR	3	Write 1 to this bit to trigger HDCP 2.2 random generator	W1P	-
	RegHDCP2Sel	4	HDCP function selection 0: HDCP 1.4, 1: HDCP 2.2	RW	0
	RegEnHDCP2RxIDSRMChk	5	HDCP 2.2 RxID SRM check 0: Disable, 1: Enable	RW	0
	RegEnHDCP2ListSRMChk	6	HDCP 2.2 KSV List SRM check 0: Disable, 1: Enable	RW	0
	RHDCP2ListRdDone	7	HDCP 2.2 KSV List read done	W1P	-
43 ~ 47	AKSV[39:0]	7:0	Read Only for AKSV[39:0]	RO	0x--
48 ~ 4F	RAN[63:0]	63: 0	When RegHDCP2Sel=0, Reg48~Reg4F are used for HDCP1.4. Random number used at HDCP Authentication. When RegHDCP2Sel=1, Reg48~Reg4F are used for HDCP2.2 which are defined as followings.	RW	0x--
48	Reserved	3:0		RW	0xF
	RegStreamID_Type[3:0]	7:4	When RegHDCP2Sel=1 StreamID_Type 0x0: Type 0 content stream 0x1: Type 1 content stream 0x2~0xF: reserved	RW	0x0
4B	RHDCP2Cap	0	When RegHDCP2Sel=1 HDCP2 capability read back by HW	RO	-
	RStreamManageOK	1	When RegHDCP2Sel=1 HDCP2 content stream management status	RO	-
		7:2	When RegHDCP2Sel=1		
4C	RHDCP2RxStatus[7:0]	7:0	When RegHDCP2Sel=1	RO	0x--
4D	RHDCP2RxStatus[15:8]	1:0	RRdMsgSize[9:0]	RO	--
		2	When RegHDCP2Sel=1 RREADY	RO	-

		3	When RegHDCP2Sel=1 RREAUTH_REQ	RO	-
		7:4	When RegHDCP2Sel=1 RHDCP2RxStatus[15:12]: reserved	RO	0x-
4E	RHDCP2AuthFailStatus[7:0]	7:0	When RegHDCP2Sel=1 HDCP 2.2 authentication fail status	RO	0x--
4F	RHDCP2AuthFailStatus[15:8]	7:0	When RegHDCP2Sel=1 HDCP 2.2 authentication fail status	RO	0x--
50 ~ 57	VgenAn[63:0]	63: 0	VgenAn[63:0] These 8 bytes are generated random number. To generate random number, please refer to Reg_AnGen.	RO	0x--
58 ~ 59	ARi	15: 0	Read Only for HDCP ARi[15:0]	RO	0x--
5A	APj	7:0	HDCP APj	RO	0x--
5B ~ 5F	BKSV[39:0]	39: 0	When RegHDCP2Sel=0: These Registers are Read Only for BKSV[39:0]	RO	0x--
5B ~ 5F	RRxID[39:0]	39: 0	When RegHDCP2Sel=1: These Registers are Read Only for RRxID[39:0]	RO	0x--
60	BRi[7:0]	7:0	HDCP BRi[7:0]	RO	0x--
61	BRi[15:8]	7:0	HDCP BRi[15:8]	RO	0x--
62	BPj[7:0]	7:0	HDCP BPj[7:0]	RO	0x--
63	BCaps[7:0]	0	1: Fast reauthentication. When set to 1, the receiver is capable of receiving (unencrypted) video signal during the session re-authentication. All HDMI-capable receivers shall be capable of performing the fast re-authentication even if this bit is not set. This bit does not change while the HDCP receiver is active.	RO	-
		1	1: HDCP 1.1 Features. Support HDCP Enhanced encryption status signaling (EESS), Advance Cipher, and Enhanced Link Verification options.	RO	-
		2	reserved. must be zero.	RO	-
		3	reserved. must be zero.	RO	-
		4	FAST. 1: the device supports 400KHz transfers.	RO	-
		5	KSV FIFO ready	RO	-
		6	HDCP Repeater capability	RO	-
		7	HDMI_Reserved	RO	-
64	BStatus[7:0]	6:0	When RegHDCP2Sel=0 DEVICE_COUNT Total number of attached downstream string devices.	RO	0x--
		7	When RegHDCP2Sel=0 MAX_DEVS_EXCEEDED 1: more than 127 downstream devices or KSV fifo.	RO	-
65	BStatus[15:8]	2:0	When RegHDCP2Sel=0 Three-bit repeater cascade depth.	RO	0x-
		3	When RegHDCP2Sel=0 MAX_CASCADE_EXCEEDED Topology error indicator. 1: more than seven levels of video repeater have been cascaded together.	RO	-
		4	When RegHDCP2Sel=0 HDMI_Mode 0: DVI mode, 1: HDMI mode	RO	-
		5	When RegHDCP2Sel=0 Reserved for future possible HDMI used.	RO	-
		6	When RegHDCP2Sel=0 reserved 0	RO	-
		7	When RegHDCP2Sel=0 reserved 0	RO	-
64	RRxInfo[7:0]	7:0	When RegHDCP2Sel=1 [0]: HDCP1_DEVICE_DOWNSTREAM [1]: HDCP2_LEGACY_DEVICE_DOWNSTREAM [2]: MAX_CASCADE_EXCEEDED [3]: MAX_DEVS_EXCEEDED [7:4]: DEVICE_COUNT[3:0]	RO	0x--
65	RRxInfo[15:8]	7:0	When RegHDCP2Sel=1 [0]: DEVICE_COUNT[4] [3:1]: DEPTH[2:0]	RO	0x--

			[7:4]: Reserved		
66	RAuthFailStatus[6:0]	0	DDC NACK Timeout	RO	-
		1	Invalid BKSv	RO	-
		2	R0 Link Integrity Check Fail	RO	-
		3	Ri Link Integrity Check Fail	RO	-
		4	Pj Link Integrity Check Fail	RO	-
		5	Sync Detect Fail	RO	-
		6	DDC HANG Timeout	RO	-
	RAuthenticated	7	HDCP Authenticated When RegHDCP2Sel=0, this bit is for HDCP1.4 When RegHDCP2Sel=1, this bit is for HDCP2.2	RO	-
67 ~ 6A					
6B	RegHDCPDisR0ShortRd	0	Test only for HDCP1/HDCP2 1: disable R0 short read in HDCP1 and disable RxStatus short read in HDCP2	WP	1
	RegHDCPDisRiShortRd	1	Test only for HDCP1 1: disable Ri short read in HDCP1	WP	0
	RKSVWrTrg	2	KSV List write to FIFO trigger	W1P	-
	RKSVWrBusy	3	KSV List write to FIFO Busy	RO	-
	Reserved	4		RW	0
	Reserved	5		RW	0
	RRepWaitTxEksDone	6	Upstream TX send EKs1 done	W1P	
	RRepWaitTxMgmDone	7	Upstream TX send RepeaterAuth_Stream_Manage done	W1P	
6C					
6D					
6E		7:0			
6F	RegEnAutoReAuthen	0	HW automatically fire authentication when authentication fail. 0: Disable, 1: Enable	RW	1
	RegEnRiChk2DoneInt	1	Ri/Pj check pass to trigger authentication done interrupt 0: Disable, 1: Enable	RW	1
	RegEnSyncDet2FailInt	2	HDCP 1.2 Sync. detection fail to trigger authentication fail interrupt 0: Disable, 1: Enable	RW	0
	RegEnHDCPAutoMute	3	Auto-AVMute before HDCP authentication done 0: Disable, 1: Enable	RW	1
		7:4			
70	RegSHASel[2:0]	2:0	See SHA_Rd_ByteX Registers below	RW	0x-
	RegPrevRiSel	3	See Prev_Ri [15:0]	RW	0
	Reserved	4		RW	0
	Reserved	5		RW	0
	Reserved	7:6		RW	00
71	SHA_Rd_Byte1[7:0]	7:0	V0h[7:0] when RegSHASel=000 V1h[7:0] when RegSHASel=001 V2h[7:0] when RegSHASel=010 V3h[7:0] when RegSHASel=011 V4h[7:0] when RegSHASel=100 Mi [7:0] when RegSHASel=101	RO	0x--
72	SHA_Rd_Byte2[7:0]	7:0	V0h[15:8] when RegSHASel=000 V1h[15:8] when RegSHASel=001 V2h[15:8] when RegSHASel=010 V3h[15:8] when RegSHASel=011 V4h[15:8] when RegSHASel=100 Mi[15:8] when RegSHASel=101	RO	0x--
73	SHA_Rd_Byte3[7:0]	7:0	V0h[23:16] when RegSHASel=000 V1h[23:16] when RegSHASel=001 V2h[23:16] when RegSHASel=010 V3h[23:16] when RegSHASel=011 V4h[23:16] when RegSHASel=100 Mi[23:16] when RegSHASel=101	RO	0x--
74	SHA_Rd_Byte4[7:0]	7:0	V0h[31:24] when RegSHASel=000 V1h[31:24] when RegSHASel=001 V2h[31:24] when RegSHASel=010 V3h[31:24] when RegSHASel=011 V4h[31:24] when RegSHASel=100 Mi[31:24] when RegSHASel=101	RO	0x--

75	Aksv_Rd_Byte5[7:0]	7:0	Mi[39:32] when RegSHASel=000 Mi[47:39] when RegSHASel=001 Mi[55:48] when RegSHASel=010 Mi[63:56] when RegSHASel=011	RO	0x--
76	Prev_Ri[7:0]	7:0	Prev_Ari[7:0] when RegPrevRiSel=0 Prev_Bri[7:0] when RegPrevRiSel=1	RO	0x--
77	Prev_Ri[15:8]	7:0	Prev_Ari[15:8] when RegPrevRiSel=0 Prev_Bri[15:8] when RegPrevRiSel=1	RO	0x--

AFE Control Registers

Reg	Register Name	Bit	Definition	Type	Default
83	Reserved	2:0		RW	000
	Reg_XP_HS2	3	0: HDMI1.4 mode, 1: HDMI2.0 mode	RW	0
		7:4			
84	Reg_XP_EC1	0	when input PCLK05 frequency < 100MHz	RW	0
	Reg_XP_ER0	1	set XP_EC1=1	RW	0
	Reg_XP_GAINBIT	2	set XP_ER0=1 set XP_GAINBIT=0	RW	1
	Reserved	3		RW	0
	Reserved	4		RW	0
	Reg_XP_PWDI	5	0: TXPLL current bias is not in power-down mode. 1: TXPLL current bias in power-down mode, no current source available for all other blocks such as IPLL/TXPLL driver/ring osc... Normally, XP_PWDI should always be set to 0.	RW	0
	Reg_XP_PWDPLL	6	Power down signal for TXPLL 0: Normal operation 1: TXPLL is power downed	RW	0
	Reg_XP_RESETB	7	Low-active reset signal for TXPLL 0: TXPLL is reset 1: Normal operation	RW	1
85	Reserved	7:0		RW	0x00
86	Reserved	0		RW	0
	Reserved	1		RW	0
	Reserved	2		RW	0
	Reg_XP_ALPWDB	3	XP_ALPWDB=0, txpll and tipll pvcc10 power is off XP_ALPWDB=1, txpll and tipll pvcc10 power is on Note: do not set XP_ALPWDB=0 and XP_PWDPLL=0	RW	1
		7:4			
87	Reg_DRV_ISW[4:0]	4:0	Output data channel current level setting Default: 3mA (always on) ISW0: +1.25mA ISW1: +2.5mA ISW2: +5mA ISW3: +10mA ISW4: +4.5mA	RW	00101
		7:5			
88	Reg_DRV_RST	0	Reset signal for HDMI_TX_DRV 1: all flip-flops in the transmitter are reset.	RW	1
	Reg_DRV_PWD	1	Power down signal for HDMI_TX_DRV 1: all flip-flops in the transmitter are reset while all other analog parts are powered off.	RW	0
	Reserved	2		RW	0
	Reg_DRV_PDRXDET	3	0: Normal operation 1: Power down monitor sense (RX detection) circuits	RW	0
	Reg_DRV_PISW[1:0]	5:4	Data channel pre-driver current level setting	RW	01
	Reg_DRV_PISWC[1:0]	7:5	Clock channel pre-driver current level setting	RW	00
89	Reg_DRV_RTERM[4:0]	4:0	Termination value. This value will impact HDMI differential signals slew rate, swing and eye diagram. The Reg_DRV_TERMON shall be 1 to have this termination active. Note: [4:3] is reserved for ECO option	RW	00000
	Reg_DRV_TERMON	5	Reserved, this bit must be 0 for normal operation. 0: transmitter termination resistors are disabled 1: transmitter termination resistors are enabled	RW	0

8A	Reserved	6		RW	1
	Reg_DRV_NOPE	7	0: enable pre-emphasis ,1: disable pre-emphasis function	RW	1
	Reserved	0		RW	0
		1		RW	0
		2		RW	0
		3		RW	0
	Reserved	4		RW	0
	Reserved	5		RW	0
	Reserved	6		RW	0
	Reserved	7		RW	0
8B	Reg_DRV_ISW_C[4:0]	4:0	Output clock channel current level setting Default: 3mA (always on) ISWC0: +1.25mA ISWC1: +2.5mA ISWC2: +5mA ISWC3: +10mA ISWC4: +4.5mA	RW	00001
	Reserved	5		RW	0
	Reserved	6		RW	0
	Reserved	7		WP	0
8C					
8D					
		7			
8E					
8F					

Input Data Processing Registers

Reg	Register Name	Bit	Definition	Type	Default
Input Data Format Registers					
90					
91	Reg_InColMod[1:0]	1:0	00: RGB mode 01: YUV422 mode 10: YUV444 mode 11: YUV420 mode	RW	00
	Reg_OutColMod[1:0]	3:2	00: RGB mode 01: YUV422 mode 10: YUV444 mode 11: YUV420 mode	RW	00
	RegEnVidBlack	4	Enable video black source	RW	0
	RegBlueScrMute	5	0: Set black screen output when AVMute=1 1: Set blue screen output when AVMute=1	RW	0
	RegAutoColMod	6	1: Auto ColMode from input	RW	1
	Reserved	7		RW	0
92					
93	RegColorClip	0	0: No clip 1: clip R/G/B/Y to 16~235, Cb/Cr to 16~240	RW	0
		5:1			
	Reserved	6		RW	0
	Reserved	7		RW	1
94	RegEnFFManualRst	0	Tx2V FIFO manual reset 0: disable, 1: enable	RW	0
	RegEnFFAutoRst	1	Tx2V FIFO auto reset enable 0: disable, 1: enable	RW	1
	RegEnPLLBufRst	2	Enable to reset Tx2V FIFO when PLL unlock 0: disable. 1: enable	RW	1

	RegEnXPLockChk	3	Enable to Check XP_lock for Tx2V FIFO reset 0: disable. 1: enable	RW	0
	RegXPStableTime[1:0]	5:4	XP_Lock stable time 01: 50us, 10:100us, others:75us	RW	00
		6			
95 ~ 9B					
9C	RTxCLKDiffLog[7:0]	7:0	TxCLK difference log	RO	0x--
9D	VRFIFOValid	0	Tx2V FIFO status: valid	RO	-
	TxWFIFOvWr	1	Tx2V FIFO status: over-write	RO	-
	VRFIFOvRd	2	Tx2V FIFO status: over-read	RO	-
	Reserved	3		RO	
	Reserved	4		RO	
	Reserved	5		RO	
		7:6			
9E	RefVidParaChgSts[7:0]	0	Video parameter change status → TxRec_HFP	RO	-
		1	Video parameter change status → TxRec_HSW	RO	-
		2	Video parameter change status → TxRec_HBP	RO	-
		3	Video parameter change status → TxRec_HDEW	RO	-
		4	Video parameter change status → TxRec_VFP	RO	-
		5	Video parameter change status → TxRec_VSW	RO	-
		6	Video parameter change status → TxRec_VBP	RO	-
		7	Video parameter change status → TxRec_VDEW	RO	-
9F	RefVidParaChgSts[10:8]	0	Video parameter change status → TxRec_VFP2nd	RO	-
		1	Video parameter change status → TxRec_HVR2nd	RO	-
		2	Video parameter change status → TxRec_Interlaced	RO	-
		7:3			

Packet FIFO Registers

Reg	Register Name	Bit	Definition	Type	Default
A0	RegPktEnManuRst	0	1: enable TX Packet FIFO manual reset	RW	0
	RegPktEnAutoRst	1	1: enable TX Packet FIFO auto reset	RW	1
	RegEnQueGBD	2	1: enable queue GDB packet	RW	1
	RegEnAudFlat	3	1: enable audio flat	RW	0
	RegFixCTSSel[1:0]	5:4	fix CTS selection 00: no change, 01: CTSx2, 10: CTS/2, 11: CTS/4	RW	00
	RegEnFw3DInfo	6	1: enable FW 3DInfoFrame	RW	0
A1	RegPktSkip[7:0]	7:0	Input packet skip setting	RW	0x08
A2	RegPktSkip[15:8]	7:0		RW	0x00
A3	RegPktSkip[23:16]	7:0		RW	0x00
A4	RegPktSkip[31:24]	7:0		RW	0x00
A5	Reserved	7:0		RW	0x01
A6	Reserved	7:0		RW	0x00
A7	Reserved	7:0		RW	0x00
A8	Reserved	7:0		RW	0x00
A9	RegPkt3DInfoSum[70]	7:0	FW setting of Pkt3DInfoSum[7:0]	RW	0x00
AA	RegPkt3DInfo04PB[70]	7:0	FW setting of Pkt3DInfo04PB[7:0]	RW	0x00
AB	RegPkt3DInfo05PB[70]	7:0	FW setting of Pkt3DInfo04PB[7:0]	RW	0x00
AC	RegPkt3DInfo06PB[70]	7:0	FW setting of Pkt3DInfo06PB[7:0]	RW	0x00
AD	RegPktAVIInfoY_Fw	0	1: use FW PktAVIInfoY[2:0]	RW	0
	RegPktAVIInfoC_Fw	1	1: use FW PktAVIInfoC[1:0]	RW	0
	RegPktAVIInfoQ_Fw	2	1: use FW PktAVIInfoQ[1:0]	RW	0
	RegPktAVIInfoEC_Fw	3	1: use FW PktAVIInfoEC[2:0]	RW	0
	RegPktAVIInfoYQ_Fw	4	1: use FW PktAVIInfoYQ[1:0]	RW	0
	RegPktAVIInfoVIC_Fw	5	1: use FW PktAVIInfoVIC[7:0]	RW	0
	RegPktAVIInfoSum_Fw	6	1: use FW PktAVIInfoSum[7:0]	RW	0
	RegEnFwAVIInfo	7	1: enable FW AVIInfoFrame	RW	0
AE	RegPktAVIInfoY[2:0]	2:0	FW setting of PktAVIInfoY[2:0]	RW	000

		3			
	RegPktAVIInfoC[1:0]	5:4	FW setting of PktAVIInfoC[1:0]	RW	00
	RegPktAVIInfoQ[1:0]	7:6	FW setting of PktAVIInfoQ[1:0]	RW	00
AF	RegPktAVIInfoEC[2:0]	2:0	FW setting of PktAVIInfoEC[2:0]	RW	000
		3			
	RegPktAVIInfoYQ[1:0]	5:4	FW setting of PktAVIInfoYQ[1:0]	RW	00
	RegFixCDCTS[1:0]	7:6	00: no change, 01: CD10to8, 10: CD12to8, 11: Rsvd	RW	00
B0	RegPktAVIInfoVIC[7:0]	7:0	FW setting of PktAVIInfoVIC[7:0]	RW	0x00
B1	RegPktAVIInfoSum[7:0]	7:0	FW setting of PktAVIInfoSum[7:0]	RW	0x00
B2	RXVPktAVIInfoSum[7:0]	7:0	Received PktAVIInfoSum[7:0]	RO	0x--
B3	RXVColorDepth[2:0]	2:0	Received ColorDepth[2:0]	RO	---
	RXVAVMute	3	Received AVMute	RO	-
		7:4			
B4 ~ BF					

HDMI Control Registers

Reg	Register Name	Bit	Definition	Type	Default
C0	RegHDMIMode	0	Set TX Mode 0: DVI mode 1: HDMI mode	RW	0
	RegEnH2Enc	1	HDMI 2.0 encoding 0: Disable, 1: Enable	RW	0
	RegEnH2Scr	2	HDMI 2.0 scrambling 0: Disable, 1: Enable	RW	0
	RegEnPktBlankGB	3	Guard band for packet transmission in blank period 0: Disable, 1: Enable	RW	0
	Reserved	4		RW	0
	Reserved	5		RW	1
	RegH2ClkRatio	6	HDMI2.0 TMDS bit clock ratio 0: 1/10, 1: 1/40	RW	0
		7			
C1	RegAVMute	0	Set AVMute 0: not AVMute, 1: AVMute	RW	1
	Reserved	1		RW	0
	Reserved	2		RW	0
	Reserved	3		RW	0
	RegColorDepth[2:0]	6:4	HDMI output color depth 000: 8-bit per color component 100: 8-bit per color component 101: 10-bit per color component 110: 12-bit per color component	RW	000
	RegAutoColorDepth	7	1: Auto ColorDepth from input	RW	1
C2	RegOESSCycle[4:0]	4:0	OEES cycles period The keep_out clock cycle in DVI mode. For definition of keep_out, please reference HDCP1.3a specification Appendix D.	RW	0x08
	Reserved	5		WP	0
	Reserved	6			0
	RegEncPause	7	Pause HDCP Encryption 0: encryption enable 1: encryption pause	RW	0
C3	Reserved	0		RW	0
	Reserved	1		RW	0

	Reserved	2		RW	0
	Reserved	3		RW	0
	RegPktGenCtrlEn	4	Enable General Control packet 0: disable, 1: enable	RW	0
	RegPktGenCtrlRpt	5	Repeat General Control packet 0: send once, 1: one for each field	RW	0
		7:6			
C4 ~ CE					
CF	Reserved	1:0		RW	00
	Reserved	3:2		RW	10
	Reserved	5:4		RW	00
		7:6			
D0 ~ D1					
D2	Reserved	4:0		RW	00000
	Reserved	5		RW	0
	Reserved	0		RW	0
	Reserved	1		RW	0
D3	VTxHLineNum[7:0]	7:0	VTxHLineNum[13:0], TMDS Clock base counted Horizontal line pixel count	RO	0x---
D4	VTxHLineNum[13:8]	5:0		RO	0x---
		7:6			
D5	VTxBlankNum[7:0]	7:0	VTxBlankNum[12:0], TMDS Clock base counted Horizontal blank pixel count	RO	0x---
D6	VTxBlankNum[12:8]	4:0		RO	0x---
		7:5			
D7	VPackPhase[1:0]	1:0	VPackPhase[1:0]	RO	--
	VDefPhase	2	VDefPhase	RO	-
	VAVMuteSts	3	AVMute status	RO	-
		7:3			

Video Record Registers

Reg	Register Name	Bit	Definition	Type	Default
D8	TxRec_HFP[7:0]	7:0	Horizontal front porch pixel number	RO	0x---
D9	TxRec_HFP[13:8]	5:0			
		7:6			
DA	TxRec_HSW[7:0]	7:0	Horizontal SYNC width pixel number	RO	0x---
DB	TxRec_HSW[13:8]	5:0			
		7:6			
DC	TxRec_HBP[7:0]	7:0	Horizontal back porch pixel number	RO	0x---
DD	TxRec_HBP[13:8]	5:0			
		7:6			
DE	TxRec_HDEW[7:0]	7:0	Horizontal DE width pixel number	RO	0x---
DF	TxRec_HDEW[13:8]	5:0			
		7:6			
E0	TxRec_HVR2nd[7:0]	7:0	Horizontal VSYNC rising position of 2nd field	RO	0x---
E1	TxRec_HVR2nd[13:8]	5:0			
		7:6			
E2	TxRec_VFP[7:0]	7:0	Vertical front porch line number	RO	0x---
E3	TxRec_VFP[12:8]	4:0			
		7:5			
E4	TxRec_VSW[7:0]	7:0	Vertical SYNC width line number	RO	0x---
E5	TxRec_VSW[12:8]	4:0			
		7:5			
E6	TxRec_VBP[7:0]	7:0	Vertical back porch line number	RO	0x---
E7	TxRec_VBP[12:8]	4:0			
		7:5			
E8	TxRec_VDEW[7:0]	7:0	Vertical DE width line number	RO	0x---
E9	TxRec_VDEW[12:8]	4:0			
		7:5			
EA	TxRec_VFP2nd[7:0]	7:0	Vertical front porch line number of 2nd field Note: for Even VTotal interlaced mode, VFP=VFP2nd+1 otherwise VFP=VFP2nd	RO	0x---
EB	TxRec_VFP2nd[12:8]	4:0			
		7:5			

EC	TxInterlaced	0	Interlaced mode	RO	-
	TxHSPol	1	Horizontal SYNC polarity	RO	-
	TxVSPol	2	Vertical SYNC polarity	RO	-
		7:3			
ED					
EE					
EF					

Misc. Registers

Reg	Register Name	Bit	Definition	Type	Default
F1					
	RegDDCDrv[1:0]	5:4	Driving setting for DDCSCL and DDCSDA	RW	01
	RegDDCSMT	6	DDCSCL/DDCSDA Schmitt trigger option	RW	0
	RegHPDPD	7	1: enable internal HPD pull-down resistor	RW	0
F3					
FC	RegEnCRCLK	0	0: power-down CRCLK and disable CEC function 1: enable CRCLK for CEC function	RW	0
		7:1			

Chap 15 CEC Registers in Bank1

CEC Control Registers

Reg	Name	Bit	Definition	Type	Default
100 ~ 105					
106	Reg_Tx_Int_Mask	0	0: Interrupt Enable	RW	1
	Reg_Rx_Int_Mask	1	1: Interrupt Mask	RW	1
	Reg_RxFail_Int_Mask	2		RW	1
	Reg_TxDone_Int_Mask	3		RW	1
	Reg_RxDone_Int_Mask	4		RW	1
	Reg_TxFail_Int_Mast	5		RW	1
	Reg_CmdOvFlw_Int_Mask	6		RW	1
	Reg_DataOvFlw_Int_Mask	7		RW	1
107		1:0			
	Reserved	4:2		RW	000
	RegCECIntStsSel	5	0: read back interrupt status after mask 1: read back interrupt status before mask	RW	0
		7:6			
108	Reg_CECInt_En	0	CEC interrupt enable 1: Enable 0: Disable	RW	0
		1			
	Reg_CEC_Rst	2	Reset CEC block 1: Enable 0: Disable	RW	0
	Reg_CEC_SMT	3	Schmitt trigger of CEC IO 1: Enable 0: Disable	RW	1
	Reg_CEC_Force	4	Force CEC output regardless of normal function	RW	0
	Reg_CEC_OE	5	Force CEC output value	RW	0
	Reserved	6		RW	1
	Fire_Frame	7	1: Fire CEC command out	RW	0
109	Reg_En100ms_Cnt	0	Used as a reference 100ms time interval for CEC calibration.	RW	0
	Reg_NACK_En	1	Acknowledge from follower to initiator 1: NACK 0: ACK	RW	0
	Reg_Pulse_Sel	2	Select illegal bit as error bit 1: Enable 0: Disable	RW	0
	Reg_ACKTrig_Sel	3	Acknowledge for broadcast mode 1: ACK 0: NACK	RW	0
	Reg_Retire	4	Retry to fire CEC command. 1: Enable 0: Disable	RW	0
	Reg_RxSelf_Sel	5	Initiator received CEC bus data. 1: Disable 0: Enable	RW	1
	Reg_Region_Sel	6	Select region for error bit 1: Whole region 0: Region form state Start to IDLE	RW	0
	Reg_DataBit_Sel	7	Select data bit 1: Increase 0.1ms 0: Normal	RW	0
10A	Reg_BusFree_Sel[1:0]	1:0	Select bus free bit time. 00: Automatic 01: 3 bit time. 10: 5 bit time 11: 7 bit time	RW	00
	Reg_BitEnd_Sel[1:0]	3:2	Select logic 0 and logic 1 output bit time. 00: Standard 01: Logic 0 maximum and logic 1 minimum 10: Logic 1 maximum and logic 0 minimum	RW	0

	Reg_ArBit_Sel	4	Select bit time for arbitration lose. 1: 3 bit time 0: 5 bit time	RW	0
	Reg_HWRtyDis	5	1: Disable HW 5 times Retry of TX	RW	0
	Reg_EnCECRx	6	1: Enable CEC RX function	RW	0
		7			
10B	Reg_Data_Min	7:0	Minimum data bit time	RW	0x14
10C	Reg_Timer_Unit	7:0	CEC timer unit, nominally 100us. This value should be decided from MS_Count.	RW	0x59
10D		3:0			
	Reg_CEC_IOPU	4	0: Normal, 1: Pull-up	RW	1
	Reg_CEC_Drv[1:0]	7:6	00: 2.5mA, 01:5mA, 10:7.5mA, 11:10mA	RW	01
10E ~ 10F					

CEC Initiator Registers

Reg	Name	Bit	Definition	Type	Default
110	Reg_Tx_Header	7:0	CEC initiator command header	RW	0x00
111	Reg_Tx_Opcode	7:0	CEC initiator command opcode	RW	0x00
112	Reg_Tx_Operand1	7:0	CEC initiator command operand1	RW	0x00
113	Reg_Tx_Operand2	7:0	CEC initiator command operand2	RW	0x00
114	Reg_Tx_Operand3	7:0	CEC initiator command operand3	RW	0x00
115	Reg_Tx_Operand4	7:0	CEC initiator command operand4	RW	0x00
116	Reg_Tx_Operand5	7:0	CEC initiator command operand5	RW	0x00
117	Reg_Tx_Operand6	7:0	CEC initiator command operand6	RW	0x00
118	Reg_Tx_Operand7	7:0	CEC initiator command operand7	RW	0x00
119	Reg_Tx_Operand8	7:0	CEC initiator command operand8	RW	0x00
11A	Reg_Tx_Operand9	7:0	CEC initiator command operand9	RW	0x00
11B	Reg_Tx_Operand10	7:0	CEC initiator command operand10	RW	0x00
11C	Reg_Tx_Operand11	7:0	CEC initiator command operand11	RW	0x00
11D	Reg_Tx_Operand12	7:0	CEC initiator command operand12	RW	0x00
11E	Reg_Tx_Operand13	7:0	CEC initiator command operand13	RW	0x00
11F	Reg_Tx_Operand14	7:0	CEC initiator command operand14	RW	0x00
120	Reg_Tx_Operand15	7:0	CEC initiator command operand15	RW	0x00
121	Reg_Tx_Operand16	7:0	CEC initiator command operand16	RW	0x00
122	Reg_Logical_Addr	3:0	CEC target logical address	RW	0x0F
		7:4			
123	Reg_Out_Num	4:0	CEC output byte size in a frame	RW	0x00
		7:5			
124 ~ 12F					

CEC Follower Registers

Reg	Name	Bit	Definition	Type	Default
130	Ref_Rx_Header	7:0	CEC follower command header	RO	0x00
131	Ref_Rx_Opcode	7:0	CEC follower command opcode	RO	0x00
132	Ref_Rx_Operand1	7:0	CEC follower command operand1	RO	0x00
133	Ref_Rx_Operand2	7:0	CEC follower command operand2	RO	0x00
134	Ref_Rx_Operand3	7:0	CEC follower command operand3	RO	0x00
135	Ref_Rx_Operand4	7:0	CEC follower command operand4	RO	0x00
136	Ref_Rx_Operand5	7:0	CEC follower command operand5	RO	0x00
137	Ref_Rx_Operand6	7:0	CEC follower command operand6	RO	0x00
138	Ref_Rx_Operand7	7:0	CEC follower command operand7	RO	0x00
139	Ref_Rx_Operand8	7:0	CEC follower command operand8	RO	0x00
13A	Ref_Rx_Operand9	7:0	CEC follower command operand9	RO	0x00
13B	Ref_Rx_Operand10	7:0	CEC follower command operand10	RO	0x00
13C	Ref_Rx_Operand11	7:0	CEC follower command operand11	RO	0x00
13D	Ref_Rx_Operand12	7:0	CEC follower command operand12	RO	0x00
13E	Ref_Rx_Operand13	7:0	CEC follower command operand13	RO	0x00
13F	Ref_Rx_Operand14	7:0	CEC follower command operand14	RO	0x00

140	Ref_Rx_Operand15	7:0	CEC follower command operand15	RO	0x00
141	Ref_Rx_Operand16	7:0	CEC follower command operand16	RO	0x00
142	Ref_In_Cnt	4:0	CEC follower received bytes.	RO	0x0
		7:5			

CEC Misc. Registers

Reg	Name	Bit	Definition	Type	Default
143	Ref_Out_Cnt	4:0	CEC initiator output bytes.	RO	0x--
		7:5			
144		0			
	Ref_Bus_Status	1	Bus status. 0: Busy 1: Free	RO	-
	Ref_Out_Status	3:2	Output status. 00: Received ACK 01: Received NACK 10: Retry, if no ACK, NACK or arbitration lose. 11: Fail	RO	--
	Ref_Err_Status	5:4	Error status. 00: No error occurs. 01: Received data period < minimum data bit period. 10: Illegal held-low period. 11: Both	RO	--
	Ref_Ready_Fire	6	Bus ready for firing a CEC command.	RO	-
		7			
145	Ref_MS_Count	7:0	MS_Count[7:0]	RO	0x--
146	Ref_MS_Count	7:0	MS_Count[15:8]	RO	0x--
147	Ref_MS_Count	3:0	MS_Count[19:16]	RO	0x-
		7:4			
148	Reserved	3:0		RO	0x-
	Reserved	4		RO	-
	Reserved	5		RO	-
	Reserved	7:6			
149	Reserved	4:0		RO	0x--
	Reserved	7:5			
14A	Reserved	3:0		RO	0x-
	Reserved	7:4			
14B	Reserved	7:0		RW	0x00
14C	Tx_Int	0	CEC initiator output byte interrupt. W: Write 1 clear this interrupt	W1C	0
	Rx_Int	1	CEC follower received byte interrupt. W: Write 1 clear this interrupt	W1C	0
	RxFail_Int	2	CEC received fail interrupt. W: Write 1 clear this interrupt	W1C	0
	TxDone_Int	3	CEC output finish interrupt. W: Write 1 clear this interrupt	W1C	0
	RxDone_Int	4	CEC received finish interrupt. W: Write 1 clear this interrupt	W1C	0
	TxFail_Int	5	CEC initiator output fail interrupt. W: Write 1 clear this interrupt	W1C	0
	CmdOvFlw_Int	6	CEC command FIFO over flow interrupt. W: Write 1 clear this interrupt	W1C	0
	DataOvFlw_Int	7	CEC data FIFO over flow interrupt. W: Write 1 clear this interrupt	W1C	0
14D	RxCmd Byte0[7:0]	7:0	Command FIFO Rx_Header	RO	0x00
14E	RxCmd Byte1[7:0]	7:0	Command FIFO Rx_Opcode	RO	0x00
14F	RxCmd Byte2[7:0]	4:0	Command FIFO In_Cnt	RO	0000
		6:5	Command FIFO Error_Status	RO	00
		7	Command FIFO CEC Rx_Fail	RO	0
150	RxData[7:0]	7:0	Data FIFO Rx_Operand	RO	0x00
151	RxCmdValid	0	Read command FIFO valid	RO	0
	RxCmdFull	1	Command FIFO full	RO	0
	RxCmdOvWr	2	Write command FIFO over flow	RO	0
	RxCmdOvRd	3	Read command FIFO over flow	RO	0
	RxDataValid	4	Read data FIFO valid	RO	0

	RxDataFull	5	Data FIFO full	RO	0
	RxDataOvWr	6	Write data FIFO over flow	RO	0
	RxDataOvRd	7	Read data FIFO over flow	RO	0
152	RxCmdStg[3:0]	3:0	Rx command FIFO stage	RO	0000
		4			
	RxCmdFIFORst	5	CEC command FIFO reset 1:Reset	RW	0
	RxDataFIFORst	6	CEC data FIFO reset 1:Rest	RW	0
	RxCECFIFOEn	7	CEC FIFO 1: Enable 0:Disable	RW	0
153	RxDataStg[5:0]	5:0	Rx data FIFO stage	RO	000000
		7:6			
154	Ref_TxFailStatus[2:0]	0	1: receive ACK	RO	-
		1	1: receive NACK	RO	-
		2	1: retry	RO	-
		7:3			